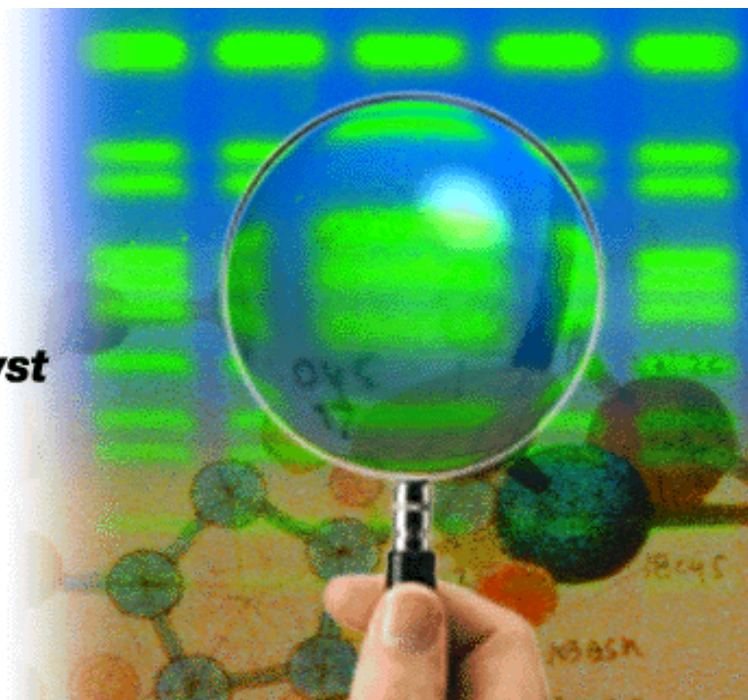




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Hercules, CA 94547

Multi-Analyst

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Build 33
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Multi-Analyst®/PC

PC Software for Bio-Rad's Image Analysis Systems

Version 1.1

Instruction Manual

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1. Introduction to Multi-Analyst[®]/PC

1.1 Introduction

Multi-Analyst[®]/PC data analysis software is a powerful analytical tool for controlling and analyzing images and data from Bio-Rad's image analysis systems. The Multi-Analyst/PC software uses a series of pull down menus and tool palettes to allow one to easily scan, display, analyze, and output data from many types of samples. The determination of valuable information such as sample concentration, relative mobility, and molecular weight, is easily performed with the software's graphical interface. It is recommended that the first-time user follow the tutorial sections of this manual to become acquainted with some of the Multi-Analyst/PC functions and features.

1.2 Image Analysis Process

1.2.1 Overview

An overview of the image analysis process is diagrammed below. After an image is acquired (either from the Fluor-S MultiImager, Molecular Imager, Densitometer, or Gel Documentation system) and saved in a digitized form, the image can be adjusted or optimized for (1) printing, (2) performing volume analysis, (3) performing profile analysis, or (4) exporting to other software.

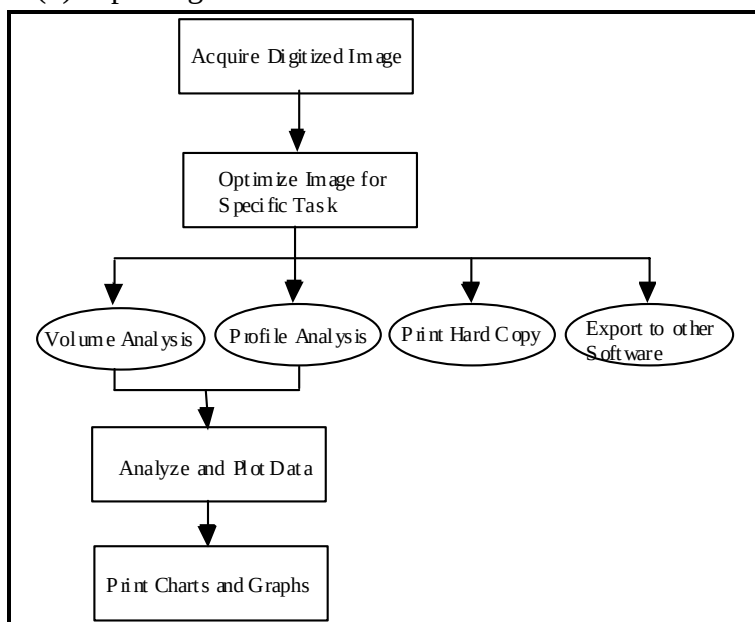


Figure 1-1 Image Analysis Process

capable of converting biological signals into digital data. The digital data are then displayed on the computer in a two-dimensional format using the Multi-Analyst software. The intensity of the various signals (i.e. bands or spots) is directly proportional to the intensity of the gray

Data obtained by volume analysis or profile analysis may be used for molecular weight or concentration determination, subjected to regression analysis, analyzed by other functions, or exported to other software. Each step of the image analysis process will be described in detail in the tutorial and the sections following.

1.2.2 Digital data

The Gel Doc, Densitometer, Molecular Imager and Fluor-S MultiImager systems are light and/or radiation detectors that are

levels displayed on the computer monitor. The stronger the signal, the darker the image displayed on the monitor.

The signal displayed on the monitor is generated from the assembly of thousands of tiny individual screen pixels, each of which processes its own intensity level or count (Figure 1-2). In order for a signal to be visible, the counts of these congregated pixels must be higher than those pixels that represent the background. The total count of a signal is determined by summing up the individual intensity or count of all the pixels (Equation 1). The mean count or intensity of a signal is the total intensity divided by the total number of pixels (Equation 2).

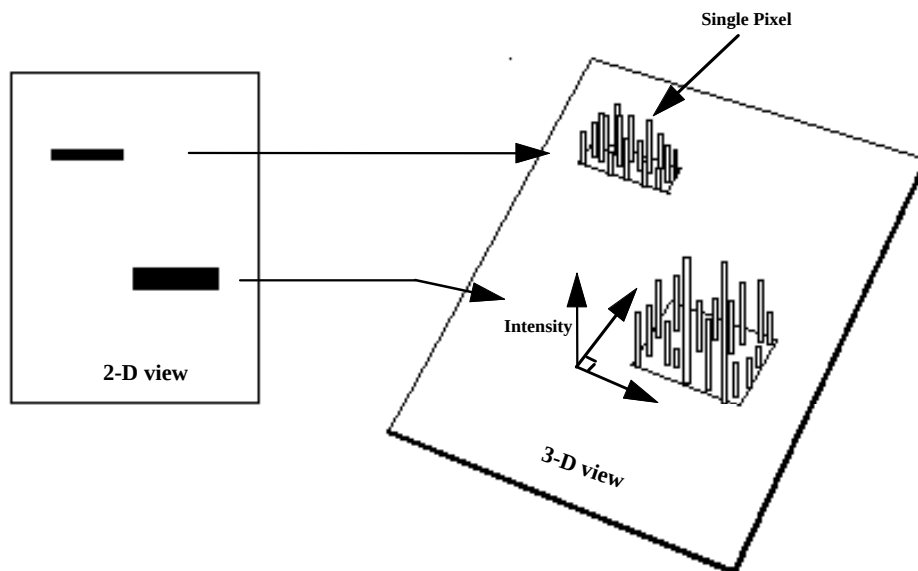


Figure 1-2 Pixel representation of digital signals

Equation 1

Total PDU or O.D. = $\sum (I_1 + I_2 + \dots)$, where I = intensity or count of each pixel

Equation 2

Mean PDU or O.D. = Total PDU or intensity / total number of pixels

The digitized data provide three important pieces of information necessary for qualitative and quantitative analysis (See Figure 1-2):

- X-coordinate which indicates the signal horizontal position
- Y-coordinate which indicates the signal vertical position
- Z-coordinate which indicates the signal intensity

The x- and y-coordinates define the exact location of the signal, whereas the z-coordinate establishes the intensity of the signal. The units of the x- and y-coordinates are millimeters or centimeters. The units of the z-coordinate (intensity) are "Optical Density" (O.D.) when using any of the Densitometers, and O.D. is also used with the Gel Doc 1000 with a white light source, or the Fluor-S MultiImager with white light illumination. When using the UV light source with either the Gel Doc 1000 or the Fluor-S MultiImager intensity is expressed

as light intensity or “counts”. “Pixel Density Units” (PDU) is used to express intensity when using the Molecular Imager systems.

1.2.3 Acquiring Images

Multi-Analyst/PC software allows the acquisition of images from Bio-Rad’s Fluor-S MultiImager systems, Densitometer (GS-690 and GS-700), Gel Documentation (Gel Doc 1000), and Molecular Imager (GS-505 and GS-525) systems. This integrated image analysis software provides a fast method to acquire images from any of these instruments, thus allowing analysis of radioactive, chemiluminescent, chemifluorescent, fluorescent and colorimetric samples. Multi-Analyst/PC will also open TIFF images acquired from Multi-Analyst®/Macintosh, Molecular Analyst®/PC, Molecular Analyst®/Mac.

1.2.4 Understanding Documents, Templates, and Experiments

Multi-Analyst/PC is set up for labs to be as streamlined as possible and to provide as much automation control of the instruments as possible to ensure Good Laboratory Practice. In this light, it is necessary to understand the difference between Documents, Notebooks, Templates, Windows (also known as a Views), as these key terms will be used throughout the manual.

1.2.4.1 Documents

Documents include everything about your experiment: all of the settings used to capture the image, the image generated (whether single-channel or multi-channel), object overlays such as lanes or volume objects, settings used to generate reports, reports that you have generated through your data analysis. All of this data are stored in the *.BIF files. There are no other files or associations.

1.2.4.2 Experiment Notebook

The *Experiment Notebook* contains all of the experiment settings used to capture your image along with all of the settings for standards, report headings and layouts, and any comments you might want to type in. It is divided into tabbed sheets, each containing a different set of information (not all tabs appear in all notebooks): General, Image Description, Tiff Description, Acquisition, Profile Analysis, Profile Report, Volume Report, Lane and Band Finding.

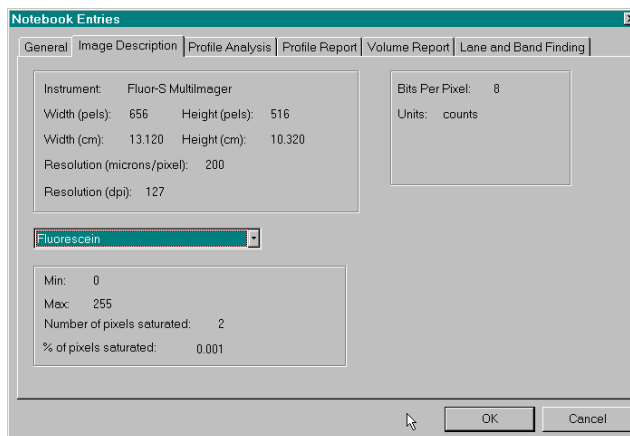


Figure 1-3 Experiment Notebook

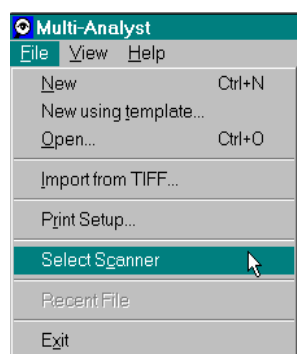
1.2.4.3 Templates

Templates can be considered a subset of an *Experiment Notebook*. As a matter of fact, if you want to exactly reproduce an experiment, all you need to do is to open any Bio-Rad document and save it as a Template. All of the settings used to capture the original image are stored (the imaging machine type as well as area scanned, time of exposure, etc., the standard layout, the equations used to calculate the standard curves, and the lanes used for the standard, and the report layouts). All of the settings needed to replicate the experiment are there. The only items the Template does not save are the images and the overlay information.

Templates are in fact the Key to Bio-Rad's Multi-Analyst software. With a Template you are able to:

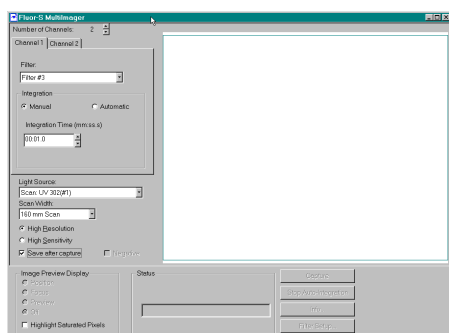
- Have streamlined profile analysis of single and dual channel fluorescent images through reuse of document templates
- Have exact replication of instrument settings
- Quickly perform repetitive experiments
- Create user templates to aid in multi-user environments
- Calculate MW and % calculation with no user intervention

1.2.5 Two methods to acquire images using Multi-Analyst/PC



New in the *File* menu will bring up the acquisition screen for the default Bio-Rad imaging device with acquisition settings filled in from the default document template. If the default template file is not found, an empty template will be used.

Figure 1-4 File Menu



New using template... allows acquisition using previously created and saved templates. This feature makes it easy to repeat experiments using the same settings.

Figure 1-5 Fluor-S Acquisition Window

1.2.6 Optimizing Images

Images from Multi-Analyst/PC can be optimized using a variety of tools. The color palette, transformation tool, and other settings can adjust faint or saturated images to aid in visualization and printing.

1.2.7 Image Display

The image display window is used to display the set of images corresponding to an experiment/document. For multi-channel images there will be two ways of viewing these images: one image at a time or several images at a time via side by side comparison.

Once displayed, images from Multi-Analyst/PC can be optimized using a variety of tools. The color palette and transformation tool, can adjust faint or saturated images to aid in visualization and printing.

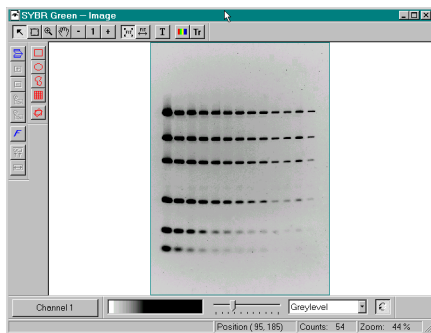


Figure 1-6 Single Image View

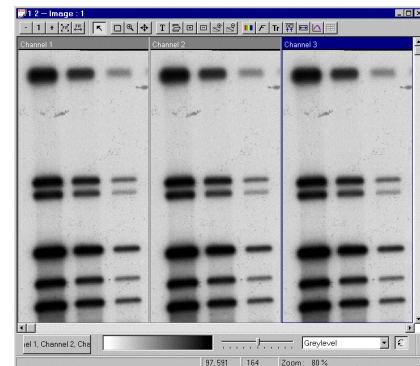


Figure 1-7 Multiple Image View

1.2.8 Color Palette

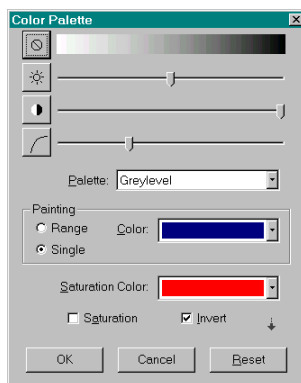


Figure 1-8 Color Palette Control Box

With this Color Palette, adjustment can be made to the way each channel is displayed. The three sliders respectively (from top to bottom) control the brightness, the contrast, and the color distribution of the palette. They all have a button on their left to reset their values. Below the sliders is a combo-box used to select from a variety of color palettes chosen to map closely to commonly used stains and fluorophores to allow printing of color images which closely resemble the color of the samples. Some examples include: Coomassie, Ethidium, and FITC.

Below the Painting area is a combo-box used to select the saturation color. That is the color that will be used to show saturated pixels in the image if the Saturation box is checked.

1.2.9 Transformation Control

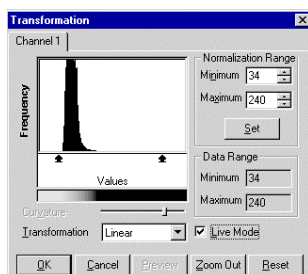


Figure 1-9 Transformation Control

Image transformation is important for visualizing very faint signals and allows a greater range of control for re-mapping the image for viewing than does the color palette.

1.2.10 Automated Lane and Band Finding

Once the image is captured, lanes and bands can be found automatically.

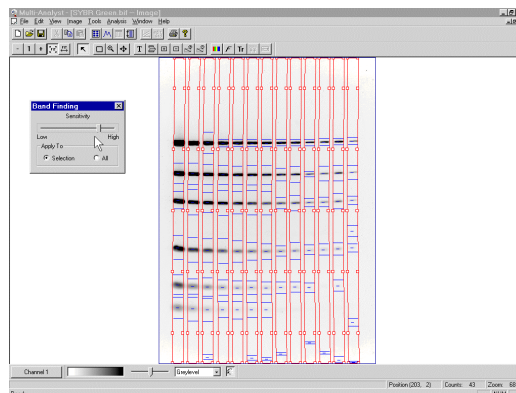


Figure 1-10 Automatic Lane and Band Finding

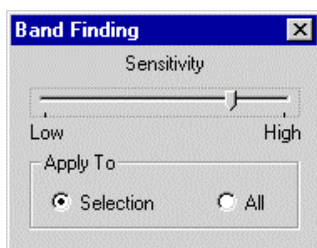


Figure 1-11 Band Finding Sensitivity Adjustment Panel

Once bands are identified, the Band Sensitivity can be easily adjusted with close to real time updates appearing on the screen. Simply choose to apply adjustments to all lanes or selected lanes. Move the slider to adjust the band finding sensitivity so that bands of interest are found.

1.2.11 Profile Analysis

Profile analysis determines relative differences in intensities and positions of bands or spots in an image. Using the profile tools, a two-dimensional graph or profile is generated and values such as “Percent of Total” and “Molecular Weight” can be calculated.

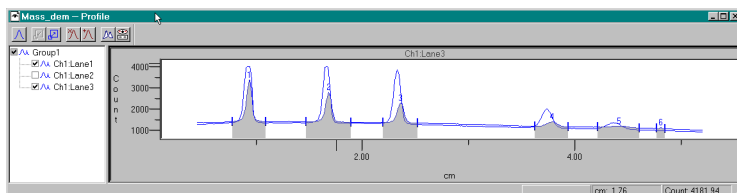


Figure 1-12 Profile Window

1.2.12 Profile Analysis - Data Table

To view the data simply select Profile Report from the View menu. Image, Profile, and Profile Data windows can be sized as desired for display.

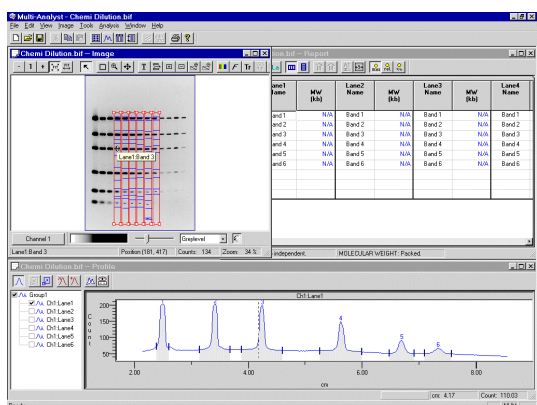


Figure 1-13 Profile Analysis Data Table

1.2.13 Volume Analysis

Volume analysis fully exploits the three-dimensional character of the digitized data. It allows accurate quantitation of unknown samples by interpolation from known standards.

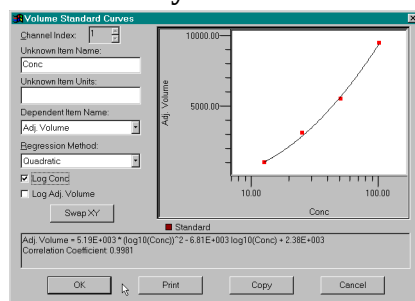


Figure 1-14 Regression Curve for Calculation of Concentration

1.2.14 Printing

Multi-Analyst/PC will print images and reports to a variety of printers. There is also a convenient *Export to Excel* function which allows rapid export of data and images into Excel.

1.2.15 Printing of Images in Actual Size

Images can be easily printed in actual size by selecting the Actual Size button.

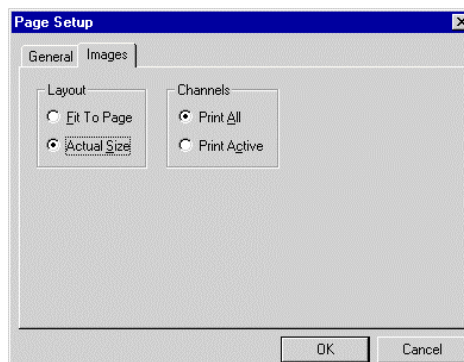


Figure 1-15 Selection of Actual Size Printing of Images

1.2.16 Print Preview

The print preview window shows what will be printed in a WYSIWYG manner.

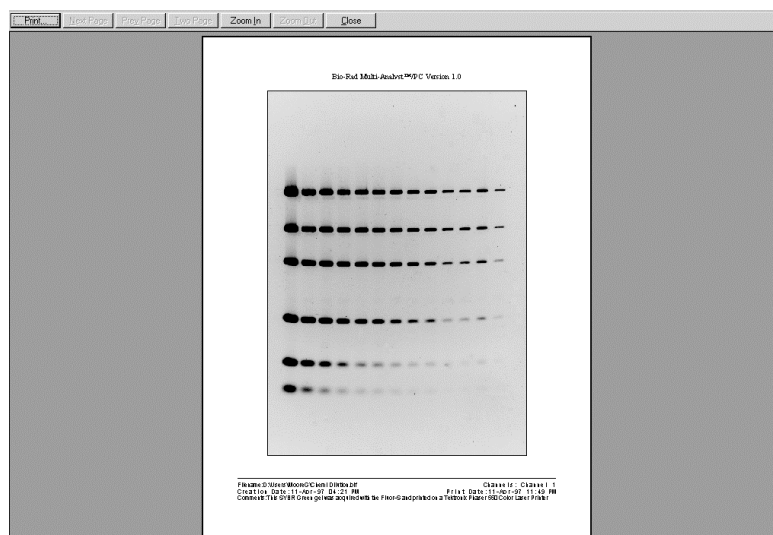


Figure 1-16 Print Preview

1.2.17 Data Export

The results obtained from Multi-Analyst/PC software can be exported to a variety of other software applications. For example, clicking on Export to Excel, your active notebook or

window will be completely copied into Excel, ready for further analysis. In addition, with simple Copy and Paste commands, the numerical data can be easily transferred to other applications for further analysis, graph and chart generation. Images may be exported as TIFF format images for insertion into applications such as Microsoft Word®, Excel® and PowerPoint®

1.3 System Requirements

Note: Please see the Release notes included with Multi-Analyst/PC for up to date information on system and SCSI card requirements.

The minimum computer requirements to run Bio-Rad's Multi-Analyst/PC software are:

1.3.1 Computer

The minimum specification to run the Bio-Rad Multi-Analyst/PC software with the Fluor-S MultiImager, Molecular Imager, and Densitometer is a Pentium 166 MHz or higher, 32 MB of RAM, 2 GB Hard Disk, 17" Monitor (1024 x 768 x 256 colors), Windows NT 4.0, or Windows 95 CD-ROM, 3.5" Diskette Drive.

Recommended specification: Pentium Pro 200 MHz, 64 MB RAM or more, 4 GB Hard Disk or larger, 17" Monitor (1600 x 1200 x 256 colors), Windows NT 4.0, or Windows 95 Microsoft Excel, CD-ROM, 3.5" Diskette Drive. An external storage device is also recommended.

Compatibility and performance of computers not purchased from Bio-Rad are the sole responsibility of the user. Please check with your local Bio-Rad representative regarding compatibility of your specific computer.

1.3.2 SCSI Communication

To use Multi-Analyst/PC software to acquire images from the Fluor-S MultiImager, Molecular Imager, and Densitometer, a SCSI card must be installed in the computer. Bio-Rad recommends certain specific SCSI cards. See the release notes for specific cards that are compatible with the Fluor-S MultiImager, Molecular Imager, and Densitometer.

To properly communicate via SCSI with the Fluor-S MultiImager, the file Wnaspi32.dll must be installed in the System 32 folder within the Windows folder. Wnaspi32.dll is supplied with EZ-SCSI from Adaptec Corp. For more information regarding Adaptec cards and drivers please visit the Adaptec web site at www.adaptec.com.

2. Getting Started

2.1 Hardware Protection Key

(ALL POWER MUST BE TURNED OFF PRIOR TO ATTEMPTING ANY CONNECTION TO THE HARDWARE PROTECTION KEY).

A hardware protection key (HPK) is required for use of the Multi-Analyst/PC software. A single-user HPK is included with each copy of Multi-Analyst/PC software. Connect the PC HPK to any free parallel port at the rear of the PC. For an additional cost, multi-user (5 or 10 user) network HPKs are also available through Bio-Rad. The Multi-User Hardware Protection Key, Catalog Number 170-7714, allows up to 5 users to use the software over the network at the same time.

To install network hardware protection keys, first connect the hardware key as described above. Next, install or reinstall Multi-Analyst/PC software and select “Install Hardware Key Driver” from the “Bio-Rad Program Group”. Once installed, reboot the computer.

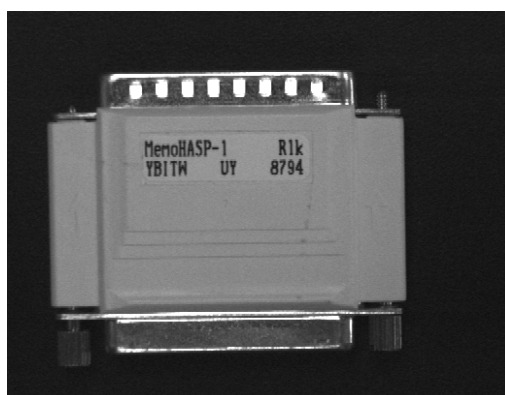


Figure 2-1 Hardware Protection Key

2.2 Installing Multi-Analyst/PC

It is assumed that you are familiar with the Windows NT 4.0 or Windows 95 Operating System. For more information on how to logon to Windows NT see your network administrator or refer to your Windows NT Basics and Installation Manual which came with your Windows NT software. Windows NT software is not included with your Bio-Rad Multi-Analyst/PC software and thus must be purchased separately.

To install Multi-Analyst/PC:

1. Insert the Multi-Analyst/PC “DISK 1” into the floppy disk drive. Select Run and type in a:setup. Note: the exact drive designation will depend on your particular computer configuration.

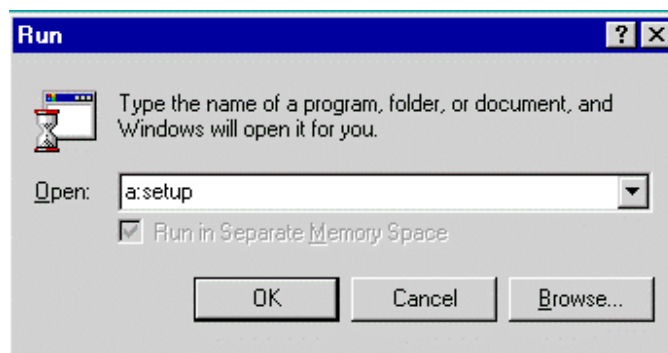


Figure 2-2 Run Setup Window

2. The program will then automatically prompt for the proper disks.
3. The screen will display a dialog box indicating the status of the installation procedure. the Multi-Analyst installation program will create an application folder titled Multi-Analyst.
4. The installation program also installs the application Multi-Analyst/PC, Release Notes, and the Install Hardware Drivers program.
5. Note: It is important to Install the Hardware Key Drivers so that the Hardware protection key will be recognized. You must have administrator access to your computer to install the HPK drivers. Select this item and launch this application which will install the hardware protection key (HPK) drivers so that your key will be recognized.

2.3 Start Up



Multi-Analyst 1.0

**Figure 2-3
Multi-Analyst
Icon**

Start the Multi-Analyst/PC program by double clicking on the Multi-Analyst icon.

After the program is loaded, it will display the title screen and the menu bar will appear. When the application starts up, it will first check the system configuration. If the configuration does not meet the minimum requirements, an appropriate dialog box will be displayed to the user and the application will quit. For questions regarding the compatibility of your computer please contact your local Bio-Rad representative.

Your Multi-Analyst/PC is now ready to use.

2.4 Instrument Installation

In order to acquire an image through Bio-Rad's Imaging systems, the host computer must be properly installed to the Bio-Rad instrumentation. For more information, it is recommended you refer to the appropriate Hardware manual for the instrument or contact your local Bio-Rad office.

2.4.1 Fluor-S MultiImager

The host computer must be connected to the Fluor-S MultiImager system via the PC's SCSI interface. (ALL EQUIPMENT MUST BE TURNED OFF PRIOR TO ATTEMPTING ANY SCSI CONNECTION).

1. Make sure the SCSI ID is unique. Each SCSI device is known by its number and conflicts will occur if duplicate ID's are used.
2. Attach one end of the SCSI-PC cable to the SCSI port on the back of PC.
3. Connect the other end of the SCSI cable to the SCSI port on the back of the Fluor-S MultiImager system.
4. The SCSI terminator must be placed at the end of the SCSI connection. Place the SCSI terminator at the back of the Fluor-S if it is the only SCSI device installed.
5. Turn on the Fluor-S, wait for it to initialize, then turn on the PC. Please see the Fluor-S MultiImager Installation manual for more details.

2.4.2 GS - 525/ 505 Molecular Imager

Installation of Bio-Rad's Molecular Imager Systems is completed by a field service technician. The Molecular Imager must be turned on first before turning on the computer.

2.4.3 Gel Doc 1000

The host computer must be connected to the Gel Doc 1000 via Bio-Rad's Frame Grabber board interface. (ALL EQUIPMENT MUST BE TURNED OFF PRIOR TO ATTEMPTING ANY CONNECTION).

WARNING!!! DO NOT SWITCH ANY CURRENTLY INSTALLED PCI CARDS INTO DIFFERENT SLOTS. IF YOU DO, YOU MAY HAVE TO REINSTALL YOUR ENTIRE SYSTEM OPERATING SOFTWARE!!!!!!

- 1) Insert the frame grabber card into an available PCI slot.
- 2) Plug the 9-pin DB video input connector of the integrating cable into the frame grabber card.
- 3) Connect the BNC video output connector (labeled Camera-Video) and the 8-pin "MINI DIN" integration control connector (labeled Camera-AUX) into the camera "Video" and "AUX" ports, respectively.
- 4) If using a Mitsubishi P68 or P90W video printer, connect the 1/8" male jack labeled "Printer Remote" into the remote outlet on the back of the printer and connect the BNC connector labeled "Video In" into the printer's video-in port.

SEE APPENDIX D/E TO SETUP THE VIDEO PRINTERS FOR THE PC.

2.4.4 GS-700 Densitometer/ GS-690 Densitometer

The host computer must be connected to the Model GS-700 Densitometer or GS-690 Densitometer via the PC's built-in SCSI interface. (ALL EQUIPMENT MUST BE TURNED OFF PRIOR TO ATTEMPTING ANY CONNECTION).

- 1) Make sure the SCSI ID is unique. Each SCSI device is known by its number and conflicts will occur if duplicate ID's are used
- 2) Attach one end of the SCSI-PC cable to the SCSI port in the PC.
- 3) Connect the other end of the SCSI cable to the SCSI terminator to the Densitometer if the Densitometer is the last item in the SCSI chain.
- 4) Place the SCSI-terminator at the end of the connection.
- 5) Turn On the Densitometer, then turn on the PC.

3. Multi-Analyst/PC Basics

3.1 Multi-Analyst/PC Desktop

Once you start the software, the Multi-Analyst/PC Banner information will be displayed for a few seconds.

Clicking the mouse will clear the banner. A Menu bar will then be displayed at the top of the screen. Clicking a menu title will display the commands you will use to drive the software. There are also Tool Bars to help rapidly access the same top level menu commands.

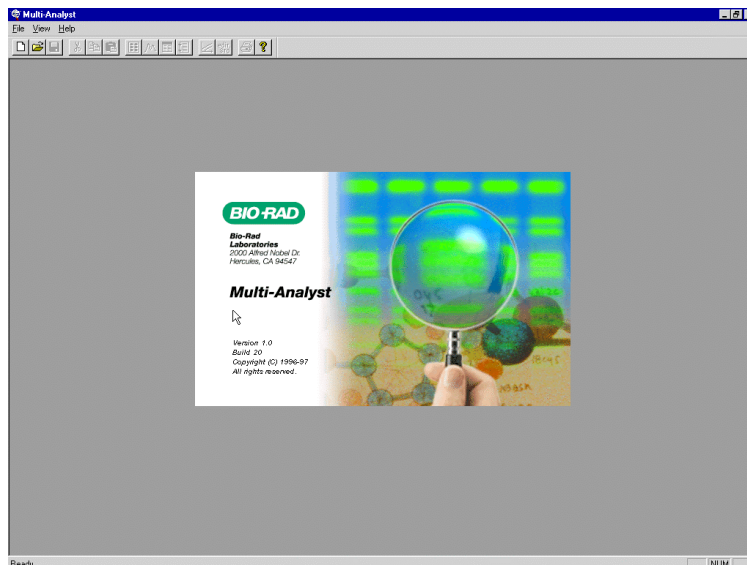


Figure 3-1 Splash Screen

3.2 Using Tool Tips

You can use Tool Tips to help familiarize yourself with Multi-Analyst/PC tool bars. To see Tool Tips, simply place the cursor over the tool and wait a few seconds (cursor must not be moving or the Tool Tips will not activate). A small dialog box will open and give a brief description of the function of that tool.

3.3 Acquiring Images

It is assumed that your Bio-Rad Imaging instrument is connected and suitable driver software is installed. For more information, please see Section 2 Getting Started and also read the Hardware Manual for further information on how to use and set up your Bio-Rad Imaging Equipment.

3.3.1 Multiple Scanner Support using Multi-Analyst/PC

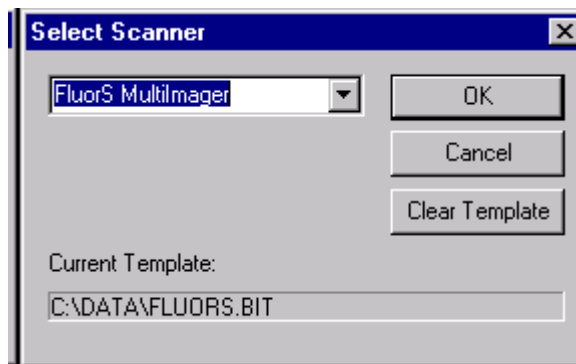


Figure 3-2 Select Scanner

file with a different scanner than the default scanner). There is also a button for you to clear the template file.

Under the File menu is an item menu called Scanner Setup. Selecting this item will bring up the dialog in Figure 3-2. In this dialog, you select the default scanner type. The default scanner type controls which scanner type will be used when you click New. The current default template is also displayed in this dialog. If you change the scanner type, the default template file will be cleared (as it would create an inconsistency to have a template

When you use the menu item *File...New using template*, the scanner type is read from a template file. This will not reset the default in this dialog window. However, you can specify any template to be the default, and then the default scanner type is reset to match the scanner type in the default template.

When Multi-Analyst/PC is first installed, the scanner type is not initialized. In this case, Multi-Analyst will automatically bring up the “Select Scanner” dialog for you when he or she selects New.

3.4 Handling of Multi-Color Images

Multi-Analyst software has multi-channel image handling capability and provides several ways to display multi-channel images.

3.4.1 Channel Selection

When a displayed image is a multi-channel image file, the channel tool box is displayed at the bottom of the image window.



This tool box allows you to handle the multiple channel image display. When the channel selection box is clicked, a drop-down menu is displayed providing several options.



The first part of this pull-down menu defines which channel is to be displayed on the image window. You can drag the mouse up to an appropriate channel and release the mouse to select the channel.

3.4.2 Side by Side display

This allows you to display multiple channel images simultaneously. Once Side by Side is selected from the pull-down menu, a Side by Side Channel Display dialog box is presented to allow selection of the method of display and the choice of channels.

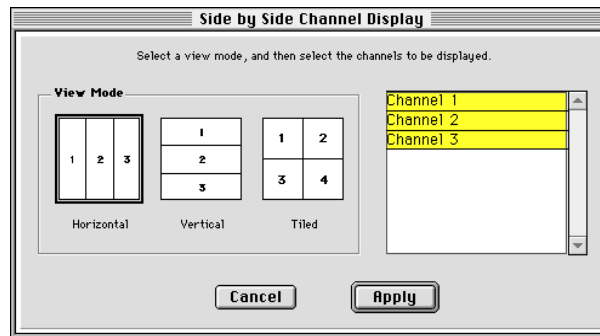


Fig. 4-29 Multi-Channel Selection Window

In this side by side display mode, the display area is automatically aligned in each channel. Exactly the same part of each image is displayed. To scroll through a larger image, use the scroll bar along the horizontal and the vertical axis. All images are scrolled together and remain in registration.

4. Acquisition Using the Fluor-S Multimager

It is assumed that your Bio-Rad imaging instrument is connected and suitable driver software is installed. For more information, please see Chapter 2, “Getting Started”, and also read the Hardware Manual for further information on how to set up and use your Bio-Rad imaging equipment.

4.1 Opening the Fluor-S Acquisition Window

To display the Acquire Window for the Fluor-S Multimager, click on the *File* menu and choose *Select Scanner...* then use the drop-down to identify your instrument and click *OK*. If your default scanner is already set, click on the *File* menu and choose *New*. If you want to use a specific template, click on the *File* menu and choose *New using template* then click on your Fluor-S template file and click *Select..* The Acquire Window will appear as shown below.

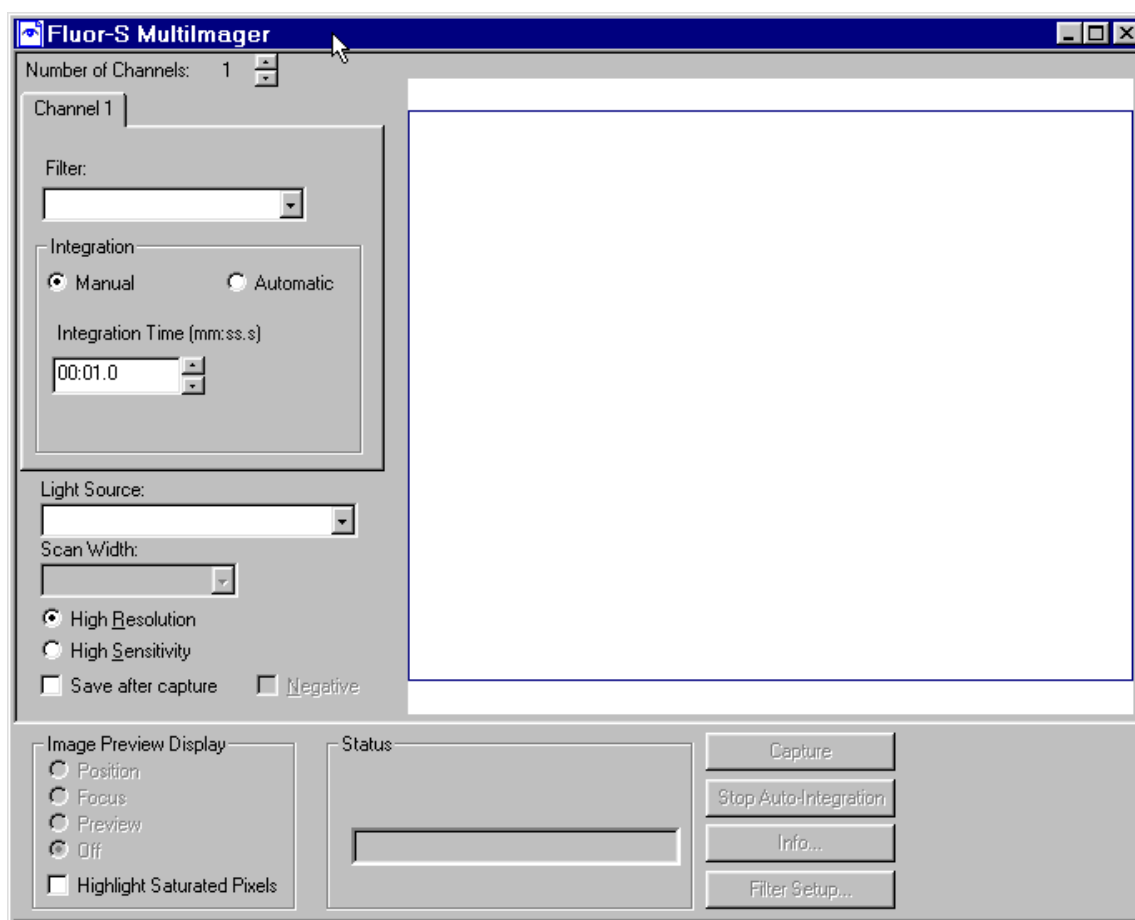


Figure 4-1 Fluor-S Acquisition Window

4.2 Selecting Channels and Filters

Once the acquisition window is displayed, select the number of channels you wish to acquire. Many applications may require only one channel. Multiple color fluorescence applications will require multiple channels.

1. Click on the up arrow or down arrow on the Number of Channels to select the number of channels.
2. Click on the Filter pull-down menu to specify the particular emission filter for each channel. Refer to Table 1 for the selection of particular filters.
3. If you are going to capture more than one channel, you will need to click on the Filter pull-down menu and select another filter after selecting Channel 2.
4. Select the appropriate list source for your gel.

	Recommended Light Source				
	UV Light		White Light		
	Trans	Epi	Trans	Epi	Emission Filter
Ethidium Bromide	YES				520 Long Pass
SYBR Green I	YES				520 Long Pass
Fluorescein	YES				530 Band Pass
Texas Red	YES				610 Long Pass
Radiant Red	YES				610 Long Pass
SYPRO Orange	YES				520 Long Pass
Coomassie Brilliant Blue			YES		None
Silver Stain			YES		None
Copper Stain			YES		None
Zinc Stain			YES		None
Chemiluminescence	No Light Source is required USE HIGH SENSITIVITY MODE				None
TSA - Fluorescein		YES			530 Band Pass
Attophos		YES			520 Long Pass
Digital Documentation				YES	None
X-ray Film			YES		None

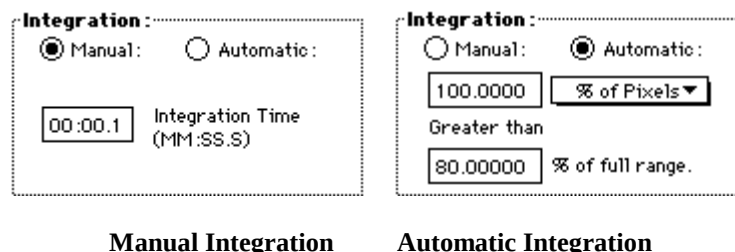
Table 1 Fluor-S Recommended Light Source and Emmission Filters

4.3 Placing the sample

After selecting Acquisition mode, click on the Position/Focus radio button in the Image Preview Display settings area. The Fluor-S will start capturing a live image and updating the image every second. While Position/Focus is ON, position the sample in the center of the platen.

4.4 Selecting the Settings

- Select the appropriate scan width from the pull-down menu.
- Select acquisition mode either *High Resolution* or *High Sensitivity*.
- Select the Integration mode either Manual or Automatic and Insert the appropriate parameters. Manual acquisition is controlled by the integration time from 0.1 second to 60 minutes. Automatic acquisition is controlled by the number or percentage of pixels that are saturated.



- Click on Preview in the Image Preview Display area and make sure the image will be acquired properly. Once you have confirmed the preview acquisition, you are ready to acquire the image.
- If you selected "*Highlight Saturated Pixels*", saturated pixels will be highlighted in red. Use this feature to make sure regions of interest are not saturated.

4.5 Acquiring the Image

- Save after acquire* will open the standard PC Save As dialog box. Specify the file name you wish to use and click on Save.
- When you select *Acquire*, the system will start acquiring the image. When the acquisition is complete the image will be automatically saved (with the name specified), and displayed.

4.6 Advanced Fluor-S Settings

4.6.1 Number of Channels

This control allows you to define the number of channels to be captured. There is always one tab per channel. If a channel is removed, and then added back; the settings will revert to defaults. This selection is saved in the document template.

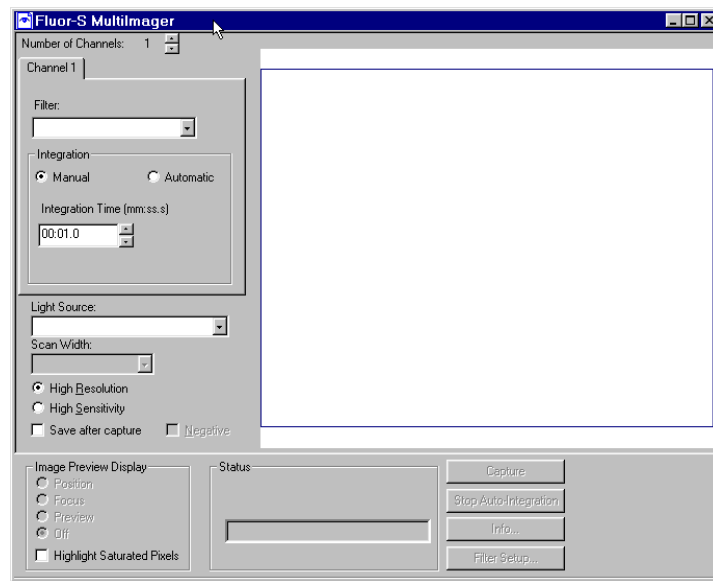


Figure 4-2 Fluor-S Acquisition Window

4.6.2 Channel Specific Parameters

4.6.2.1 Filter

The Filter combo box allows you to select a filter to be used for the selected channel. This is a selection-only list, filters can not be typed in. As the channel selection changes, the Filter combo box will be updated accordingly. Filter names are obtained from the Fluor-S and Filter names may be set via the Setup option. Filter selections are saved in the document template.

4.6.2.2 Integration Time

The Integration control allows you to define the integration period (exposure) for the selected channel. You can set a value either by typing it in directly or by using the spin control to the right of the time field. As the channel selection changes, the Integration control will be updated accordingly. The software asks the instrument for a minimum integration time. If something smaller is entered, it will be ignored and the minimum value substituted. This selection is saved in the document template.

Maximum integration time is one hour.

4.6.2.3 Auto Integration

Auto Integration is a method intended to automatically determine the exposure time which will maximize the dynamic range of the image. This selection is saved in the document template.

4.6.3 Channel Independent Parameters

4.6.3.1 Light Source

The Light Source combo box allows you to select the light source for an experiment. Light source is not a channel parameter (it would not make sense to have an image with one channel in counts and one channel in OD as could happen if you were to select a UV light source for channel one, and a white light source for channel two.) The light source selections are obtained from the instrument. This selection is saved in the document template.

For images captured with a white light source, a calibration table is generated for the data, and pixel units will be set to OD. For all other images, no calibration table will be associated, and pixel units will be set to counts. The calibration table is computed as follows:

$$\text{OD}[\text{value}] = -\text{LOG}_{10}((4096 - \text{value})/4096.)$$

So the OD value of a black pixel is $-\log(1/4096) == 3.61$.

And the value of a saturated pixel is $-\log(4096/4096) == 0$.

(Recall that in white light mode, a black pixel will have pixel value 4095 (after inversion), and a white or saturated pixel will have pixel value 0.)

4.6.3.2 Scanning Width

When the trans-illumination light source is selected, an additional option entitled scan width becomes available. The extent is the distance swept out by the light source. The region scanned is always centered horizontally on the platen. For small gels, this option allows smaller scan times by letting you pick a scan area that more closely approximates the size of

their gel. Again, the list of scan widths is obtained from the scanner. Currently, three different widths are provided. These are embedded in the Fluor-S and cannot be reprogrammed. This option is disabled for epi light sources and chemi-luminescence. For these light types, the zoom lens is used to modify the “scan area”. This selection is saved in the document template.

4.6.3.3 Resolution and Sensitivity

Resolution and Sensitivity are two mutually exclusive options. In Resolution mode, the bin factor is set to one, so you may obtain an image of maximum resolution for the CCD. This mode also has the highest gain setting for the camera. This gain setting is optimal for fluorescent and white light applications. In Sensitivity mode, the bin factor is set to 2, so a higher signal is obtained for equivalent integration time in high resolution mode. However, the number of pixels in each dimension is halved, so the spatial resolution of the image is reduced. In addition, the camera is set to low gain. This provides optimal sensitivity to low light applications such as chemiluminescence. This selection is saved in the document template.

4.6.3.4 Negative (Signal is white)

This option is enabled for white light images only. For white light images, the data is inverted by default. That is, value $\leftarrow 4095 - \text{value}$. The palette is also inverted so that the image on the screen looks the same as the real sample.

For example, 0 maps to 4095, 100 maps to 3995, and 4095 maps to 0.

For white light images in which the signal is white (high pixel values), this option must be selected for proper quantitation.

4.6.4 Runtime Control

4.6.4.1 Highlight Saturated Pixels

When selected, pixels corresponding to saturated data will be painted red (or the selected saturation color). This selection is not saved in the document template.

4.6.4.2 Preview Modes

There are two mutually exclusive modes: focus/zoom and preview. In focus mode, your selected settings are ignored, and the system is set up for fast transfer of white light images. You should select this mode to adjust the zoom lens and set the focus, since this mode gives the fastest real-time feedback. In focus mode, the bin factor is set to 2, integration time is set to $1/100^{\text{th}}$ of second, and the light source is white light epi-illumination. Since your settings are ignored while in focus mode, all scan parameters controls are disabled.

To validate capture parameters, you may want to see images from the instrument without having to save these images into a document. To do this, you should select preview mode. When selected, the instrument will begin capturing images of the currently selected channel. As soon as each image is transferred to the computer, it will replace the previous image on the screen. The channel selection control and the number of channels control is disabled during preview mode. You must stop the preview to change these items. If you change other parameters, the changes take effect after the current frame is completed. The current frame is not aborted every time you make a change. For example, if you are in the middle of a five minute integration and you change the integration time to two seconds, the software will

complete the five minute integration before starting the two second integration. If you do not want to wait for the long integration to complete, you may select “Off”, and then restart the preview mode.

To end either of these modes, you may select “Off”.

When “Off” is pressed, the last image transferred will remain on screen. When an operation (such as Capture) is initiated that requires an “Off”, the program will stop automatically to save you from having to manually switch modes.

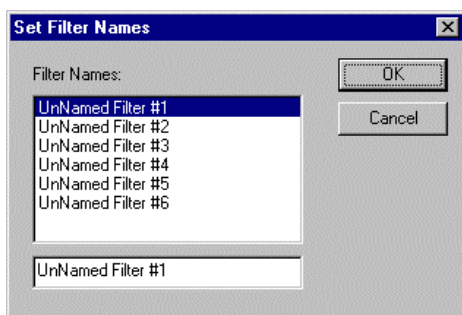
After 45 minutes, both these modes will automatically shut off to preserve the bulb.

4.6.4.3 Capture

When you are satisfied with everything, you can then capture an image. As soon as the image is captured, it will be added to the experiment document. If “Save after Capture” is checked, the program will prompt you for a filename for the document before the scan starts (a filename based on the login name and current date and time will be suggested). The document will then be saved to a disk file as soon as the image is captured.

4.6.4.4 Filter Setup

This brings up a dialog allowing you to configure the device. For example, when you change filters, you can enter new filter description strings in this window.



4.6.5 Scanner Information

The information dialog displays version information about the instrument, and allows you to enter certain information that cannot be obtained from the instrument. Specifically, since the zoom factor of the lens is not relayed back to the host software, the software does not know the physical dimensions of the area being imaged. Here you can enter the width of the image area. The

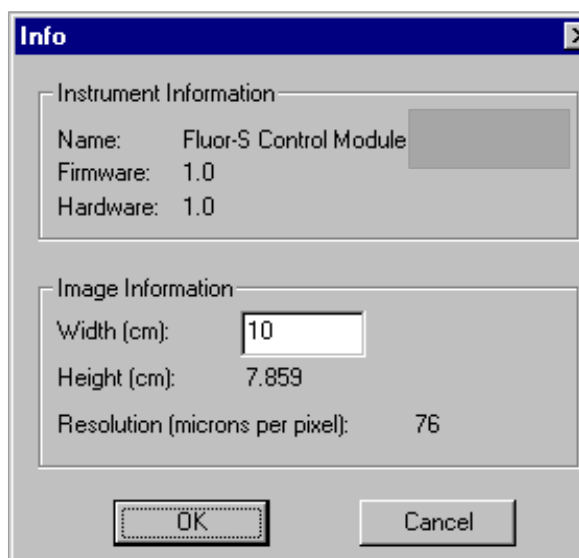


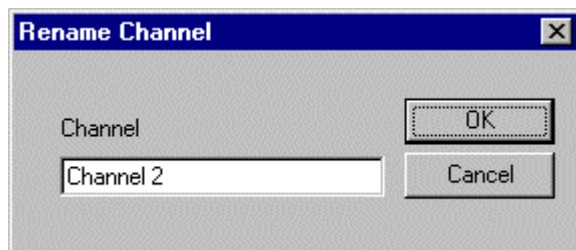
Figure 4-3 Scanner Info

height is then computed based on the aspect ratio of the image. Lastly, the resolution is computed based on the width you entered. The computed information is updated on each keystroke.

4.6.6 Channel Selection

Each channel is displayed in a different tab. To switch to a different channel, click its tab. The image preview (if any) and the acquisition parameters will be updated to correspond to the newly selected channel.

Channel names may be changed from the default “Channel X” by double clicking on a tab. The dialog pictured below will come up with the current channel name in the edit field. You may then change it to what ever you like. There is no limit on the length of the name, however past a certain length, it may not be displayed fully in some fields.



4.6.7 Image Display

The image will always be displayed so as to fit the available window space while preserving aspect ratio. Under the *Tools* menu, you can select the *Color Palette*. The palette changes will affect the currently active image only. The image is transformed using a fixed linear transform mapping the data range 0-4095 to 0-255.

Certainly no rotating, flipping, or filtering will be permitted in the acquisition window. Palette operations will be allowed to help you find your data.

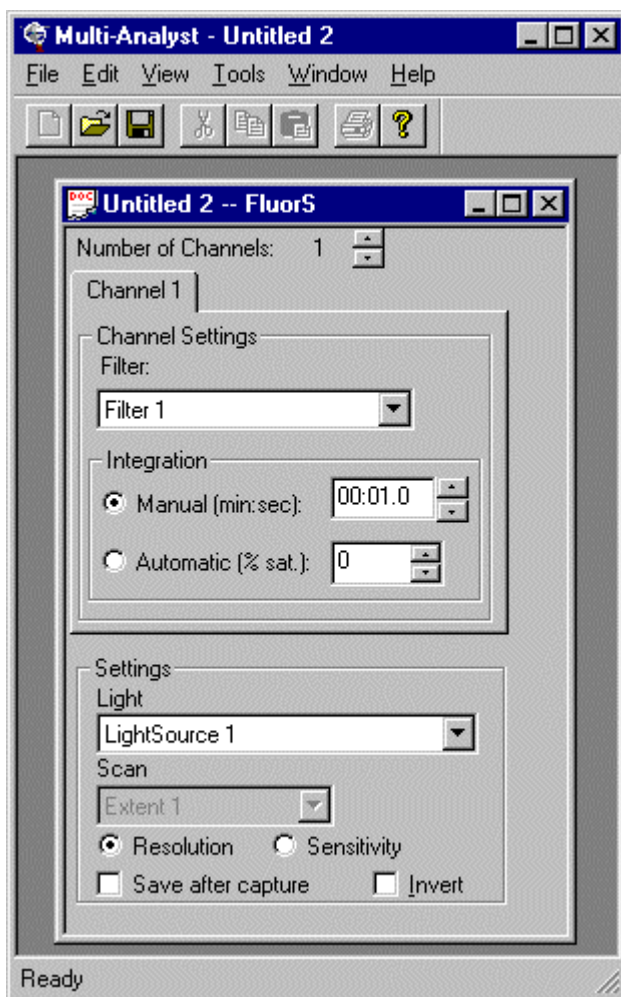
The image can not be printed from the Acquisition Window.

4.6.8 Image Processing

Dark current subtraction should only be performed if the temperature of the camera is stabilized at 10 C degrees. If this is not the case, Multi-Analyst will inform you via a message box that the instrument temperature is not stable. You will be given the option whether to continue image capture.

Dark frames will be reacquired whenever the integration time, or camera changes. Of course, a dark frame will be acquired for the first image captured after the scan window is opened. The integration time for the dark frame is the same as the integration time of the light frame.

4.6.9 Template Settings



If the Fluor-S is not found, it is still possible to create a template. The acquisition window will appear without the image pane or the runtime controls. In this window, you can define and save a set of runtime parameters. Also, the *Notebook Entries* dialog is available under the *View* menu, so you can configure other settings as well.

The choices in the drop down menus are stored from any previous sessions on the instrument. If the instrument has never been attached to the machine before, then the default factory settings are used.

5. Acquisition using the Molecular Imager

This version of **Multi-Analyst/PC** is designed for use with the Bio-Rad GS-505 or GS-525 imaging system. If you have an older model imager, please continue using the latest version of Molecular Analyst/PC.

It is assumed that your Bio-Rad Imaging instrument is connected and suitable driver software is installed. For more information, please see Chapter 2, “Getting Started”, and also read the Hardware Manual for further information on how to use and set up your Bio-Rad imaging equipment.

5.1 Opening the Imager Acquisition Window

To display the Acquisition Window for the Molecular Imager, click on the *File* menu and choose *Select Scanner...* then use the drop-down to identify your instrument and click *OK*. If your default scanner is already set, click on the *File* menu and choose *New*. If you want to use a specific template, click on the *File* menu and choose *New using template* then click on your Imager template file and click *Select..* The Acquisition Window will appear as shown below.

5.2 Setting Scan Parameters

1. The scan window will appear with pull-down menus, scanning area window, control

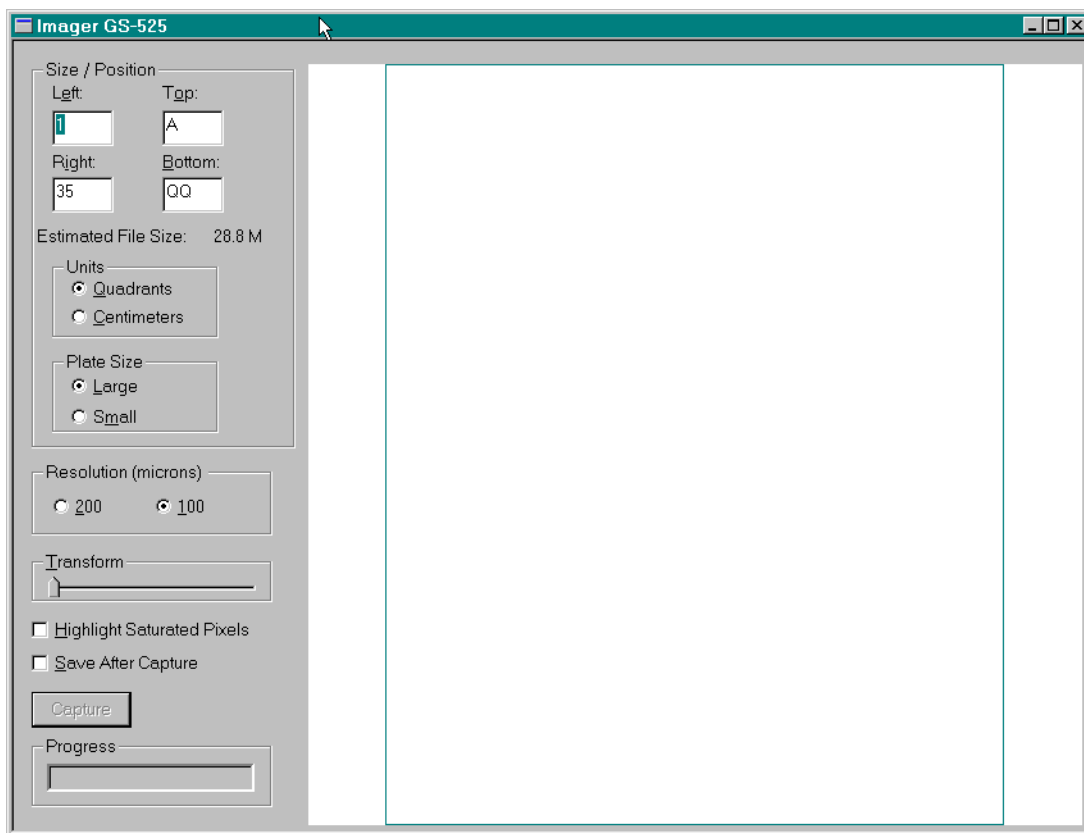


Figure 5-1 GS-525 Molecular Imager Acquisition Window

buttons, and status bar. The scan window is initially blank with its dimensions representing the full size of a phosphor screen.

2. Using the mouse, select your resolution of 200 μ or 100 μ . The 200 μ resolution is a good choice for routine imaging because files obtained at 200 μ resolution are only 25% the size of files acquired at 100 μ resolution.

5.3 Identifying the Area to Scan

Next, identify the scanning area. When the *Scan Window* is opened, the default scanning area is the entire screen size, 20 cm x 25 cm for the GS-505 or 35 cm x 40 cm for the GS 525.

To identify a more specific scanning area on the phosphorescence imaging screen:

1. Identify the area where the sample(s) was located on the loading dock's sample placement grid. Place the mouse onto the scanning area. Notice that the pointer is replaced with a square scanning tool when placed in the scanning region.
2. Place the square scanning tool at the upper left corner of the region to be scanned, click, and hold down the left mouse button, and drag down diagonally until the desired scanning area is selected.
3. Release the mouse button. As the scanning region is identified the scanning coordinates (cm or quadrants) are displayed within the *Scan Region Area* boxes designated (*Left, Right, Top, and Bottom*).

4. Additionally, as the coordinates of the area are selected, the file size of the image, and the time left to complete scanning are also displayed.

Alternatively, the sample placement grid coordinates for the sample's upper left and lower right corner can be entered. Click on the quadrants radio button and enter the grid coordinates noted from the lid of the sample loading dock.

5.4 Starting the Scan

Click the *Capture* button. As scanning progresses, adjust the brightness, contrast, or gamma correction of the image with the *Color Palette* to view faint samples which might not be visible using the default view full range (0-65,535).

Scanning may be stopped at anytime by selecting *Cancel*. When Cancel is selected, the instrument stops and the Acquisition Window remains active. The software can display any part of the already scanned area in an untitled display window by first dragging the mouse across the desired area and then clicking on *Acquire* again. The image should be saved before beginning analysis by choosing *Save* under the *File* menu.

5.5 Saving the Scan

1. Adjust the contrast of the image by selecting the appropriate contrast to best display your image from the *Transform* scroll bar. To perform image optimization functions, the scan from the sequence must be saved as an image file using the *Save* control button.
2. In the *Save* dialog box, save your image as *.bif* file.
3. The size (MB) of each file saved depends upon the total area scanned and the resolution.

5.6 Advanced Molecular Imager Menu Descriptions

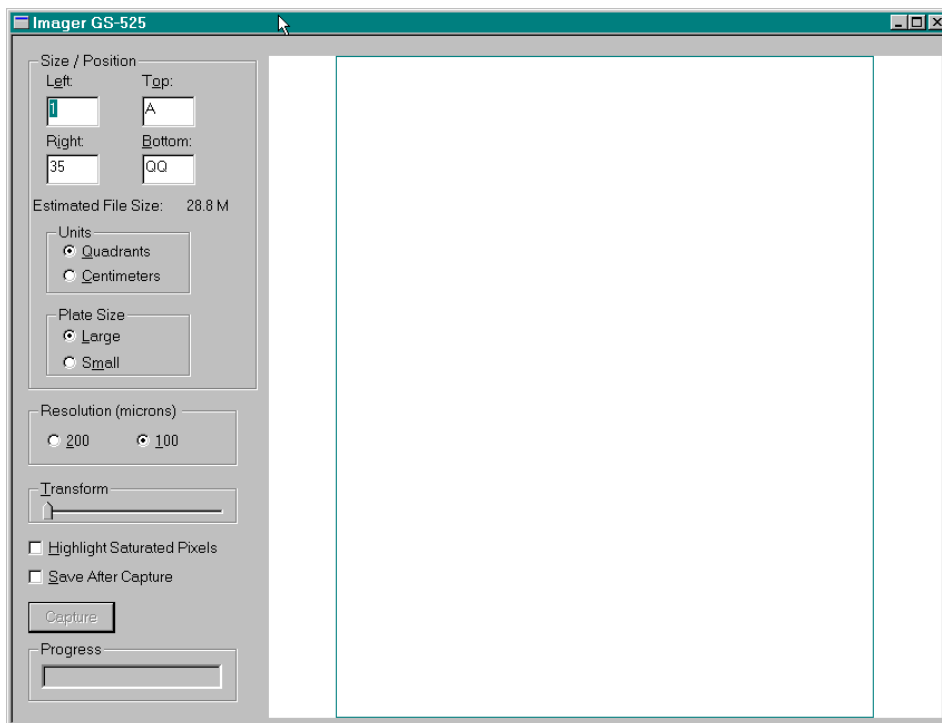


Figure 5-2 GS-525 Acquisition Window

5.6.1 Size/Position

Select the coordinates and the resolution, and then press the *Capture* button. The image is then scanned and you can monitor the progress in the progress bar. As soon as the image is fully scanned, a new document is created, and the gel image will appear in a regular Multi-Analyst image window.

Note: *As scanning is destructive, You must save your data as soon as it is acquired.*

The window is always in *Zoom to Fit* mode, which means that as you resize the dialog, the image window will be resized and zoomed accordingly.

5.6.2 Units

The units for the quadrants must be either centimeters or quadrants. In the case of centimeters, the value must be a positive whole number.

5.6.3 Plate Size

You must check the Plate size you have inserted into the GS-525 in order to facilitate coordinate validation.

5.6.4 Transform

It is possible to display and modify the gel image while it is being acquired. You may transform the image while it is being acquired using a slider. Moving the slider is analogous to using the gamma correction tool and does not change the raw data.

5.6.5 Save After Capture

If *Save after capture* is checked, the program will prompt you for a filename for the document before the scan starts (a filename based on the login name and current date and time will be suggested). The document will then be saved to a disk file as soon as the image is captured.

5.6.6 Highlight Saturated Pixels

When *Highlight Saturated Pixels* is selected, pixels corresponding to saturated data will be painted red. When several data points map to a single pixel on the display, that pixel will be considered saturated if any of the data points is saturated. This selection is not saved in the document template.

5.6.7 Notebook Entries

Since the Imager is a single-channel device, the Acquisition sheet consists of a single text pane. This information will consist of the

- Scanner type (GS-525 or GS-505)
- Resolution
- Scan Area

Detailed image information may be found as always in the Image Description page of the *Notebook Entries*.

5.6.8 Molecular Imager Template Files

The following information is saved in the template file:

- scan area (top, left, bottom, right), in centimeters
- plate size (small, large)
- resolution (100 microns, 200 microns)
- highlight saturated pixels (on, off)
- save after capture (on, off)
- device name

6. Acquisition using the Gel Doc 1000

This version of **Multi-Analyst/PC** is designed for use with the Bio-Rad Gel Doc 1000 imaging system using a PCI video capture board. If you have an older model ISA video capture board, please see your local sales representative. This version supports both RS-170 (US video standard) and CCIR (European video standard) video input signals and camera integration control.

It is assumed that your Bio-Rad Imaging instrument is connected and suitable driver software is installed. For more information, please see Chapter 2, “Getting Started”, and also read the Hardware Manual for further information on how to set up and use your Bio-Rad imaging equipment.

6.1 Opening the Gel Doc Acquisition Window

To display the Acquisition Window for the Gel Doc 1000, click on the *File* menu and choose *Select Scanner...* then use the drop-down to identify your instrument and click *OK*. If your default scanner is already set, click on the *File* menu and choose *New*. If you want to use a specific template, click on the *File* menu and choose *New using template* then click on your Gel Doc template file and click *Select..* The Acquisition Window will appear as shown

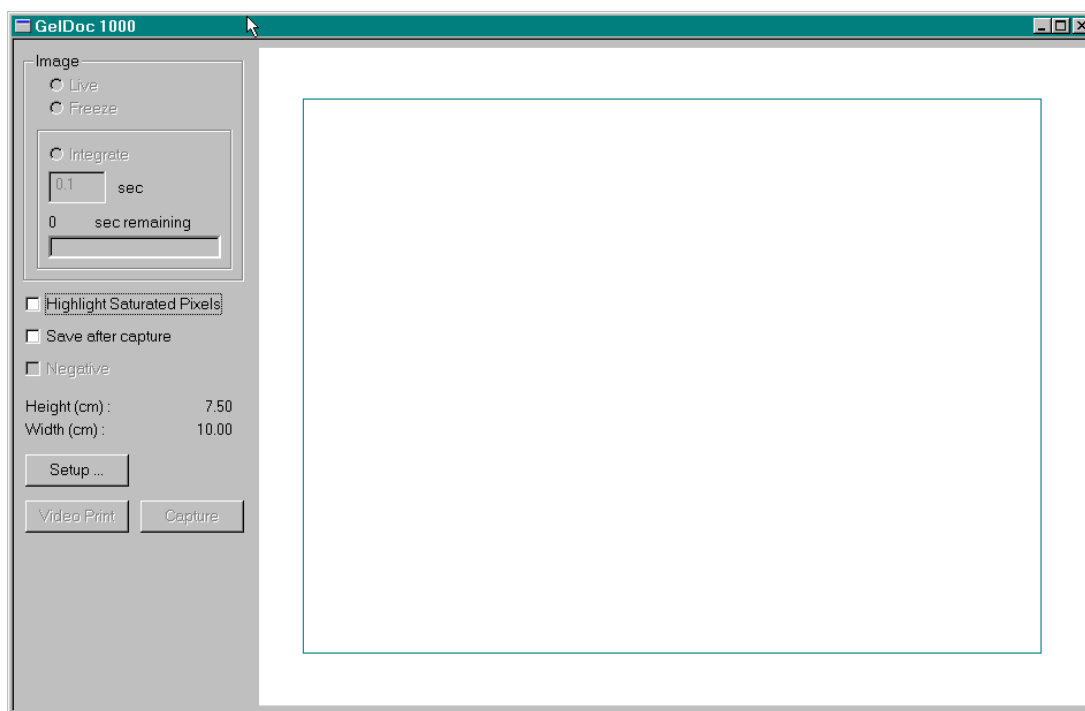


Figure 6-1 Gel Doc Acquisition Window

below. A real-time image will be displayed.

6.2 Displaying “Live” Images

When the scan window is opened, it will automatically be in “LIVE” mode and will display a live video image allowing you to zoom and focus on the sample area. If no image is visible,

check cable connections from the frame grabber board to the camera and check that the protective cap is off the lens. Also check that the transilluminator is on and working. The image should stay in focus while zooming from high to low magnification. If focus is not maintained, refer to Appendix D to correct the camera focus.

6.3 Integrating



To identify faint areas of fluorescence, click on the “Integrate” radio button. Integration time can be set from 0.1 seconds to 1 hour. (Integration has the same effect as exposure time). The integration pull-down menu only displays up to 1 second; however, any number may be entered manually. To set integration time, either click on the integration time pull-down menu and select one of the settings, or for longer periods of time, highlight the integration number inside the *Integration Time* field, type the desired integration time, then press <Enter>.

Figure 6-2 Integration Window

6.4 Printing a live or integrated image

Live or integrated images from the acquire window are only printed to a video printer. To print a live or integrated image to a video printer, select *Video Print* from the acquisition window. The video printer will automatically print the image. Please see Appendix D to adjust the Mitsubishi P68U Thermal Printer, or Appendix E to adjust the Mitsubishi P90W Thermal Printer.

6.5 Capturing or “Freezing” the image

To capture the image for observation or analysis, click on the *Capture* button. The image will be displayed in an untitled image window. It is recommended that the image be saved by using the *Save* command from the *File* menu at this point.

6.6 Did you Remember to Turn off the Light?

Multi-Analyst reminds you to turn off the UV transilluminator when you close the Acquisition Window, or if the image has been frozen on the screen for longer than a few minutes. You can disable this message in the *Setup* dialog.

This setting allows you to turn off the UV light box and prevent UV damage to the samples, including possible photobleaching, and extends the lives of the UV bulbs.

6.7 Saving the Image

Save the image as an image file by choosing *Save* from the *File* menu. Alternatively, if the *Save after Capture* is checked, the image will be saved to disk. A default filename (composed of your login name and current date and time) will be suggested. You may change the default filename if you wish.

6.8 Printing captured or saved images

Captured and saved images can also be printed either to a video or to a laser printer.

a. **Printing captured images to a Mitsubishi video printer**

To print saved or captured images to a video printer, select *Video Print* from the *File* menu. If there is no Bio-Rad frame grabber board present, this command will not be available, and data must be printed on a laser printer using the *Print* command from the *File* menu.

b. **Printing to a laser printer**

To print saved or captured images to a laser printer, select *Print* from the *File* menu.

6.9 Advanced Gel Doc Settings

6.9.1 Live Mode

Continuously displays “LIVE” gel images in the display window at a minimum rate of 3 to 5 frames per second. Your video may be faster or slower depending on your computer’s speed and the speed of its various components such as RAM, video card, and hard disk.

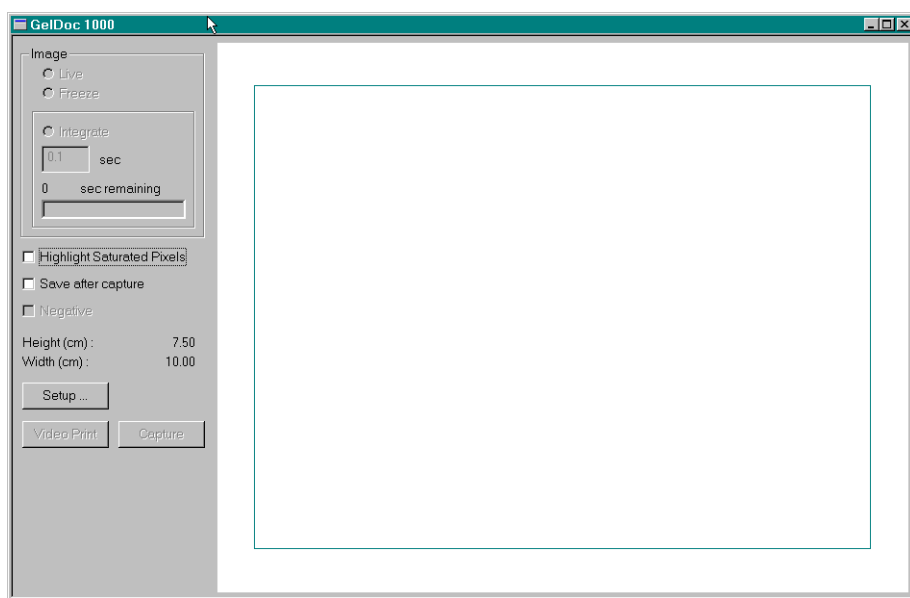


Figure 6-3 Gel Doc 1000 Acquisition window

6.9.2 Freeze Mode

Stops “Live and Integrate” video capture modes. The last displayed gel image will remain displayed in the window.

6.9.3 Integrate Mode

Continuously displays a gel image in the display window at your specified integration interval. Integration of your gel image is done on the camera CCD. You can specify integration times in seconds from 0 to 300.0 seconds. Integration steps are in increments of 1/30 second for RS-170 (US) systems and 1/25 second for CCIR systems (EU).

6.9.4 Integration Time

You can select from factory default times or enter a time between 0.0 and 300.0 seconds.

6.9.5 Negative

This option is only available when the light source is White Light. This option inverts the data so that film negatives or Polaroids can be captured and properly analyzed.

6.9.6 Highlight Saturated Pixel

To help you identify saturated data, you may choose to highlight all saturated pixels with a chosen color. The default color is red. If *Highlight Saturated Pixels* is checked, all saturated pixels will be displayed in red. This feature is useful when quantitating bands on a gel. For more accurate quantitation, either reduce integration time or close down the aperture until all red saturated values are white.

6.9.7 Save After Capture

Saves the currently displayed video frame to file.

6.9.8 Status

The status section contains a progress bar and status field representing the remaining time to update the next video frame.

6.9.9 Video Print

This option is enabled when the video capture board recognizes that a video printer is attached. When selected, the current frame being displayed will be output to the video printer. Video printers supporting remote print options will be signaled to print.

6.9.10 Info dialog

The button will display the information and capabilities of your Gel Doc 1000 Imaging System.

6.9.11 Gel Doc Setup dialog

From the setup dialog, you can specify the light source (UV or White Light) and adjust the video capture boards range and offset settings.

6.9.11.1 Light Source

This displays the currently selected Light Source (Ultraviolet or White Light). Click on the Light Source being used. When using UV Light, the signal strength is measured and displayed in counts; when using White Light, the signal strength is measured and displayed in OD units.

6.9.11.2 Image Size

This contains the image size specified in centimeters. The image size is used to calculate image pixel size. Because the Gel Doc 1000 comes with an adjustable zoom lens, pixel size can't be automatically calculated by the Multi-Analyst software without your input. Image Size always defaults to the largest possible viewing area (zoomed out). For the most accurate quantitation it is necessary to enter in the scan dimensions taken from the imaging area. This is done by simply using a fluorescent ruler to measure the imaging area and entering the information.

6.9.11.3 Set DAC...

The effects of DAC on the acquired image are similar to brightness and contrast controls on a video monitor. This feature should be rarely used because it alters the raw signal coming from the camera. Please use the default values for Range and Offset, and use the *Color Palette* or *Transformation* sliders to adjust brightness and contrast.

6.9.11.4 Manipulating the Video Display Color Palette

During video acquisition, the Contrast, Brightness, and Gamma values of the Color Palette are modifiable by using the standard *Color Palette*... Note: The image transform range is equal to 0 to 255 with a linear transformation.

6.9.11.5 Zoom

You can select *Best Fit*, *View at 100%* or *View at 50%* from the menu. In addition, you may use the zoom tool, just as in the image window. When the image is larger than the window, scroll bars will be available for navigating the image.

6.9.11.6 Template Support

When no Gel Doc is present, a message box will pop up informing you that no Gel Doc could be found, then the normal scan window will come up. However, all control buttons will be disabled. Instrument defaults are stored in Multi-Analyst, so that you can still set up a template.

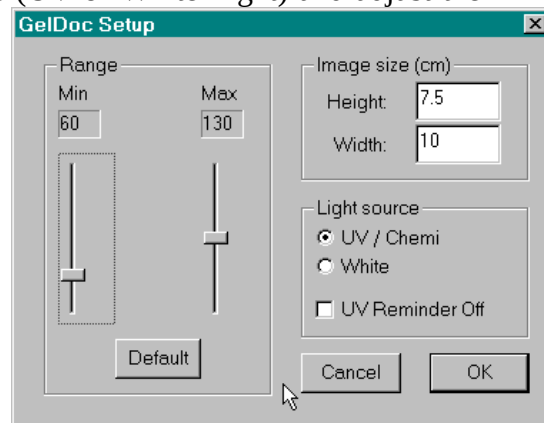
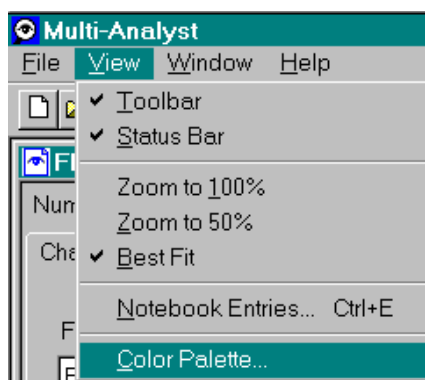


Figure 6-4 Gel Doc Setup Window



6.9.11.7 Notebook Entries

Since the Gel Doc is a single channel device, the Acquisition page will consist of a single text pane. This information will consist of the

- Light Source
- Integration Time
- Camera Type (RS-170 or CCIR)

Detailed image information may be found as always in the *Image Description* page of the *Notebook Entries*.

7. Acquisition using the GS-700/GS-690 Densitometer

This version of **Multi-Analyst/PC** is designed for use with the Bio-Rad GS-700 or Bio-Rad GS-690 Densitometer imaging systems. If you have an older model Densitometer, please continue using the latest version of Molecular Analyst/PC.

It is assumed that your Bio-Rad imaging instrument is connected and suitable driver software is installed. For more information, please see Chapter 2, “Getting Started”, and also read the Hardware Manual for further information about how to set up and use your Bio-Rad imaging equipment.

7.1 Opening the Densitometer Acquisition Window

To display the Acquisition Window for the GS-700 Densitometer or the GS-690 Densitometer, click on the *File* menu and choose *Select Scanner...* then use the drop-down to identify your instrument and click *OK*. If your default scanner is already set, click on the *File* menu and choose *New*. If you want to use a specific template, click on the *File* menu and choose *New using template* then click on your Densitometer template file and click *Select..*. The Acquisition Window will appear as shown below.

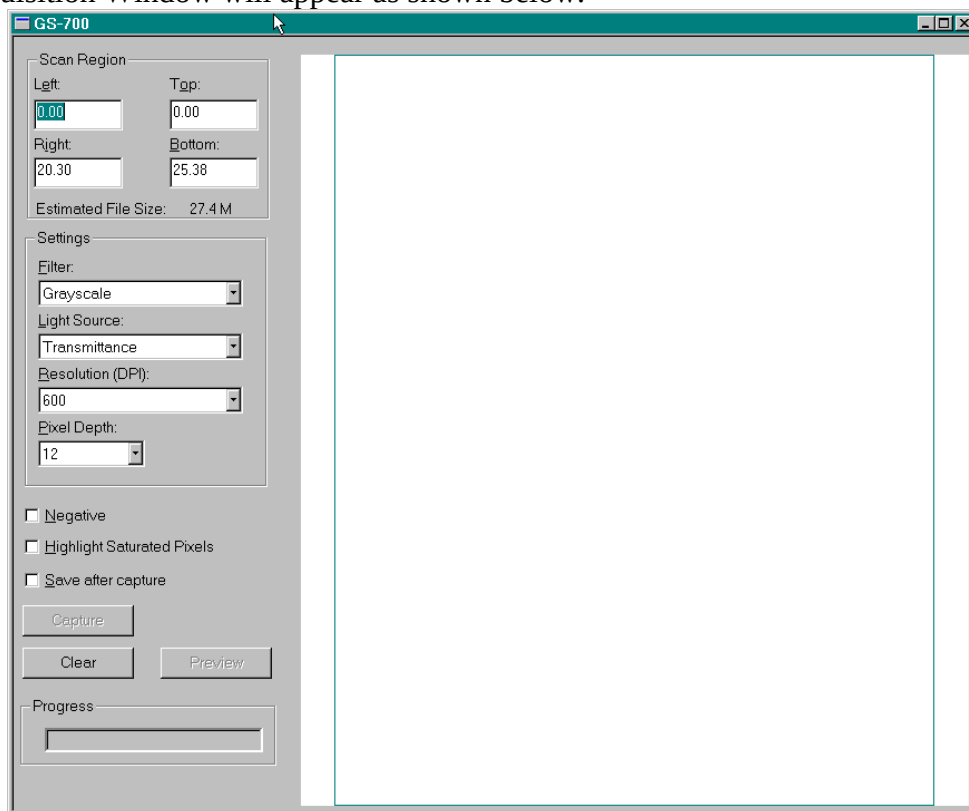


Figure 7-1 Densitometer Acquisition Window

7.2 Select Densitometer Scan Settings

The scan window is initially blank with its dimensions representing the full size of the scanning area in Reflectance mode.

1. Select the appropriate *Light Source*. Reflectance mode allows the scanning of opaque samples such as blots, gels dried on filter paper, or photographs, and has dimensions of 21 cm x 35 cm. Transmittance mode allows the scanning of transparent samples such as films, gels, or slides, and has dimensions of 20 cm x 25.5 cm.
2. Select the proper *Filter* for your sample from the *Filter* drop-down. Each filter corresponds to a wavelength range that is sensitive to specific dyes and stains. See the following table for further details.

<u>Filter Color</u>	<u>Wavelength</u>	<u>Application Examples</u>
Red	(595nm-750nm)	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	(520nm-570nm)	Coomassie R-250, Basic Fuchsin
Blue	(400nm-530nm)	Crocein Scarlet
Gray Scale	(400nm-750nm)	Silver Stains, Copper Stains, Film

Table 2 Densitometer Filter Chart

3. Select the appropriate *Pixel Depth* (8-bit or 12-bit). The 8-bit mode allows pixels to be assigned gray levels in the range from 0 (white) to 255 (black), while in 12-bit mode each pixel is assigned a value in the range from 0 to 4095. The 12-bit data provide more accurate quantitation, but take up more file space than 8-bit data.
4. If scanning negatives of film strips or photographs in the Transparent mode, click on the *Negative* box in the Acquire Window.
5. To identify saturated pixels during the scan, click on the *Highlight Saturated Pixels* box.
6. Move the cursor into the scan window and identify the location of the samples by clicking and dragging the mouse from the upper left to the lower right coordinates of the sample. Release the mouse button. The resulting dashed rectangle identifies the area which will be scanned. The *Scan Region* will show the coordinates of the area selected, the file size of the image, and the time left to complete scanning.

7.3 Preview Scanning Area

The scanning area may be previewed by performing a low-resolution (24 dpi) scan. **The preview scan is designed to provide a rapid preview of the sample and its location, and always scans at 24 dpi.** No data are saved during previewing, but then again, no data are lost.

1. If the exact position of the sample is unknown, simply select the whole scanning area with the mouse. Scanning areas may be modified and moved easily with the mouse.
2. Click on the *Preview* button to initiate preview scanning. As scanning progresses, the brightness, contrast, or gamma correction of the image may be adjusted with the *Color Palette*. This allows imaging of faint samples which might not be visible using

- the full range default view (0-255 for 8-bit scanning, and 0-4095 for 12-bit scanning). Because the software is processing extremely large amounts of data during the acquisition process, response to mouse clicks tends to be somewhat slower. When the preview scanning is complete, specific areas of the sample may be rescanned at a higher resolution using the *Capture* button.
3. Scanning may be stopped with the *Cancel* button, and another preview scan or higher resolution scan performed.
 4. The *Clear* button can also be used to clear any image in the acquisition window in preparation for further scanning.

7.4 Initiating Image Acquisition

Images may be acquired and analyzed at any desired resolution.

1. Using the mouse, outline a specific area within the preview scan to scan at a higher resolution.
2. Choose a high resolution value (i.e. 400 dpi) and then click on the *Capture* button. The Densitometer will immediately start scanning. Images may also be adjusted with the *Color Palette* when scanning.
3. Once scanning has completed, the program will automatically display the image in an untitled Image display window.
4. Save the image before beginning analysis. From the *File* menu, select *Save*. Enter a name in the Save File dialog box and click *OK*.

7.5 The Scan Process

After scanning is initiated, the status bar will indicate the percentage of the scan that has been completed. The scan window display will be updated as the scanning process continues. Once the image has been fully scanned, it will be displayed in the scanning area at 100% of the current window size. Alternatively, after displaying the preview image, selecting the *Show All* button displays the area scanned in proportion to the whole scanning area. This feature is useful to allow rapid re-identification and rescanning if the desired target area was missed. Note that at any point in the scanning process, scanning can be stopped and images either saved or rescanned using a different resolution, using a different filter, or different pixel depth.

7.6 Scanning Negatives (Ethidium Bromide Photographs)

Clicking the *Negative* check box allows the scanning of negatives or samples which have light bands on dark background (i.e. ethidium bromide photographs). This information can also be entered after the image has been saved using the *Image Info* command from the *View* menu. Selecting the *Options* button in the *Image Info* dialog box allows the selection of either a positive (dark bands on light background) or negative (light bands on dark background) image.

7.7 Advanced Densitometer Acquisition Control Settings

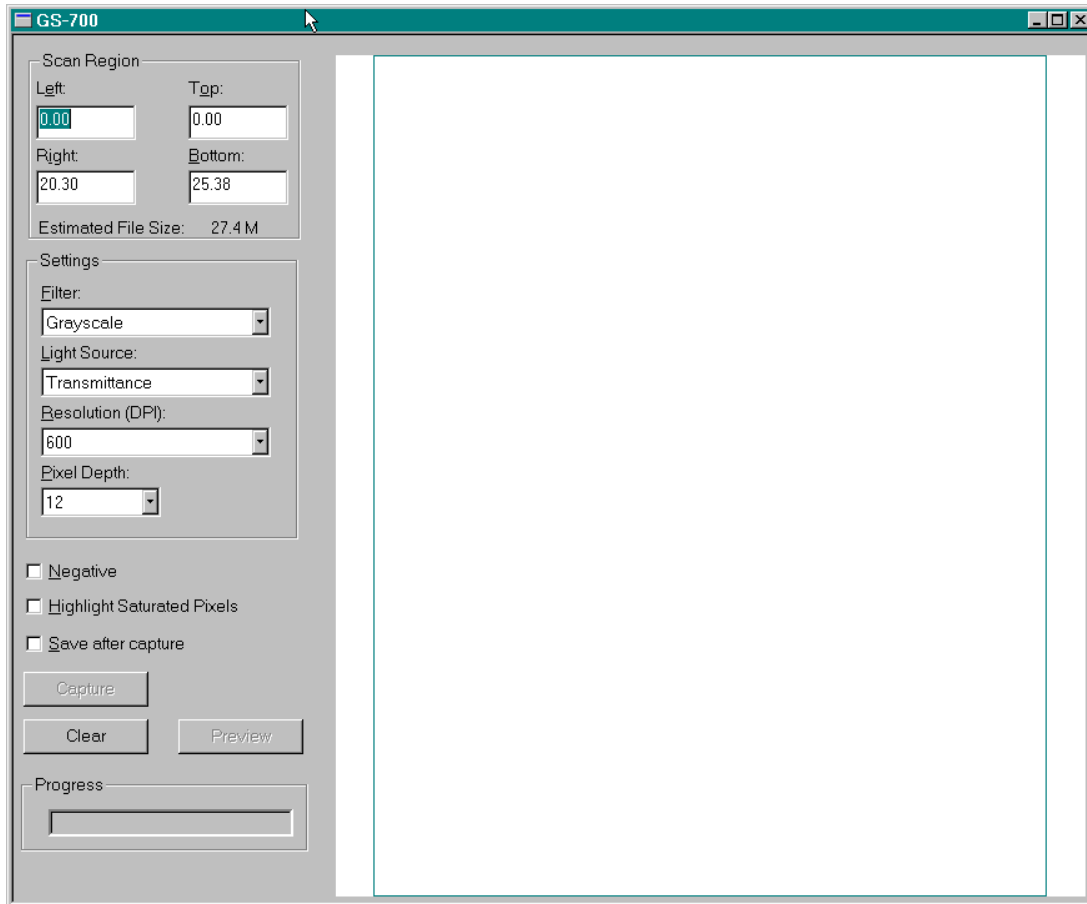


Figure 7-2 Densitometer Acquisition Window

7.7.1 Filters

The filter settings are: Grayscale, Red, Green, Blue.

7.7.2 Light Source

The light source settings are: Transmittance and Reflectance.

7.7.3 Resolution

Resolution settings are in dots per inch (DPI).

The GS-700 can scan at resolutions of 1200, 900, 600, 450, 300, 150, 72, 36, and 24 DPI.

The GS-690 can scan at resolutions of 800, 600, 400, 300, 200, 100, 72, 36, and 24 DPI.

7.7.4 Pixel Depth

Pixel depth can be either 12 bits per pixel or 8 bits per pixel for the GS-700. It is fixed at 8 bits for the GS-690.

7.7.5 Scan Region

Scan dimensions: In reflectance mode, the scanning area is 21.08 cm by 35.56 cm. In transmittance mode, the scanning area is 20.32 cm by 25.40 cm. At maximum resolution and scan area, this will generate an image of ~164 million pixels, which in turn will require ~328 MB of storage space. (Multi-Analyst uses 12 bit samples unpacked in 16 bit words.)

The estimated file size is updated on the fly and takes into account all of the parameters that affect file size, namely, scan area, bits per pixel, and resolution.

7.7.6 Save After Capture

If “Save after Capture” is checked, the program will prompt you for a filename for the document before the scan starts (a filename based on the login name and current date and time will be suggested). The document will then be saved to a disk file as soon as the image is captured.

7.7.7 Highlight Saturated Pixels

When *Highlight Saturated Pixels* is selected, pixels corresponding to saturated data will be painted red (or the selected saturation color). When several data points map to a single pixel on the display, that pixel will be considered saturated if any of the data points is saturated.

7.7.8 Preview Mode

In preview mode, the whole screen will be scanned at low resolution (24 dpi) using the current light source and filter settings.

7.7.9 Capture

When you click *Capture*, the whole scanning process begins. The *Capture* button will change to a *Stop* button and will remain enabled. All other controls will be disabled for the duration of the scan. When the scan is complete, the image will be stored inside the current document, and the scan view attached to a new document. If the scan is stopped while in progress, the Densitometer is stopped, and whatever image data has been captured thus far is stored inside the document. If no image data has been acquired, no new document is created. While scanning is in progress, the image in the scan window will be updated at regular intervals to show the progress of the scan. A progress bar is displayed showing the percent of the image acquired so far. After a sub-image has been captured you may again select a sub-region and repeat the process. You may keep “drilling-down” in this manner until the minimum scan area has been selected.

7.7.10 Clear

The *Clear* button may be used to clear the current image and reset the scan area to the maximum area for the selected light source. Changing the Light Source has the same effect as using the *Clear* button.

7.7.11 Template Settings

The following settings are stored with the template:

- Scan area
- Light source
- Scanner type
- Filter
- Resolution
- Pixel depth
- Negative if checked
- Highlight saturated pixels if checked
- Save after capture

7.7.12 Notebook Entries

The Densitometer is a single-channel device. The *Acquisition* page will consist of the following information:

- Color Filter selection
- Scanner type (GS-700 or GS-690)
- Resolution
- Scan Area requested
- Light Source selection

Detailed image information may be found as always on the *Image Description* page of the *Notebook Entries*.

8. Automatic and Semi-Automatic Lane and Band Finding

8.1 Lane and Band Finding

Lane and Band Finding mode detects lanes and bands on a image within a specified region of interest based on pattern recognition algorithms. The Semi-Automatic mode utilizes help from you in defining a typical lane width to assist the pattern recognition algorithms in lane determination.



Figure 8-1 Lane Finding Menu

8.2 Automatic Lane Finding

The *Auto Lane and Band Finding* command will find lanes and bands automatically in the active image.

To help find lanes and bands more accurately, it is recommended that the region of interest be specified using the Image Selection Tool. Drag to encompass the region of interest. This provides the algorithm with a smaller area of information to process and increases the accuracy and speed of the lane and band finding.

Note: The Automatic mode requires that an image contains 4 or more lanes within a selected region. For images with 3 lanes or less, use the manual lane definition tools.

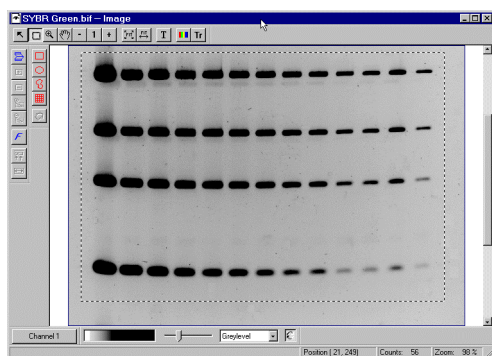


Figure 8-3 Selected Region

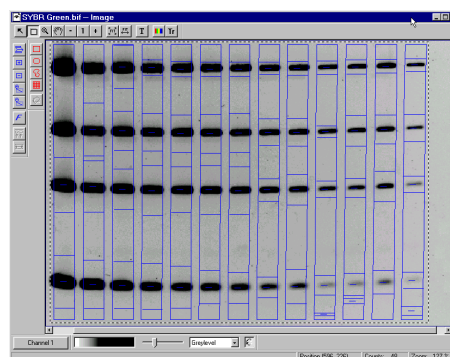


Figure 8-2 Found Lanes

Automatic Lane and Band Finding is very effective for many typical 1-D electrophoresis gels which have regular spacing between lanes. It is extremely difficult to design an algorithm which will work for all gel types and handle all types of gel artifacts. Some manual editing after Automatic Lane and Band Finding may be required to get all of the bands of interest. The intent is that for many situations the Automatic Lane and Band Finding will help speed your analysis and you can concentrate on only a few edits. If Automatic Lane and Band Finding is not finding most of the lanes and bands on your gel try Semi-Automatic mode.

Tips for using Automatic and Semi-Automatic Lane and Band Finding

Gels which do not have space between lanes, such as sequencing gels using shark tooth combs, will not be found by Automatic Lane Finding. You must use the manual lane drawing function .

Gels must be oriented vertically with the top of the gel as the top of the image. Bands on gels placed horizontally will not be found by Automatic Lane Finding. If your image was captured in a horizontal orientation, rotate the image vertically (Edit , Rotate) before selecting Lane & Band Finding.

Bands which have saturated intensity will not be found by Automatic Band Finding. Saturated bands will have to be added manually. For this reason it is important to acquire images using the Highlight Saturated pixel feature available with the Fluor-S MultiImager, Gel Doc 1000, and Densitometers.

If Automatic Lane & Band Finding finds bands and or lanes in areas between bands or lanes (background) the data is inverted such that bands are viewed as valleys instead of as peaks.

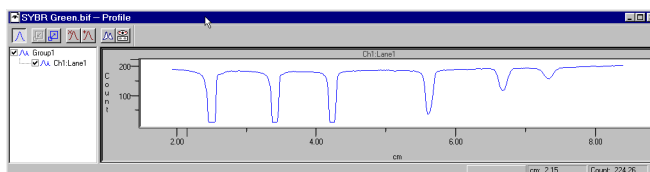


Figure 8-4 Inverted Data

Solution: Invert the raw data using the *Image... Invert Raw Data* menu item. NOTE: using the color palette to invert does not invert the data. This is only accomplished using the Invert Raw Data in the menu

8.3 Semi-Automatic Lane and Band Finding

The Semi-Automatic command will also find lanes and bands. When this command is selected, the following Find Lanes Using Template window is displayed.

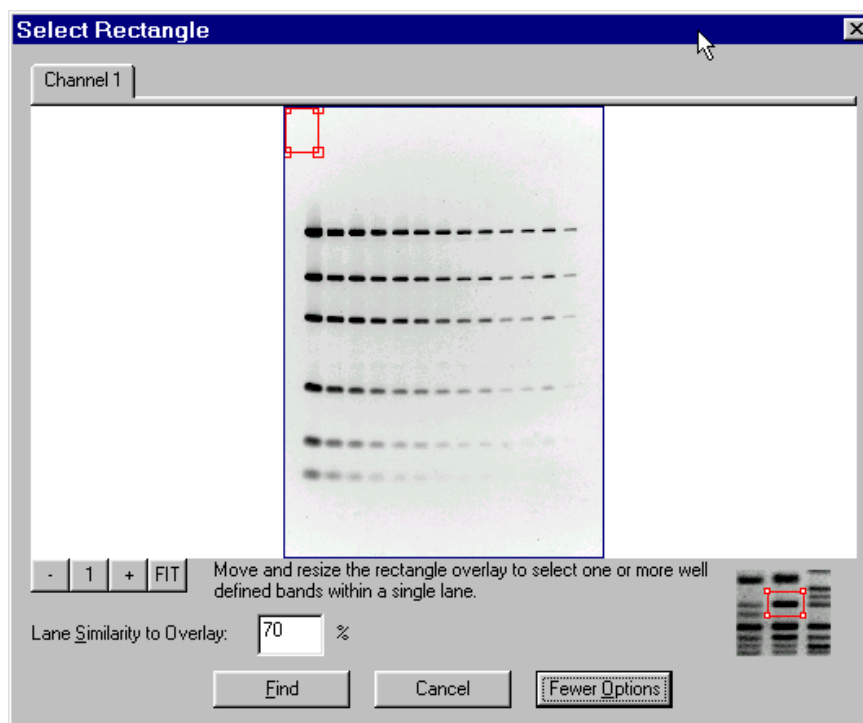


Figure 8-5 Semi-Automatic Template

- Drag the rectangle to a region of the image where typical bands are located.
- If the bands are not displayed in the window, use the horizontal or vertical scroll bars to move the gel.
- After placing the rectangle on the left top of a band or bands, click on the right bottom point of the rectangle. Resize the rectangle to the width of a lane. More than one band of the lane may be included to help provide the algorithm with as much information about the lane as possible. See below.

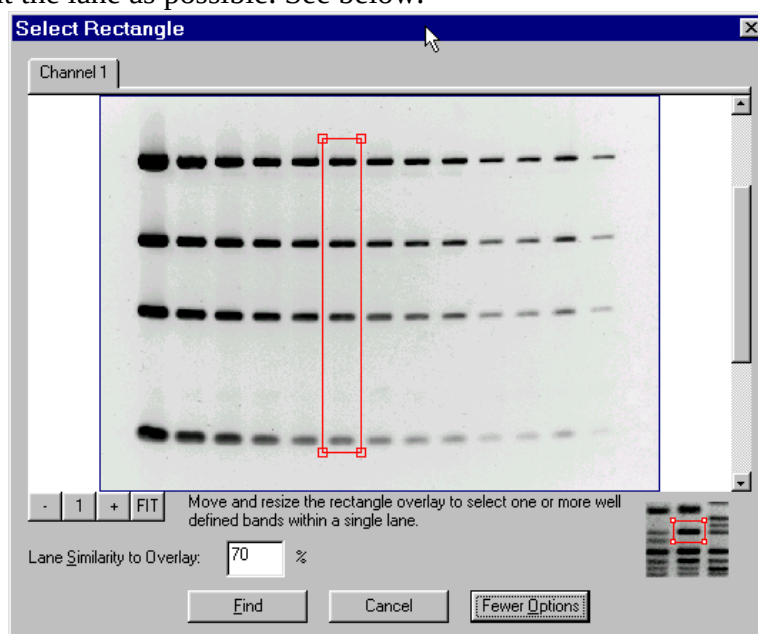


Figure 8-6 Semi-Automatic Example

- d. Click on *Find*

More Options

Selecting the *More Options* button, will open the Lane Similarity to Overlay option allowing you to set the similarity of the template overlay to the lanes and bands in the image. The default value is 70.0%. Increasing the Lane Similarity percentage makes the lane finding more discriminating - the algorithm becomes less forgiving for lanes that do not match the template overlay closely. Decreasing the Lane Similarity value will make the algorithm more forgiving and more lanes will be found.

8.4 Parameter Setting

The dialog box for Automatic Lane & Band Finding is selected by *Lane & Band Settings* in the *Analysis* menu. Multi-Analyst's *Automatic Lane & Band Finding* will work correctly for many 1-D electrophoresis gels without changing the default parameters. However, changing these settings allow improvement of the automated band finding. Tip: If Automatic Lane and Band Finding has found most of the lanes and bands correctly it may be useful to simply manually edit the remaining bands.

Shown below is the Lane and Band Settings dialog box:

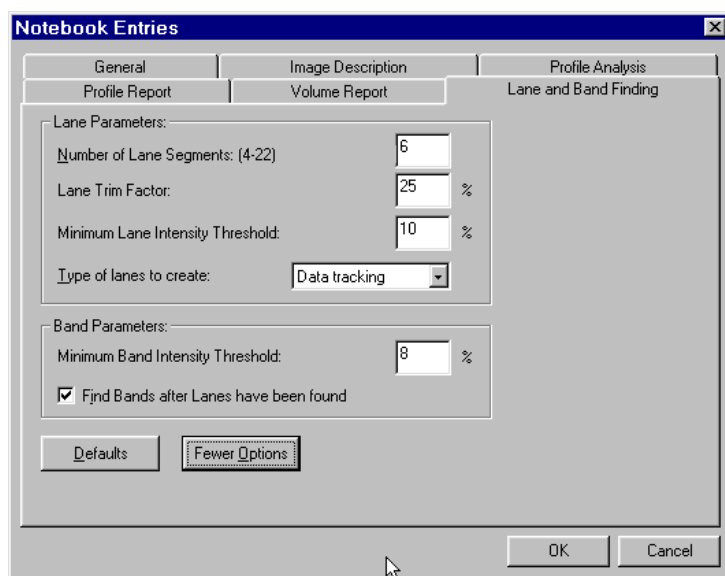


Figure 8-7 Lane and Band Settings

Number of Lane Segments

This command sets the number of anchor points (also known as nodes or segments) which will allow a lane to bend and thus to track the curvature of lanes. Increasing the number makes the lane tracking more flexible. The default value is 5 which will be suitable for most lanes. If changes are desired, the number of segments may be changed from 4 to 22 segments.

Minimum Band Intensity Threshold

This parameter affects the minimum intensity threshold used to recognize bands. The *Find Bands* function will find bands of anything larger than x% of the maximum band intensity in each lane. If the number is increased, fewer bands will be found. If the number is decreased, more bands will be found.

More Options

Additional parameter settings are displayed when you click on the *More Options* button. Again, it is unnecessary to change these parameters for many gels.

Lane Trim Factor

This parameter is used to change the lane trimming factor. Lane Trim is the ratio of the width of the lane divided by the width of inter-lane spacing. The default value, 75%, is the default used for the most common electrophoresis combs.

Minimum Lane Intensity Threshold

When finding lanes the software automatically calculates the band intensity by taking a vertical profile through the image at multiple regions of the image. The lane which has the highest intensity is recorded. The *Minimum Lane Intensity Threshold* sets the minimum threshold that will be recognized as a lane as a percentage of the maximal lane.

Type of Lanes to create

This is a pull-down menu to allow selection of the type of lane tracking mode that will be used.

- *Data Tracking* is the default for this menu. It uses the Raw tracking results and typically will give a smoother lane result. It will track the lanes according to the *Number of Lane Segments*.
- *Raw Tracking* tracks the lanes according to the *Number of Lane Segments*, and its center of gravity. This may be useful for some very irregular lane patterns.
- *Straight* averages the data to find and place a straight lane on your gel.

Find Bands after Lanes have been found

This button sets the *Automatic Lane & Band* function to find the bands automatically after finding lanes. If this is not checked, the *Automatic Lane & Band* function finds only lanes.

Defaults

This button allows an easy way to reset the factory settings for all parameters. Note: any user defined default values will be discarded if Factory Setting is selected.

More Options/Fewer Options

This button is to close the *More Options* window.

8.5 Interactive Band Finding

This menu item initiates band finding. This function is normally used when:

- After drawing lanes with *Manual Lane Drawing Tool* . Since *Automatic Lane & Band Finding* requires more than three lanes, *Find Bands* can be used after manually defining the lanes.
- When the lane/band set was moved or resized to fit to an actual lane after performing *Automatic Lane & Band Finding*, it may be better to select *Interactive Band Finding* again for a recalculation of the band finding algorithm.
- If the *Automatic Lane & band Finding* was performed and *Find Bands after Lanes have been found* option in *Lane & Band Setting* is not selected, you may select *Interactive Band Finding*.

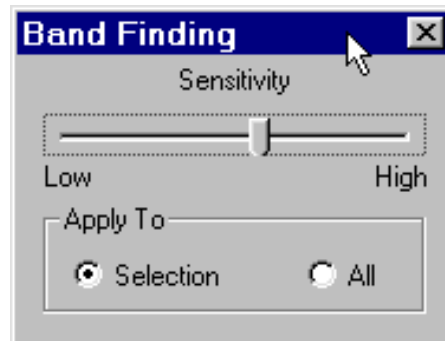


Figure 8-8 Band Finding

☐ Find Bands after Lanes have been found

8.6 Using the Lane and Band Tools

The Lane and Band Tools allows manual creating and editing of lane and bands. Note: If using these tools with a multi-channel image, drawing the Lane and Band in a lane in one channel will automatically draw the same lane in the other channel.



8.6.1 Lane Drawing

This tool allows an easy method of manually drawing a multi-segment lane. To use this tool:

- a. Select the tool. Move the cursor to the image window. The cursor will be identified as a .
- b. Click on the top left of the lane.
- c. Drag the mouse down the lane. You will see a red line rectangle.
- d. Let go of the mouse button at the location where you need to change the direction of the lane (this establishes an anchor point).
- e. Click whenever you need to change the direction of the lane (each click creates an additional anchor point).
- f. **Double click at the bottom of the lane to finish and draw the resulting lane.**



8.6.2 Add Anchor point

This function is used to add a new anchor point pair after the manual lane drawing or automatic lane finding has been performed. To add the new anchor point pair:

1. Click on the *Add Anchor point* tool
2. Move the cursor into the lane rectangle.
3. The cursor is changed to .
4. Move the cursor vertically and click at the place where you would like to add anchor point pair.
5. Adding Anchor point pairs is for fine tuning for the particular lane.
6. After adding an Anchor point, you will move the points to track the lane. To do this:
7. Click on the Select Tool.
8. Move the cursor on the anchor point which you would like to move.
9. The cursor is changed to .
10. While the cursor is changed to above, click on the anchor point and drag it in order to track the lane correctly.



8.6.3 Delete Anchor point

This tool is used to delete an existing anchor point pair after manual lane drawing or automatic lane finding has been performed. To delete the anchor point pair:

1. Click on the *Delete Anchor point* tool and move the cursor to select the lane of interest.
2. Move the cursor to right on the point you would like to delete.
3. The cursor is changed to .
4. Now click on the point. The point will be deleted and the lane will be made into a straight line.



8.6.4 Band Drawing

This tool is used to manually add a band in a lane. To use this tool:

1. Click on the *Band Drawing* tool and move the cursor to the lane and click to select the lane of interest.
2. Move the cursor into the selected lane.
3. The cursor is changed to .
4. Place the cursor on top of the band.
5. Click and drag down to the bottom of the band.
6. Release the mouse.



8.6.5 Band Deleting

This tool is used to delete a band in a lane after *Automatic Band Finding* has been performed. To use this tool:

1. Click on the *Band Deleting* tool and move the cursor in the lane and click to select the lane.

2. Move the cursor into the band in the selected lane.
3. The cursor is changed to .
4. Click on the band to delete.

When the above tools are selected, the cursor is changed to  and . These allow resizing of the lane width  and the lane length . Just click and drag to resize. The cursor changes to  to allow the lane to be moved

9. Volume Analysis

Volume analysis is a way to quantitate the intensity of the data mathematically.

9.1 Volume Integration Tool

There are several tools displayed with an image. They are used to make volume integration objects as follows:

9.1.1 Rectangle



This tool allows you to identify a spot or band to be integrated..

To use this tool:

1. Click and select.
2. Move the cursor in the image area. The cursor will be changed to $+$.
3. Place the cursor at left top of the band.
4. Drag the cursor down to bottom left.
5. Release the mouse.

9.1.2 Ellipse



This tool allows you to identify a spot to be integrated. To use this tool:

1. Click and select.
2. Move the cursor in the image area. The cursor will be changed to $+$.
3. Place the cursor at left top of the band.
4. Drag the cursor down to bottom left.
5. Release the mouse.

9.1.3 Freehand



This tool allows you to identify any object to be integrated. To use this tool:

1. Click and select.
2. Move the cursor in the image area. The cursor will be changed to .
3. Place the cursor over your object.
4. Click and hold the mouse to drag around the object.
5. When you release the mouse, the object “box” will automatically close to complete the area.

9.1.4 Grid



This tool provides a pre-defined or user defined grid template which superimposes a grid on top of the image in the active image window.

9.1.4.1 Defining the grid template:

1. Select grid type: standard or custom.
2. If it is custom, you need to insert the numbers for Rows and Columns.
3. Select a shape if it is either Rectangular or Elliptical.
4. Click OK after making change.

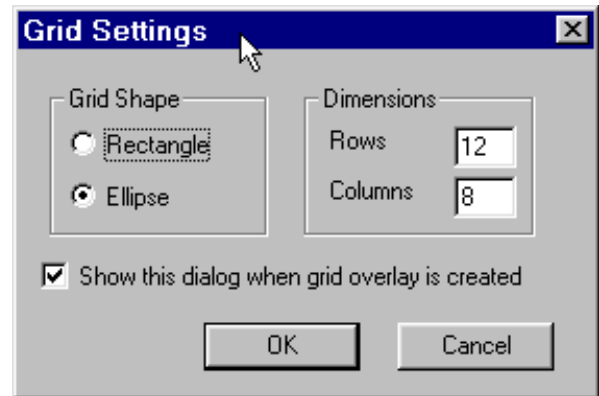


Figure 9-1 Grid Settings

9.1.4.2 Placing the grid on the image:

1. Click and select the grid tool.
2. Move the cursor into the image area.
3. The cursor will be changed to $\oplus$.
4. Place the cursor at the top left of the area to be quantitated.
5. Click and hold the mouse and drag the mouse down to the bottom right.
6. Release the mouse to make the grid.

9.2 Useful object operations:

9.2.1 To copy a volume integration tool:

1. Move the cursor into the object.
2. The cursor will be changed to $\oplus$.
3. Press and hold <Option> key.
4. Drag the object to the next location you wish to quantitate.
5. Once the object is moved, release the mouse button.
6. The object will be copied.

9.2.2 To resize a volume integration tool:

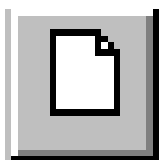
1. Move the cursor to a corner of a volume object.
2. The cursor will be changed to $\boxtimes$.
3. Click and drag the corner to resize the object.
4. Release the mouse.

10. Image Menus

Note: The following Menu items are shown when an Image window is the active window. When a Profile Window or a Profile Report window is active the menus change to offer functions useful with the respective views of the data. See the Sections on Profile Menu and Profile Report Menus for detail on any differences from the Image Menu.

10.1 File Menu - Image Window Active

10.1.1 New



Creates a new document indicating the start of a new experiment. New will open the acquisition screen for the default template allowing image capture. See New Using Template... below.

10.1.2 New Using Template...

Creates a new document with a saved template. A previously saved template may contain as much or as little experimental information as you wish. You may store the acquisition settings that work best for your particular samples. Templates also save preferred lane and band finding settings, etc.

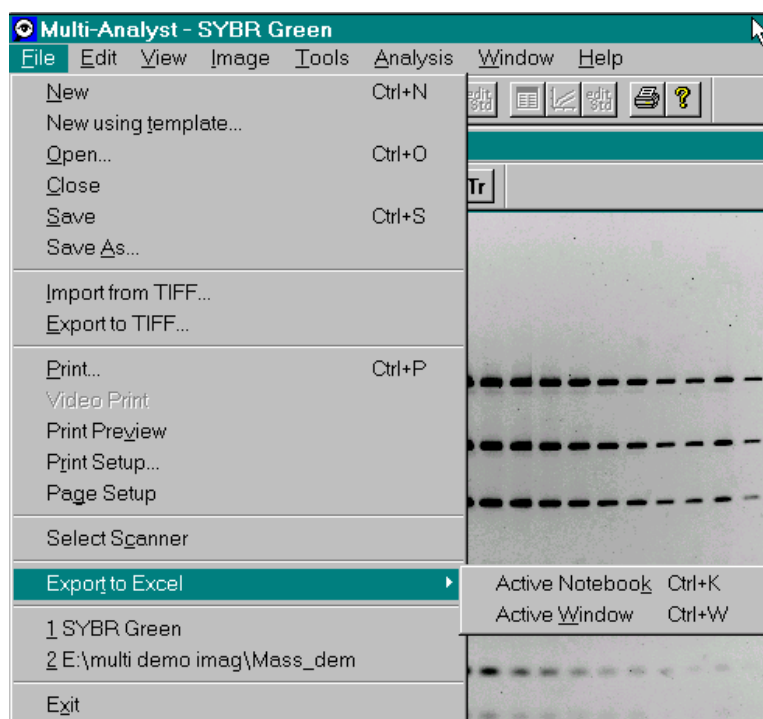


Figure 10-1 File Menu - Image Window Active

10.1.3 Open...

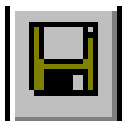


Opens a saved document.

10.1.4 Close

Closes the active document.

10.1.5 Save



Saves the active document.

10.1.6 Save As...

Saves the active document with a new name. The default file extension is .bif (Bio-Rad Image File). To save a template you must change the file type to template (.bit)(Bio-Rad Image Template)

10.1.7 Import From TIFF...

Allows import of images that have been saved in a TIFF format.

10.1.8 Export to TIFF...

Provides a way to export images as a TIFF format file for use in other applications such as Microsoft Word, PowerPoint, Adobe Photoshop or other packages designed to handle images.

10.1.9 Print



Prints the active document. Provides standard print functions.

10.1.10 Video Print

Prints the active document image only. File creation date and filename is printed on the image. Will be grayed out if no Gel Doc frame grabber card is installed.

10.1.11 Print Preview

Display a full page view in WYSIWYG for previewing before printing. Also provides the ability to zoom in and out to examine detail such as the footer information.

10.1.12 Print Setup...

Change the printer setup settings.

10.1.13 Page Setup

Change the printing options. Provides the ability to print images in actual size.

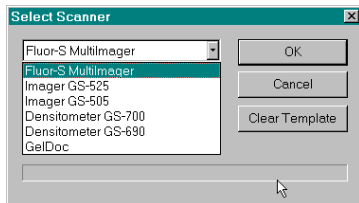


Figure 10-2 Scanner Settings

10.1.14 Select Scanner

Selects the Bio-Rad imaging device to use or to develop templates.

10.1.15 Scanner Menu

All currently supported Bio-Rad Devices are listed for template creation.

10.1.16 Export to Excel - Active Notebook

Export the active notebook to Excel. Notebook refers to a complete set of data. This would include the Image, Profile and Profile Report.

10.1.17 Export to Excel - Active Window

Exports the active Window to Excel. In contrast to the Active Notebook, this function will only export the window that is currently active. For example, you can export only the profile report and not the profile or image if you wish.

10.1.18 <Recently Opened File List>

This is a list of recently opened files.

10.1.19 Exit

Quit the application, prompts to save documents.

10.2 Window Edit Menu - Image Active

10.2.1 Undo

Undo the last action. NOT AVAILABLE IN THIS VERSION!!

10.2.2 Cut

Cut the selection and put it on the Clipboard. NOT AVAILABLE IN THIS VERSION!!

10.2.3 Copy Image

Copies the image and puts it on the Clipboard.

10.2.4 Paste

Insert the Clipboard contents. NOT AVAILABLE IN THIS VERSION!!

10.2.5 Delete

Deletes the selected overlays.

10.2.6 Select All

Selects all the overlays.

10.2.7 Delete All

Combines Select all and Delete All into a single command. Will select all the objects, make them all active, ask for user verification of desire to delete. If click OK, then deletes all objects.

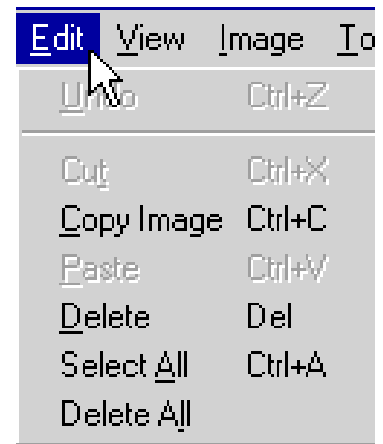


Figure 10-3 Edit Menu Image Active

10.3 View Menu - Image Window Active

10.3.1 Toolbar

Show or hide the toolbar at the top of the image



10.3.2 Status Bar

Show or hide the status bar at the bottom of the screen. The status bar gives long description of all icons and active menu items.

10.3.3 Image Toolbars

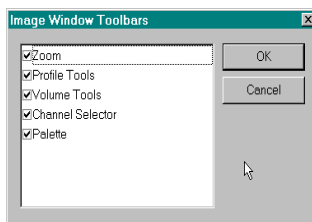


Figure 10-5 Image Toolbars Menu

Brings up a menu which allows selection of which toolbars to display in the Image window. Useful to find missing toolbars!

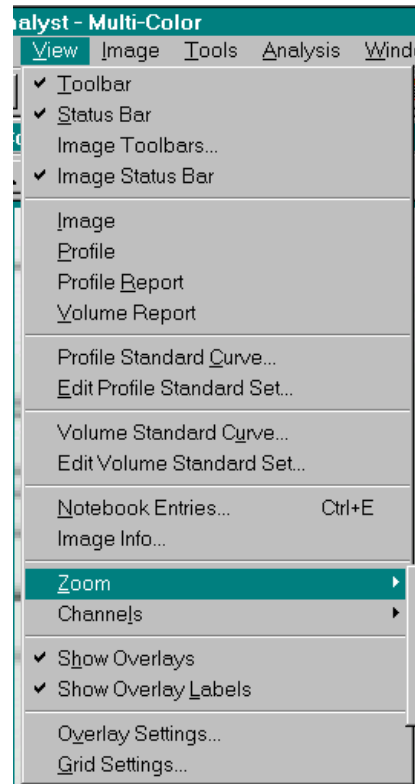
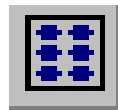


Figure 10-4 Image View Menu

10.3.4 Image Status Bar

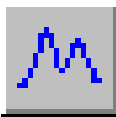
Show or hide the image Status Bar. The image Status Bar gives valuable data about your image pixels, lanes, and bands and other quantitative values.

10.3.5 Image



View the image. Displays the image on top of the open windows. Very useful for small monitors, or when many windows are open.

10.3.6 Profile



View the profile. Displays the profile on top of the open windows. Very useful for small monitors, or when many windows are open. Only available when a lane has been identified (drawn).

10.3.7 Profile Report



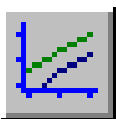
View the profile Report. Only available when a lane has been identified and the bands have been found.

10.3.8 Volume Report



View the volume Report. Only available when volume object boxes have been drawn around bands of interest.

10.3.9 Profile Standard Curve...



Show Profile Standard Regression Curve Window. Only available if a lane has been identified and standards have been set.

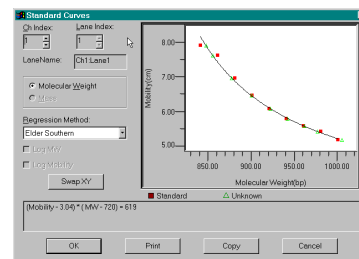


Figure 10-6 Profile Regression Window

10.3.10 Edit Profile Standard Set...



Show Profile Standard Set entry Window.

Band Name	Mol Weight	Mass
StdBand 1	1000.00	100.00
StdBand 2	700.00	70.00
StdBand 3	500.00	50.00
StdBand 4	200.00	20.00
StdBand 5	100.00	10.00

Figure 10-7 Edit Profile Standard

10.3.11 Notebook Entries...



Set or view the experimental variables which were used in the experiment. This includes:

General information such as

date of acquisition, user name, and image comments,

Image Description which includes type of file TIFF etc., bit depth etc.

Profile Analysis such as type of background used, and standards assigned

Profile Report which allows customization of the fields to be displayed in the report

Lane and Band Finding parameters

Volume Report such as type of background used, standards assigned and fields shown.

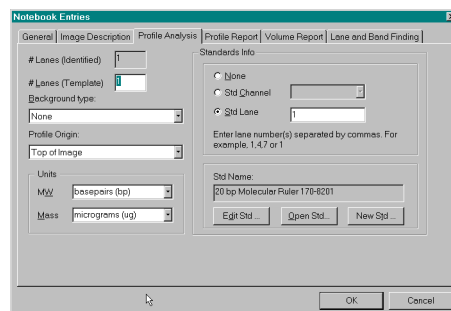


Figure 10-8 Notebook Entries

10.3.12 Image Info...

Provides access to image comment fields in the notebook which will be printed in the footer for images.

10.3.13 Zoom



Zoom Out

Zooms the image to half its current size.



Zoom to 100%

Zooms the image to 100% of its original size.



Zoom In

Zooms the image to twice its current size.



Best Fit

Displays the entire image fit within the image window



Fit To Width

Displays the image to fit the width of the window

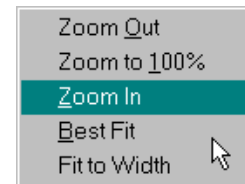


Figure 10-9 Zoom Menu

10.3.14 Channels

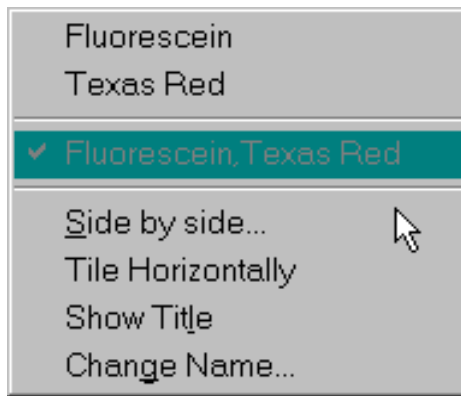


Figure 10-10 Channel Menu

throughout Multi-Analyst

Channels - <channel name>

Displays the channel names used

Channels - Side by side...

View images of the different channels side by side

Channels - Tile Horizontally

Tile the image channels horizontally

Channels - Show Title

Show channel title at the top of each image.
Handy for keeping track of multiple channel image sources.

Channels - Change Name...

Change the channel name which will appear

10.3.15 Show Overlays

A toggle which allows overlays to be shown or hidden from view.

10.3.16 Show Overlay Labels

Shows or hides the labels for the overlays.

10.3.17 Overlay Settings...

Sets the default colors used for active Selected objects, Standard lanes, Unknown lanes and Default text

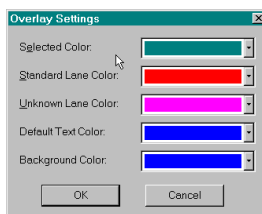


Figure 10-11 Overlay Color Settings Menu

10.3.18 Grid Settings

Menu to input the number of lanes and rows for creating custom volume grid tools.

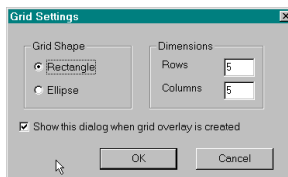


Figure 10-12 Grid Settings Menu

10.4 Image Menu - Image Window Active

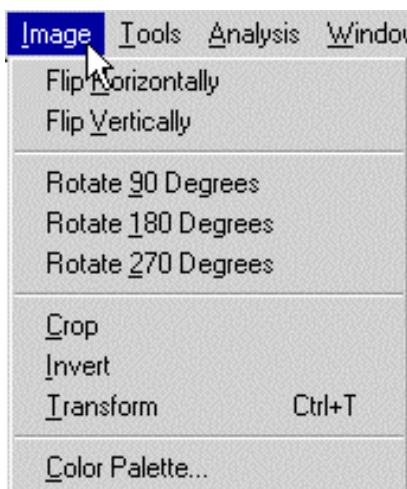


Figure 10-13 Image Menu

10.4.1 Flip Horizontally

Flips the image around a horizontal axis

10.4.2 Flip Vertically

Flips the image around a vertical axis

10.4.3 Rotate 90 Degrees

Rotates the image 90 degrees (clockwise)

10.4.4 Rotate 180 Degrees

Rotates the image 180 degrees (clockwise)

10.4.5 Rotate 270 Degrees

Rotates the image 270 degrees (clockwise)

10.4.6 Crop

Crops the selected part of the image

10.4.7 Invert

Inverts all the pixel values of the **raw** image (min <-> max). Use this function if the profile data is inverted or reacquire using the Negative check box in the acquisition screen. THIS MENU ITEM DOES NOT HAVE AN ICON EQUIVALENT!!!

10.4.8 Transform



Determines how the raw data is transformed into display data for viewing purposes. Brings up the Transform menu dialog box.

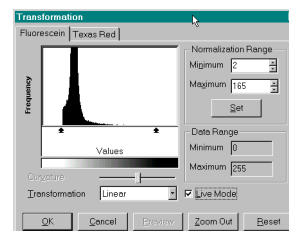
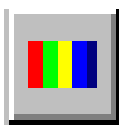


Figure 10-14 Transform Dialog Box

10.4.9 Color Palette...



Allows adjustment of the displayed image by activating the color palette menu. Adjust brightness, contrast and apply pseudocolor to images. Pre-set pseudocolors including Coomassie, EtBr etc.

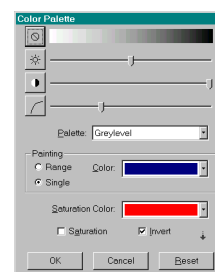


Figure 10-15 Color Palette Menu

10.5 Tools Menu - Image Window Active

10.5.1 Select Overlays

Activates the pointer tool to allow selection of object overlays.



10.5.2 Rotate Overlays

Rotates the selected object volume box.
DOES NOT ROTATE LANE
OBJECTS!!!



10.5.3 Selector

Image Selector tool. Allows definition of a region of interest to be manipulated ex. select group of volume boxes.



10.5.4 Zoom

Show the magnifying glass for zooming.



10.5.5 Pan

Pan or move the Image.



10.5.6 Add Text Overlays

Create text overlays.



10.5.7 Add Lanes

Create lane overlays.



10.5.8 Add Lane Segments

Add segment to lane. Also known as lane nodes. These serve as anchor points allowing lanes to bend to fit to the curvature of a lane.

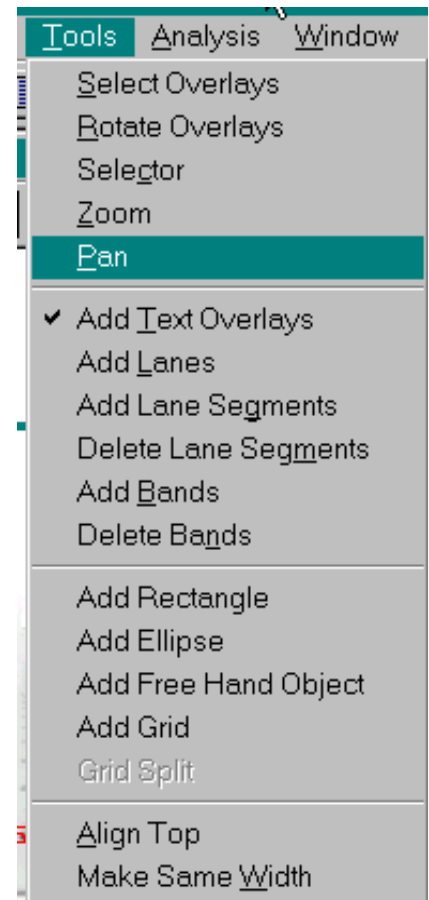


Figure 10-16 Tools Menu

10.5.9 Delete Lane Segments

Delete segment nodes from a lane.



10.5.10 Add Bands

Adds bands to a lane.



10.5.11 Delete Bands

Delete band from lane.



10.5.12 Add Rectangle

Volume Tool. Activates the draw rectangle volume object.



10.5.13 Add Ellipse

Volume Tool. Activates the draw ellipse volume object.



10.5.14 Add Free Hand Object

Volume tool. Activates the draw Free Hand volume object.

10.5.15 Add Grid

Volume Tool. Activates the draw Grid volume object.



10.5.16 Grid Split

Allows for the decomposition of a defined grid into its constituent parts.

10.5.17 Align Top

Aligns the top edges of all lanes with the selected lane.



10.5.18 Make Same Width

Resize all lanes to have the same width as the selected lane.



10.6 Analysis Menu - Image Window Active

10.6.1 Auto Lane and Band Finding

Finds all lanes on the image or selected region automatically



10.6.2 Semi-Auto Lane and Band Finding...

Finds lanes and bands using a specified band size defined by you. The region specified is one lane wide but may contain multiple bands. This information assists the lane and band finding algorithm in accurately determining the location of lanes and bands

10.6.3 Interactive Band Finding

Provides a way to adjust the band finding sensitivity for either all lanes or just selected lanes. This adjustment of band finding sensitivity is interactive and displays the results of band finding with the new setting quickly on screen.

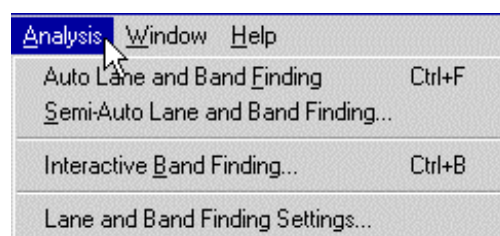


Figure 10-17 Analysis Menu

10.6.4 Lane and Band Finding Settings...

Allows access to the default lane and band finding settings used in the Auto Lane and Band Finding operation.

10.7 Window Menu - Image Window Active

10.7.1 New Window

Opens another window of the active document. This allows you to have a duplicate window open of (for example) an image and apply a green color palette to one and a red color palette to the other for the purpose of visual comparison. Or you can open multiple instances of the profile window in order to have independent axis of different profiles.

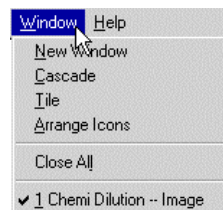


Figure 10-18 Window Menu

10.7.2 Cascade

Arranges all windows so they overlap.

10.7.3 Tile

Arranges windows as non-overlapping tiles.

10.7.4 Arrange Icons

Arrange icons at the bottom of the window

10.7.5 Close All

Close all open windows

10.7.6 <Open View List>

Displays a list of all open windows

10.8 Help Menu - Image Window Active



NOTE: HELP IS NOT CURRENTLY ACTIVE IN THIS RELEASE OF MULTI-ANALYST. PLEASE SEE THE MANUAL FOR INFORMATION ON THE VARIOUS FEATURES AND FUNCTIONS.



10.8.1 About Multi-Analyst

Displays program information, version number and copyright information

10.9 Lane Context Menu

The Lane Context Menu is accessed by positioning the cursor in a lane. With the cursor within the lane click the right mouse button. The lane will be made active and the Lane Context Menu will appear. Note: If the cursor is between a region that designates a band region (delineated by blue horizontal lines) the Band/Lane Context Menu will be activated. See Section 3.5 for the Band/Lane Context Menu

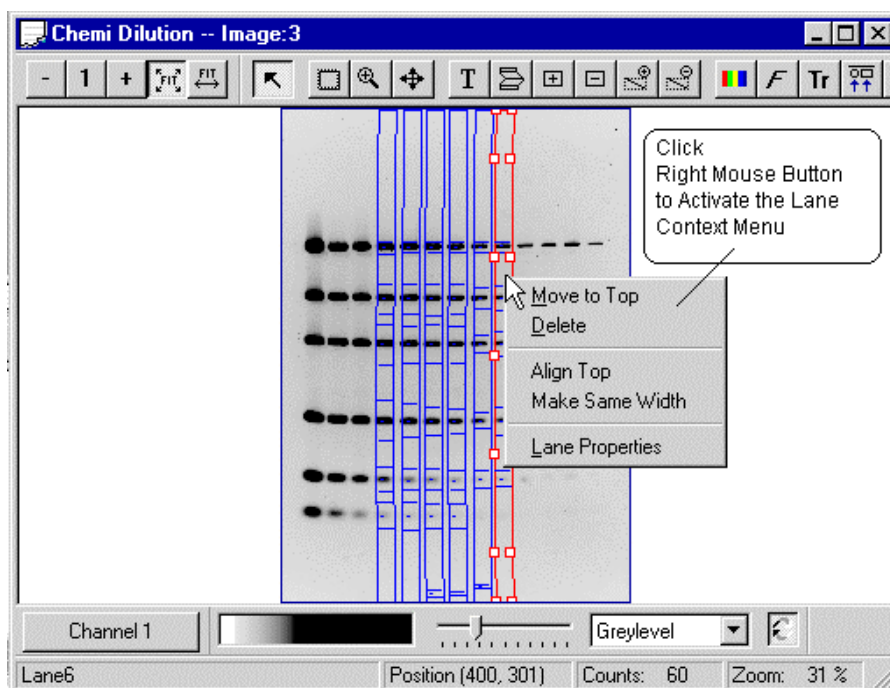


Figure 10-19 Lane Properties

10.9.1 Move to Top

Move to top of list

10.9.2 Delete

Deletes the lane

10.9.3 Align Top

Aligns the top edges of all lanes with the selected lane.

10.9.4 Make Same Width

Resize all lanes to have the same width as the selected lane.

10.9.5 Lane Properties

Provides access to Lane Properties which allows you to assign lane names, lane colors, etc.

10.10 Band/Lane Context Menu

10.10.1 Move Lane to Top

Move lane to top of list

10.10.2 Delete Band

Deletes band

10.10.3 Delete Lane

Deletes lane

10.10.4 Band Properties

Provides access to Band Properties which allows you to assign band names, and access information about the band.

10.10.5 Lane Properties

Provides access to Lane Properties which allows you to assign lane names, lane colors, etc.

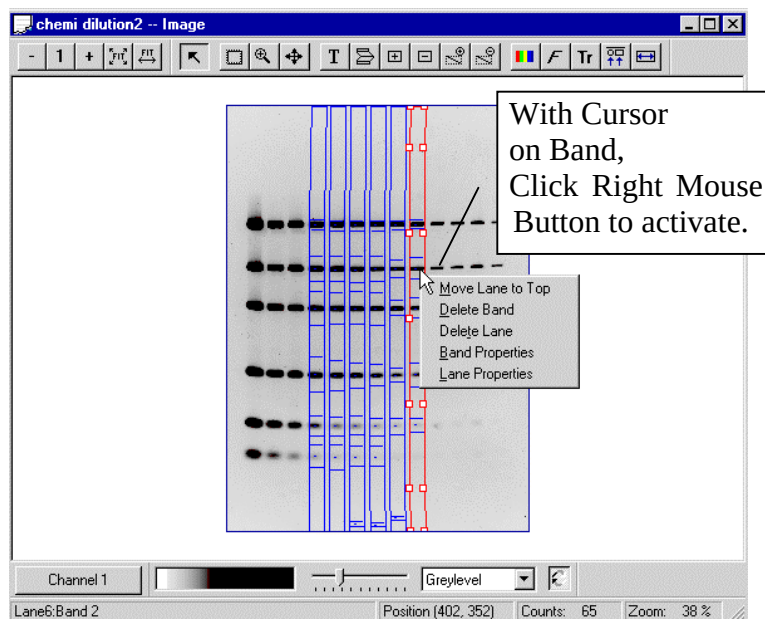


Figure 10-20 Band/Lane Context Menu

11. Profile Menus

11.1 Profile Window Menus

11.1.1 File Menu - Profile Window Active

The only difference between the Profile File Menu and the Image File Menu is that Export to Tiff... is grayed out in the Profile File Menu.

11.1.2 Edit Menu - Profile Window Active

11.1.2.1 Undo

NOT AVAILABLE IN THIS VERSION.

11.1.2.2 Cut

NOT AVAILABLE IN THIS VERSION.

11.1.2.3 Copy - View

Copies the Profile View to the Clipboard

11.1.2.4 Copy - Data

Copies the Selected Profile's XY Data To The Clipboard

11.1.2.5 Paste

NOT AVAILABLE IN THIS VERSION.

11.1.2.6 Delete

Deletes the selected lane profile from the profile view and the lane overlay is deleted from the image. The lane and profiles are automatically renumbered.

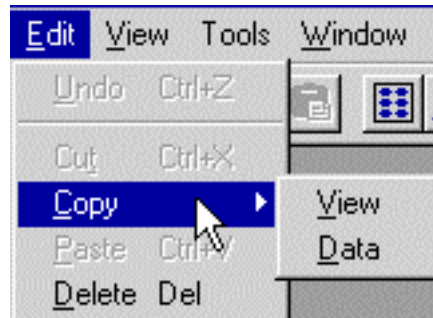


Figure 11-1 Edit Menu

11.1.3 View Menu - Profile Window Active

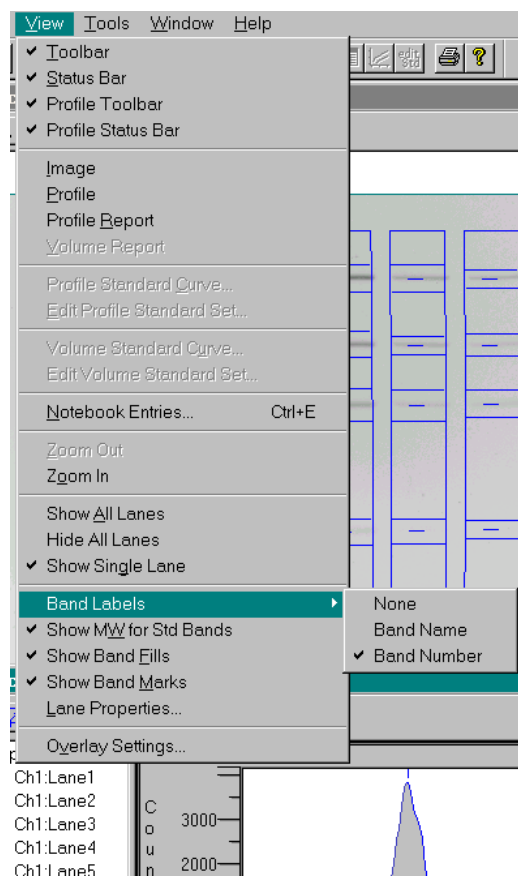


Figure 11-2 View Menu Profile Window Active

11.1.3.1 Toolbar

Displays/Hides the toolbar

11.1.3.2 Status Bar

Displays/Hides the status bar

11.1.3.3 Profile Toolbar

Displays/Hides the profile analysis toolbar.

11.1.3.4 Profile Status Bar

Displays/Hides the profile status bar.

11.1.3.5 Image

View the image

11.1.3.6 Profile

View the profile

11.1.3.7 Profile Report

View the profile report

11.1.3.8 Volume Report

View the volume report

11.1.3.9 Profile Standard Curve...

Displays the Standard Curve window

11.1.3.10 Edit Profile Standard Set...

Displays the Edit Profile standard window

11.1.3.11 Volume Standard Curve...

Displays the Volume Standard Curve

11.1.3.12 Edit Volume Standard Set...

Displays the Edit Volume Standard window

11.1.3.13 Notebook Entries...

Opens the Notebook for the active document

11.1.3.14 Zoom Out

Scales the profile view to display the entire profile graph.

11.1.3.15 Zoom In

Zooms in on an area of interest specified using a selection rectangle.

11.1.3.16 Show All Lanes

Displays all lane profiles.

11.1.3.17 Hide All Lanes

Hides all lanes. Simply select the lanes you wish to view by checking the corresponding check box next to the lane name.

11.1.3.18 Show Single Lane

When checked, a single lane is shown at a time.

11.1.3.19 Band Labels - None

Hides band labels.

11.1.3.20 Band Labels - Band Name

Shows band name as a label. Note: Band Names which will appear as labels can be easily entered by positioning the cursor over a profile peak and clicking the **right mouse** button. Select Band Properties and enter the Band Name of your choice.

11.1.3.21 Band Labels - Band Number

Shows band number as a label for example Band 1.

11.1.3.22 Show MW of Std Bands

Shows the MW of the Std Bands as a label.

11.1.3.23 Show Band Fills

Show/Hides Band fills within the profile window.

11.1.3.24 Show Band Marks

Show/Hide band tick marks within the profile window.

11.1.3.25 Lane properties...

Displays selected lanes properties.

11.1.3.26 Overlay Settings...

Display the overlay settings menu

11.1.4 Tools Menu - Profile Window Active

11.1.4.1 Add Band

Click at position of new band to add a band.

11.1.4.2 Delete Band

Click on band to delete it.

11.1.4.3 Set Profile background...

Opens Notebook Entries and allows ways of determining the method to be used for profile background settings.

11.1.4.4 Interactive Band Editing

Allows adjustment of band sensitivity settings so bands are found or not found.



Figure 11-3 Tools Menu

11.1.5 Window Menu - Profile Window Active

There are no differences between the Image Window Menu and the Profile Window Menu

11.1.6 Help Menu - Profile Window Active

There are no differences between the Image Help Menu and the Profile Help Menu

11.1.7 Lane Context Menu - Profile Window Active

11.1.7.1 Delete Lane

Deletes Selected Lane

11.1.7.2 Lane Properties

Display Lane Properties

11.1.7.3 Find Bands

"Find Bands"

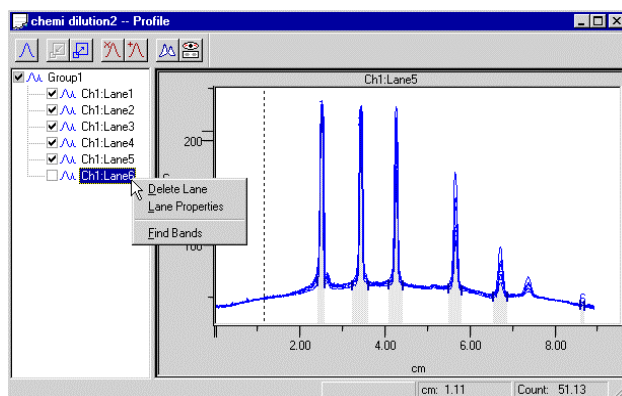


Figure 11-4 Lane Context Menu

Position cursor over desired band and click right mouse button

11.1.8 Band/Lane Context Menu - Profile Window Active

11.1.8.1 Delete Lane

Deletes Lane

11.1.8.2 Lane Properties

Display lane properties

11.1.8.3 Delete Band

Deletes Band

11.1.8.4 Band Properties

Display band properties

11.1.8.5 Find Bands

Find Bands

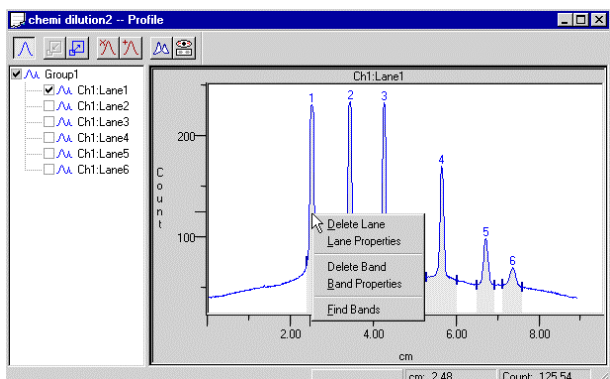


Figure 11-5 Band/Lane Context Menu

11.2 Profile Report Menus

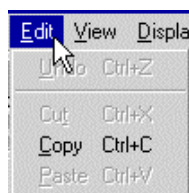
11.2.1 File Menu - Profile Report Active

This menu is the same as the Profile View Menu

11.2.2 Edit Menu -Profile Report Active

11.2.2.1 Copy

Copies the selection and puts it into the



clipboard

Figure 11-6
Edit Menu

11.2.3 View Menu -- Profile Report Active

11.2.3.1 Toolbar

Show or hide the toolbar

11.2.3.2 Status bar

Show or hide the status bar

11.2.3.3 Report Toolbar

Show or hide the report toolbar

11.2.3.4 Report Status Bar

Show or hide the report status bar

11.2.3.5 Image

View the image

11.2.3.6 Profile

View the profile

11.2.3.7 Profile Report

View the report

11.2.3.8 Volume Report

View the volume report

11.2.3.9 Profile Standard Curve...

Displays the Standard Curve window

11.2.3.10 Edit Profile Standard Set...

Displays the Edit Profile standard window

11.2.3.11 Volume Standard Curve...

Displays the Volume Standard Curve

11.2.3.12 Edit Volume Standard Set...

Displays the Edit Volume Standard window

11.2.3.13 Notebook Entries...

Opens the Notebook for the active document

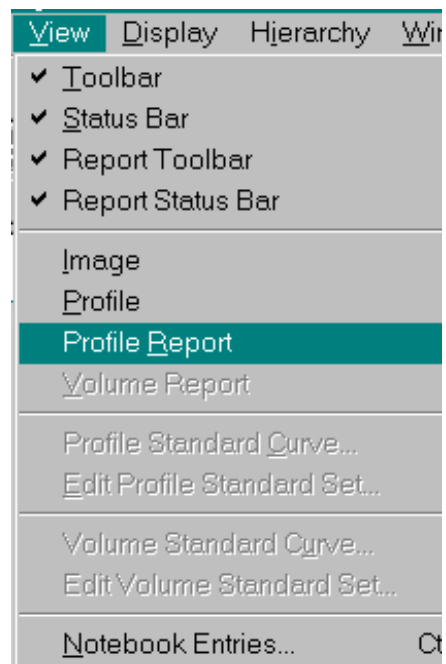


Figure 11-7 View Menu Profile Report

11.2.4 Display Menu - Profile Report Active

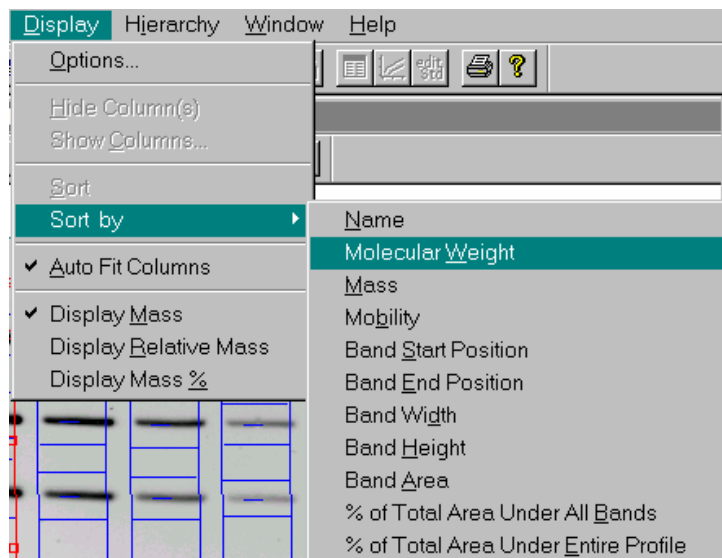
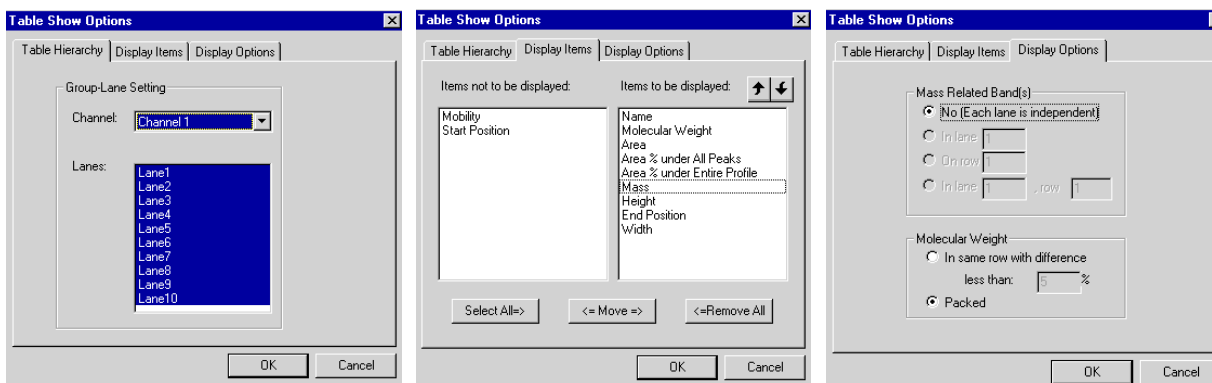


Figure 11-8 Display Menu Profile Report

11.2.4.1 Options...

Controls how Profile Report is configured (See below)



11.2.4.2 Hide Column(s)

Hide currently selected item(s) in report

11.2.4.3 Show Columns...

Show hidden item(s)

11.2.4.4 Sort

Sort by the selected item. See toolbars for sorting ascending or descending.

11.2.4.5 Sort by - Name

Sort by peak name

11.2.4.6 Sort by - Molecular Weight

Sort by molecular weight

11.2.4.7 Sort by - Mass

Sort by mass

11.2.4.8 Sort by –Mobility

Sort by band mobility

11.2.4.9 Sort by –Band Start Position

Sort by band start position

11.2.4.10 Sort by – Band End Position

Sort by band end position

11.2.4.11 Sort by – Band Width

Sort by band width

11.2.4.12 Sort by – Band Height

Sort by band height

11.2.4.13 Sort by – Band Area

Sort by band area

11.2.4.14 Sort by – % of Total Area Under All Bands

Sort by % of total area under all bands

11.2.4.15 Sort by – % of Total Area Under Entire Profile

Sort by % of total area under entire profile

11.2.4.16 AutoFit Columns

Auto fit the column width

11.2.4.17 Display Mass

Displays Mass

11.2.4.18 Display Relative Mass

Display relative mass

11.2.4.19 Display Mass %

Display mass %

11.2.5 Hierarchy Menu - Profile Report Active

These functions change the way the profile report is setup. Allowing access to your data in different ways.

11.2.5.1 Channel--Lanes

Channel--Lanes display mode. List lanes from top to bottom and columns represent the data.

11.2.5.2 Lane--Channels

Lane--Channels display mode. Produces file folder style tabs of the different lanes in each channel.

11.2.5.3 Landscape

Landscape display

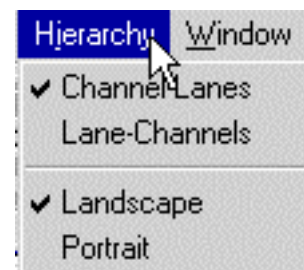


Figure 11-9 Hierarchy Menu Profile Report

11.2.5.4 Portrait

Portrait display

11.2.6 Window Menu - Profile Report Active

Same as Profile Window menu

11.2.7 Help Menu- Profile Report Active

Same as Profile Help menu

12. Volume Report Menus

12.1.1 Volume Context Menu

12.1.1.1 Move to Top

Moves the selected volume object to the top of the report

12.1.1.2 Move to Bottom

Moves the selected volume object to the bottom of the report

12.1.1.3 Delete

Deletes the selected volume object

12.1.1.4 Volume Overlay Type

Sets the selected volume objects to one of the three following types:

- Unknown
- Standard
- Background

12.1.1.5 Volume Object Properties

Opens the volume object property info box.

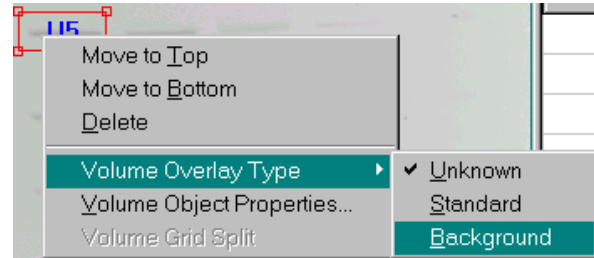


Figure 12-1 Volume Context Menu

12.1.2 Options Menu – Volume Active

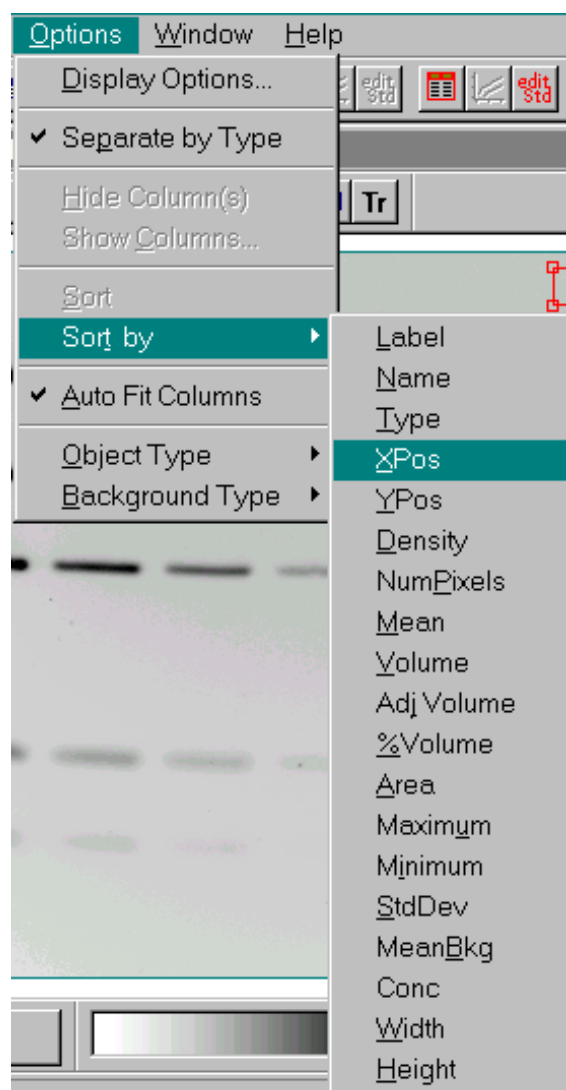


Figure 12-2 Options Menu Volume Active

12.1.3 View Menu Volume Active



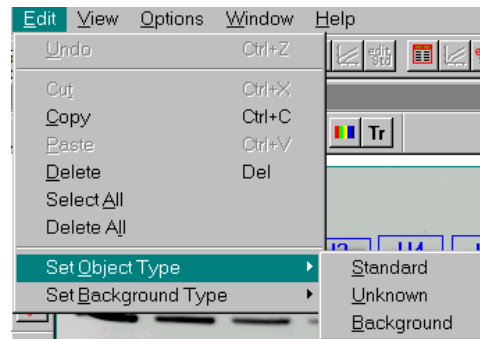
Figure 12-3 View Menu Volume Active

12.1.4 Edit Menu Volume Active

12.1.4.1 Set Object Type

Sets the active volume object to one of the following types:

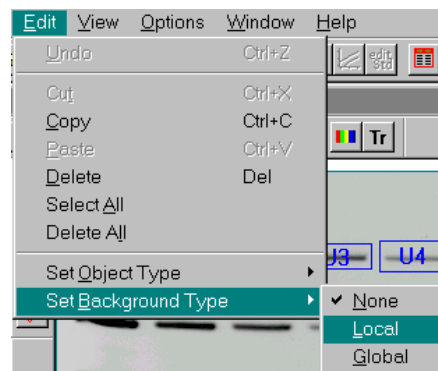
- Standard
- Unknown
- Background



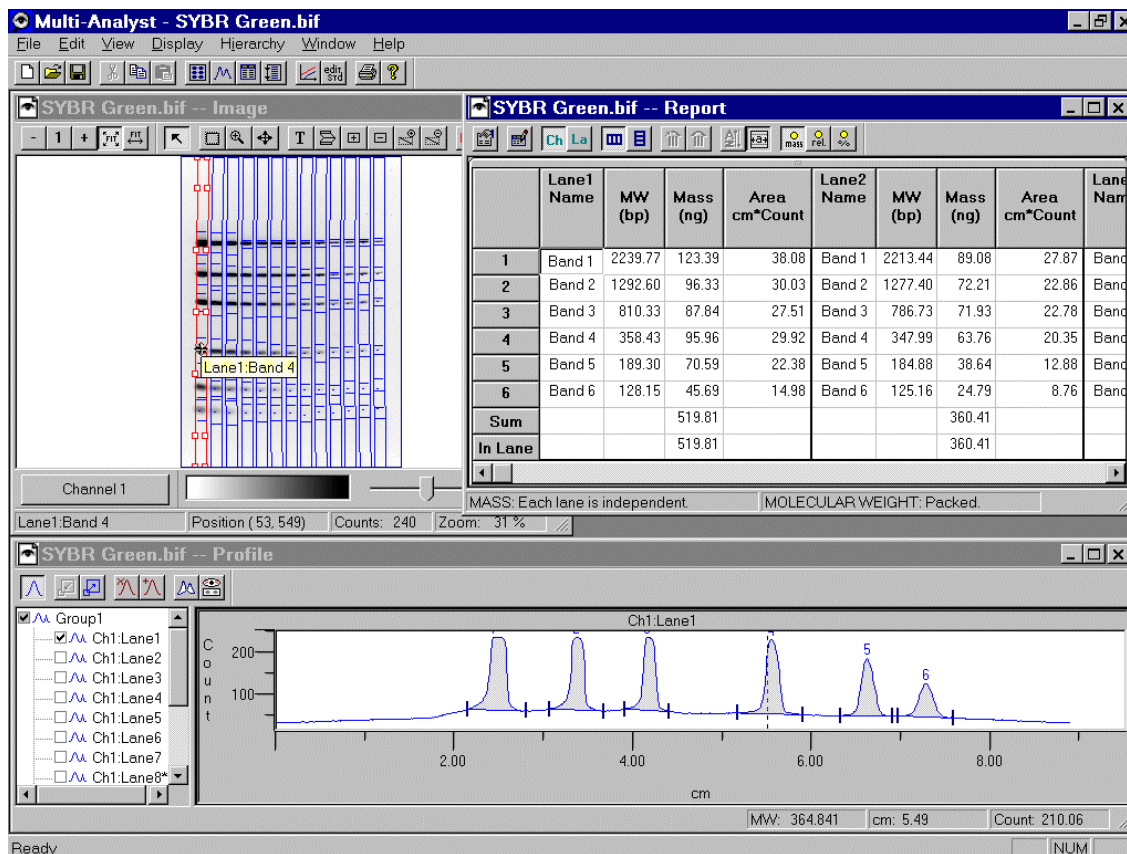
12.1.4.2 Set Background Type

Sets the background type used for calculating volumes to one of the following types

- None
- Local
- Global



13. Tutorial – Basic Operations



Multi-Analyst PC is a sophisticated image capture and data analysis package designed to allow you to quickly and easily obtain your data. This tutorial will take you through the major components of the software. We recommend that you familiarize yourself with the system via the tutorial before starting to analyze your own data.

A typical screen shot, shown above, is an information dense resource. Toolbars supplement the top level pull down menus offering rapid access to the programs features with minimal mouse clicks.

Look at the screen above for the following helpful items.

- Placing and holding the cursor over a button or lane will, after a couple of seconds, pop up a tool tip. Tool tips are brief descriptions of the items the cursor is over.
- At the bottom left of the program screen is a longer description of the button's function.
- The bottom right of the image window contains useful data about the pixel the cursor is currently over. If it is also over a lane or band, it will be described in the bottom left of the image window.
- The image window and the profile window are linked. That means that as you move the cursor over the image, a dotted line tracks the movement over the profile window.
- Molecular weight calculations are dynamic. Moving the cursor over the profile or the image will give molecular weights for the position.

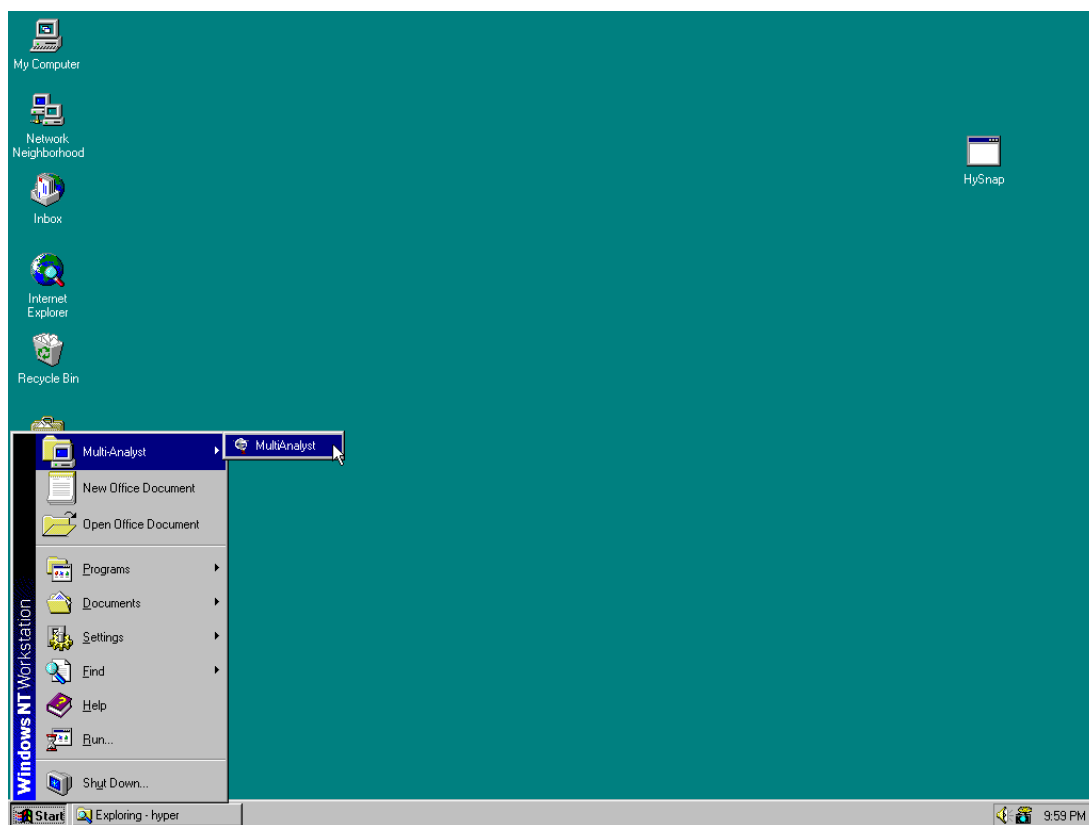
13.1 Operation of the Fluor-S MultiImager

Please see the Fluor-S MultiImager instruction manual for the proper setup and operation of the Fluor-S MultiImager.

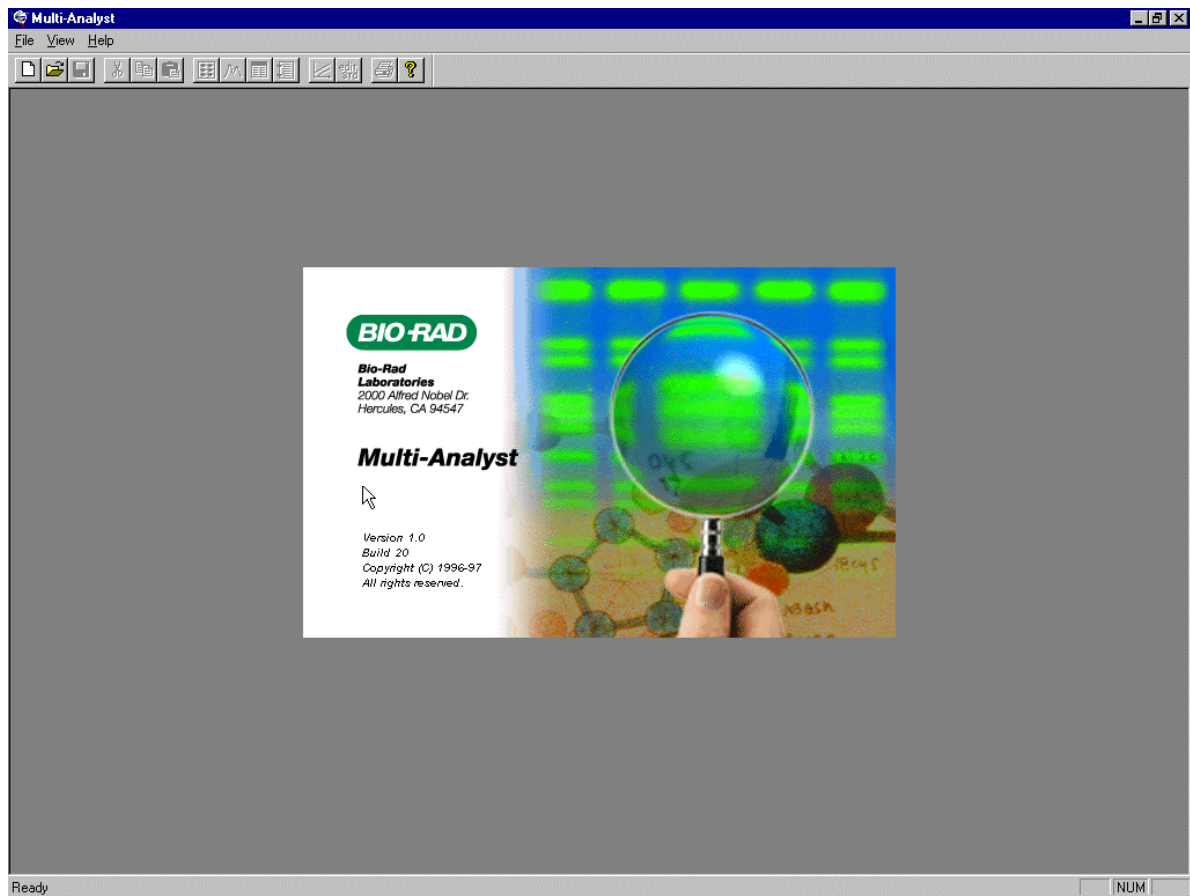
Turn on the power to the instrument, wait for it to initialize, then turn on your computer.

Follow the instructions on the software installation disks to load your software.

13.2 Launching the Program



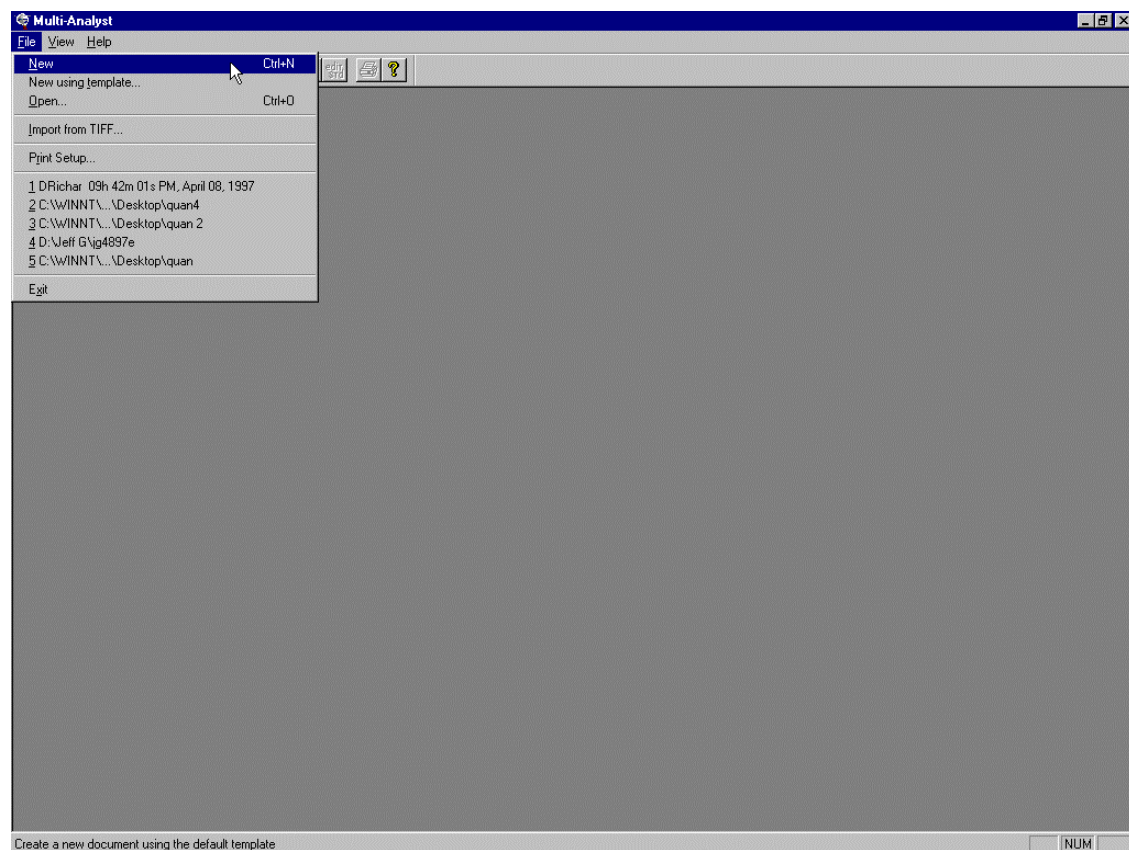
Use your start menu to launch the Multi-Analyst PC application



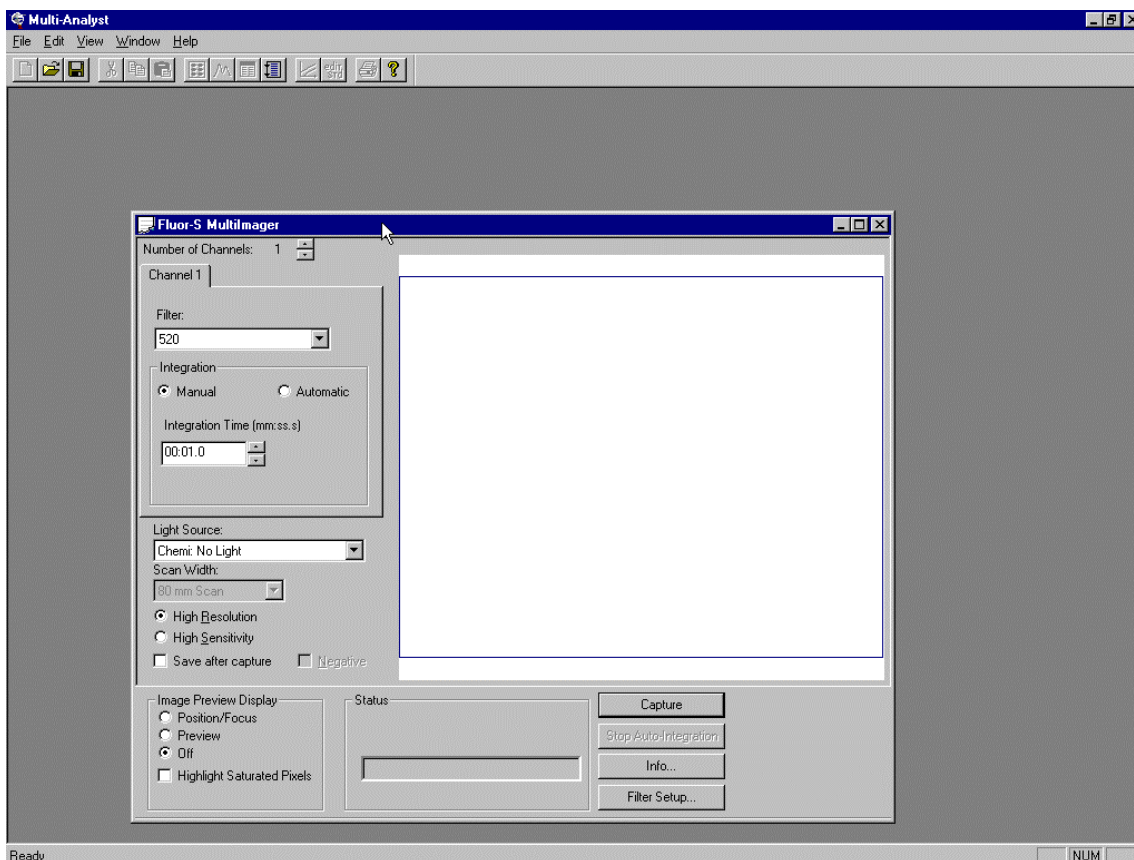
Multi-Analyst may be used to acquire your gel from your instrument or to open an already acquired image.

13.3 Acquiring your Gel Image

13.3.1 New (using default template)



Click on *New* to acquire your gel with the default template.



Default template

For single-color imaging, choose 1 as the number of channels, then select the appropriate filter for your application.

See Table 1, page five of the application guide or page 4-2 of this manual for appropriate filter and light source choices.

For Multi-color imaging, first determine the number of images to be captured by selecting the *Number of Channels*, then select Channel 1 and a filter to use for that channel. For additional images of your gel with different filters and/or light sources, select the additional number of channels and a different filter to use for each channel. Finally, you must decide between manual integration (you choose the exposure time) and automatic integration. For now, let's use the manual integration method.

Multi-Analyst allows you to set up all the experimental conditions that the gel requires ahead of time and can reuse those same conditions over and over. For the purpose of this tutorial, we will use just the *New*.

13.4 Position the sample and focus

- **IMPORTANT:** Change the filter to Clear (chemiluminescence) for maximum light for focusing
- **IMPORTANT:** The camera door must be closed in order for the MultiImager to select the proper filter.
- Select *Position/Focus* in the *Image Preview Display* of the Acquisition Module.
- Position the sample in the center of the platen. It's a good idea to also use a ruler to assist in the focusing of the camera lens.
- Now open the camera door of the imager and focus the lens and set the appropriate aperture and zoom setting
- Set *Image Preview Display* to *Off* when finished
- You may check your results without capturing your image by using the *Preview* check box.

NOTE: The MultiImager will use the filter you see displayed in the acquisition window. If the filter allows enough light through to focus, it is best to use that one for focusing. Do not use an empty filter position to focus however, since that will change the focal length and thus cause your acquired image to be out of focus.

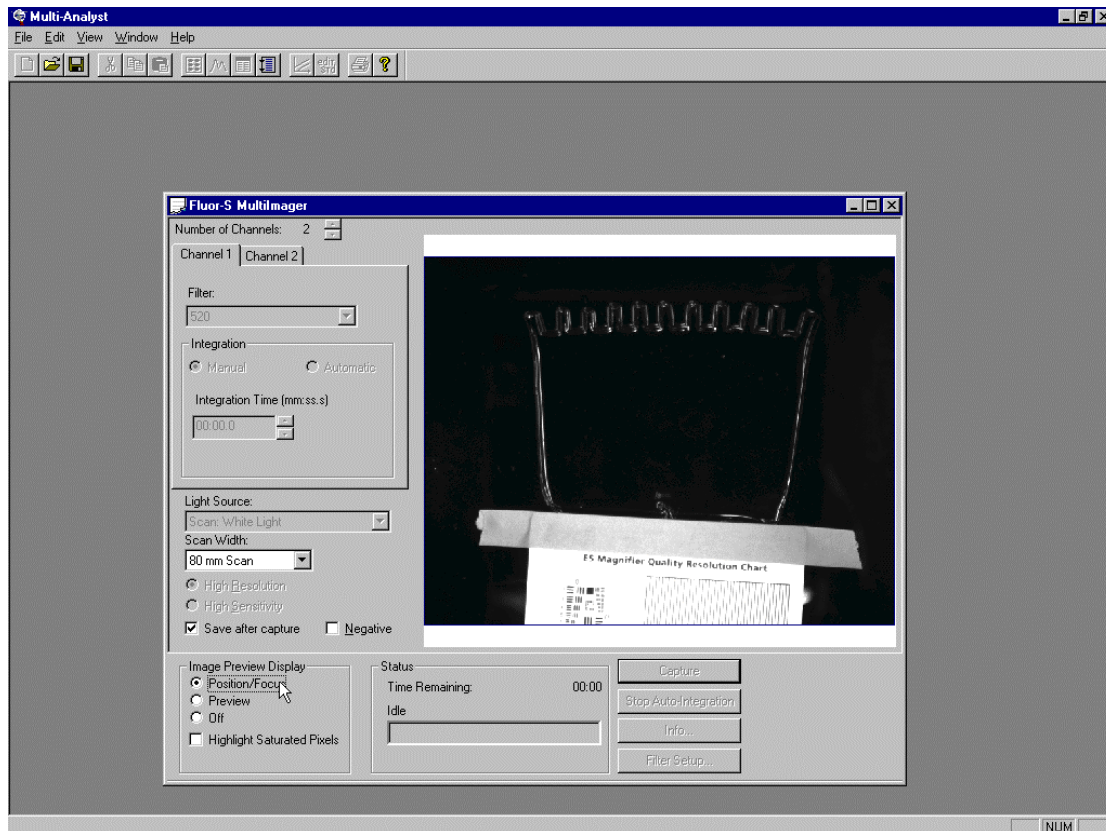
13.4.1 Adjust the aperture

For most low light applications, such as fluorescence and chemiluminescence, the aperture of the lens should be fully open, i.e. with the lowest number f-stop. This allows the maximal amount of light to the CCD chip producing the best image quality. Integration times are in the several seconds to several minutes range with the lens fully open for these low light applications.

For white light applications, significantly more light is presented to the CCD. Therefore, one limits the exposure by scanning with the shortest scan time, 1 second when using scanning illumination, as well as by decreasing the aperture of the lens (higher f-stop numbers). With white light epi-illumination, integration times can be as short as 10 milliseconds. Close down the aperture of the lens for further control of light level.

For accurate quantitation, the signal from the region of interest should not saturate the CCD. In order to easily determine if an integration time is resulting in saturation of the CCD, the *Highlight Saturated Pixels* button in the *Image Preview Display* is useful. Turn on the *Highlight Saturated Pixels* and set the integration time and aperture controls to keep the region of interest just below saturation.

13.4.2 Zoom



White light epi-illumination is used to position the sample in the center of the imaging area and to focus the image. A ruler can be placed next to the sample to help focus the image. The Fluor-S MultiImager sample platen can accommodate samples as large as sequencing gels of 35 x 43 cm. With the standard 20-40 mm zoom lens, samples from 15 x 12.5 cm to 30 x 25 cm can be imaged. For imaging samples smaller than 15 x 12.5 cm, or for imaging small regions of larger samples at high resolution, other F-mount lenses of longer focal length can be used. When using scanning illumination, the maximum size limit for the imaging area is 30 x 25 cm, the size of the transparent region of the platen.

FOR MAXIMUM RESOLUTION, ZOOM IN (MAKE YOUR GEL IMAGE BIGGER) AS FAR AS POSSIBLE ON YOUR GEL. REMEMBER, RESOLUTION IS DEPENDENT ON YOUR ZOOM FACTOR, THE HIGHER THE ZOOM, THE BETTER THE RESOLUTION.

Close the camera door when you are satisfied with the zoom, aperture and focus settings.

13.5 Select instrument settings

13.5.1 Light Source

In the *Light Source* menu, use the pull downs to select your light source for each channel to be used. The recommended illumination source for many applications is shown in Table 1 of the Application Guide, Bulletin No. 2132.. When using *Scan: White Light*, *Scan: UV(302) #1*, *Scan: UV(302)#2*, select a *Scan Width*.

When scanning samples less than 30 cm across, the scanning area should be restricted to one of the three smaller subsets of the 30 cm scanning width for more efficient and rapid data collection in the scanning illumination mode.

13.5.2 High Resolution or High Sensitivity.

In *High Resolution* mode, images are ~3.0 MB in size per channel. In *High Sensitivity* mode (where the data from four pixels are combined) the image files are ~750KB in size per channel. *High Resolution* is the default scan mode. *High Sensitivity* mode reduces your resolution by a factor of two but increases the instruments sensitivity to weak samples.

13.5.3 Select Save after Capture

If you want the program to automatically save your images after acquiring them, check this box. The default file name given will be your log-in file name followed by the date and the time. You are given the option to change this if you wish.

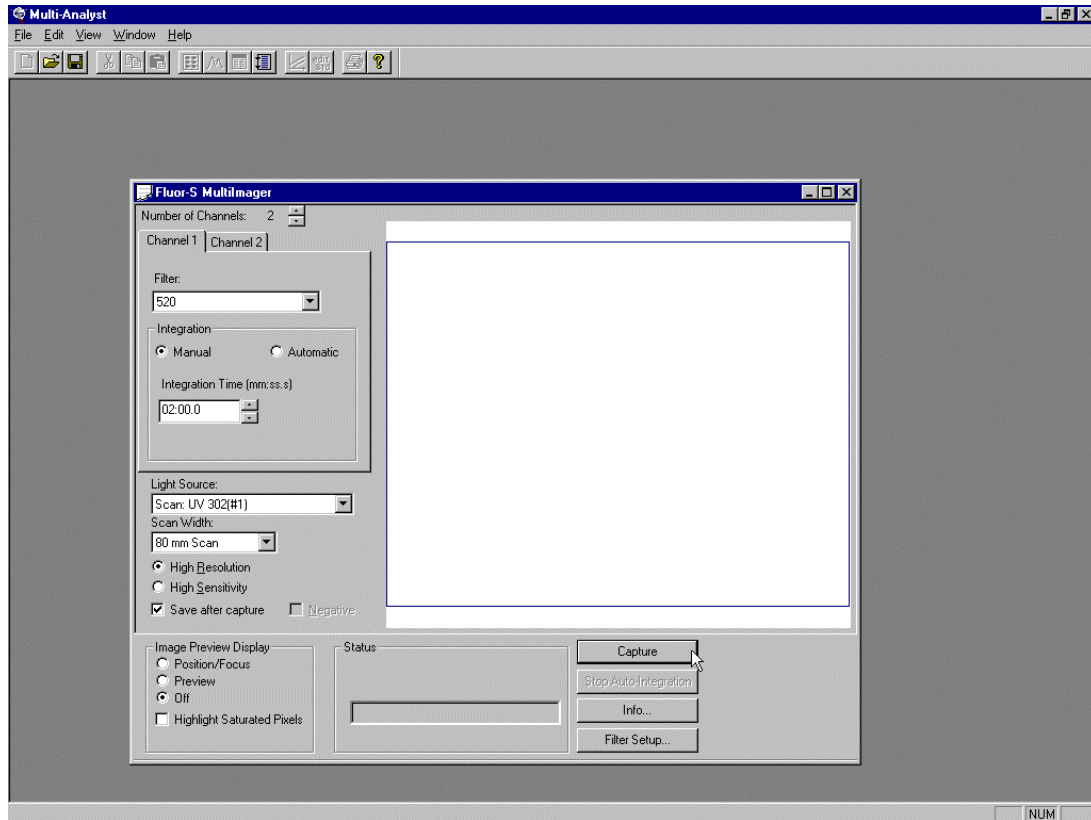
13.5.4 Negative

If you wish to invert the data set at the time of capture, check this box. A common application of this is imaging an etbr Polaroid photograph that has white bands on a black background. In order to get peaks instead of valleys at the band positions, the pixel values must be remapped in reverse order.

13.5.5 Set integration time

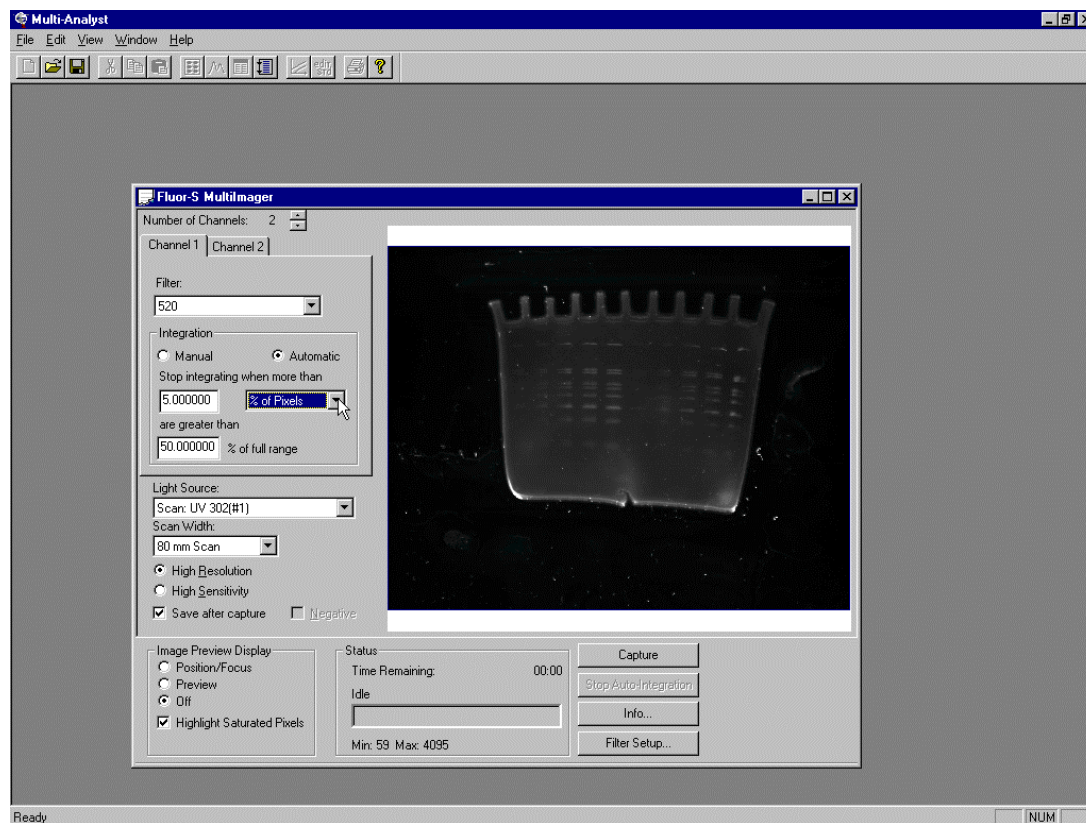
In the *Integration* window, select either *Manual* or *Automatic* integration.

13.5.5.1 Manual integration



For Manual integration, set a value in the *Integration Time (mm:ss.s)* that is appropriate for your stain and light source.

13.5.5.2 Automatic Integration

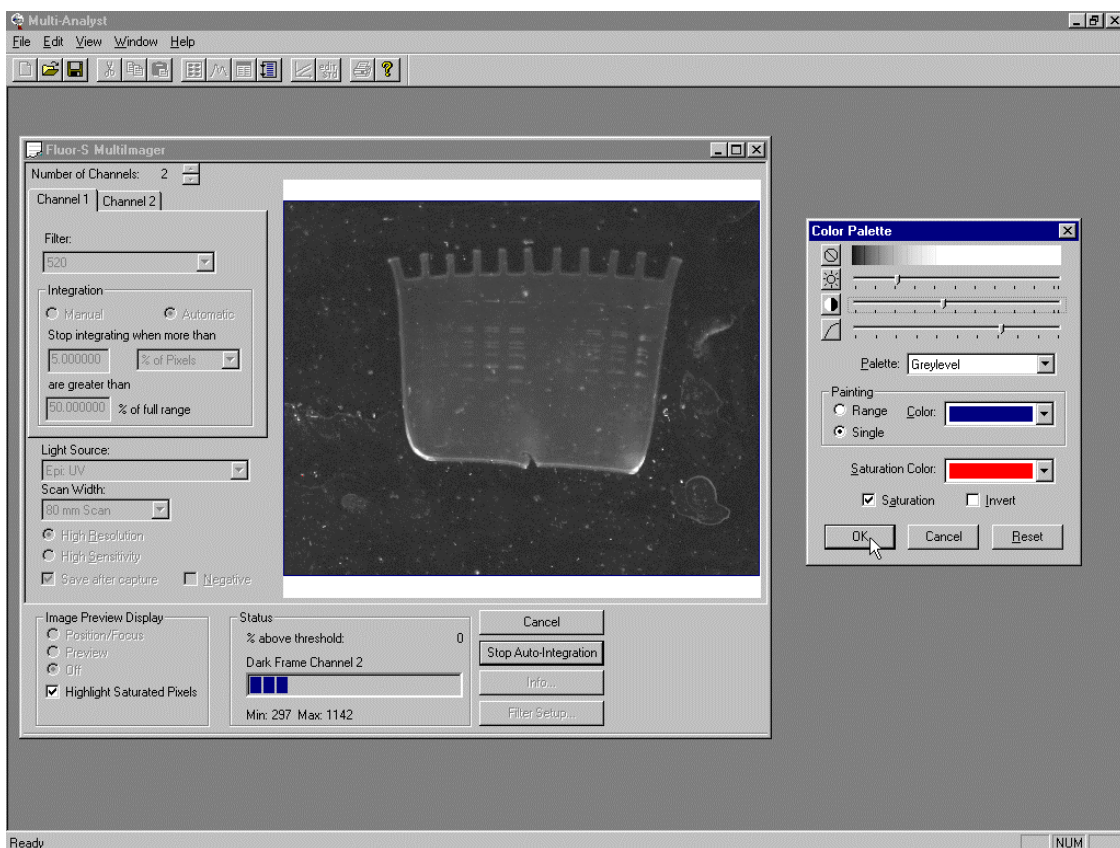


In Automatic integration mode, the instrument collects an image, checks the pixel values against the parameters you set in the acquisition window. If the conditions have not yet been reached, the system doubles the integration time, sums the result with the prior acquisition and then repeats its calculations. Image collection stops when the summed images reach the set level of signal below saturation. For multi-channel imaging, each channel can be previewed independently to determining the optimal integration time for each channel.

13.6 Acquire the Image

You may acquire the image of your gel by clicking on the *Capture* button on the Fluor-S MultiImager control module. For this tutorial, we will use the *Manual* mode.

During manual acquisition mode, the time remaining for image capture is presented in the Status window. Acquire will automatically capture all channels in order and subtract the dark current. Remember that the actual acquisition time is twice the scan time. This is because the dark current scan must be performed for the same amount of time as the acquisition scan in order to get the optimal dark current reading from the camera.



During your image acquisition, you can open the *Color Palette* window (found under the view menu) and modify the image to better see your bands during the scan.

You may also, at any time stop the auto integration and save the image as it is currently displayed. If you are doing multiple channel imaging, the software will continue on to the next channel.

13.7 Save your Gel

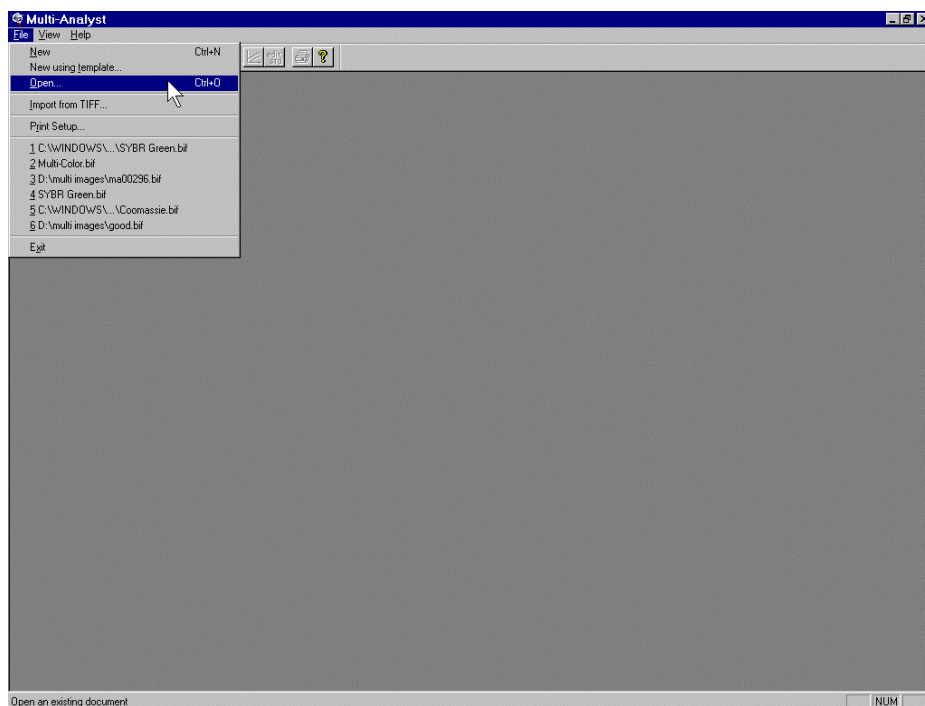
If you check the *Save after acquisition*, you will be presented with a menu asking you to name your gel. By default, Multi-Analyst will attempt to name your gel with a default naming convention that is useful for tracking your gel. There is no need to enter a file extension (.bif) since Multi-Analyst will do that for you. You may, of course, choose to enter in your own label for the gel. There are no limitations on your gel name beyond those found in normal Windows NT naming schemes.

Once you have captured an image, Multi-Analyst will automatically open it for immediate work. You can choose to begin working with your gel or capture more gels using the Acquisition window.

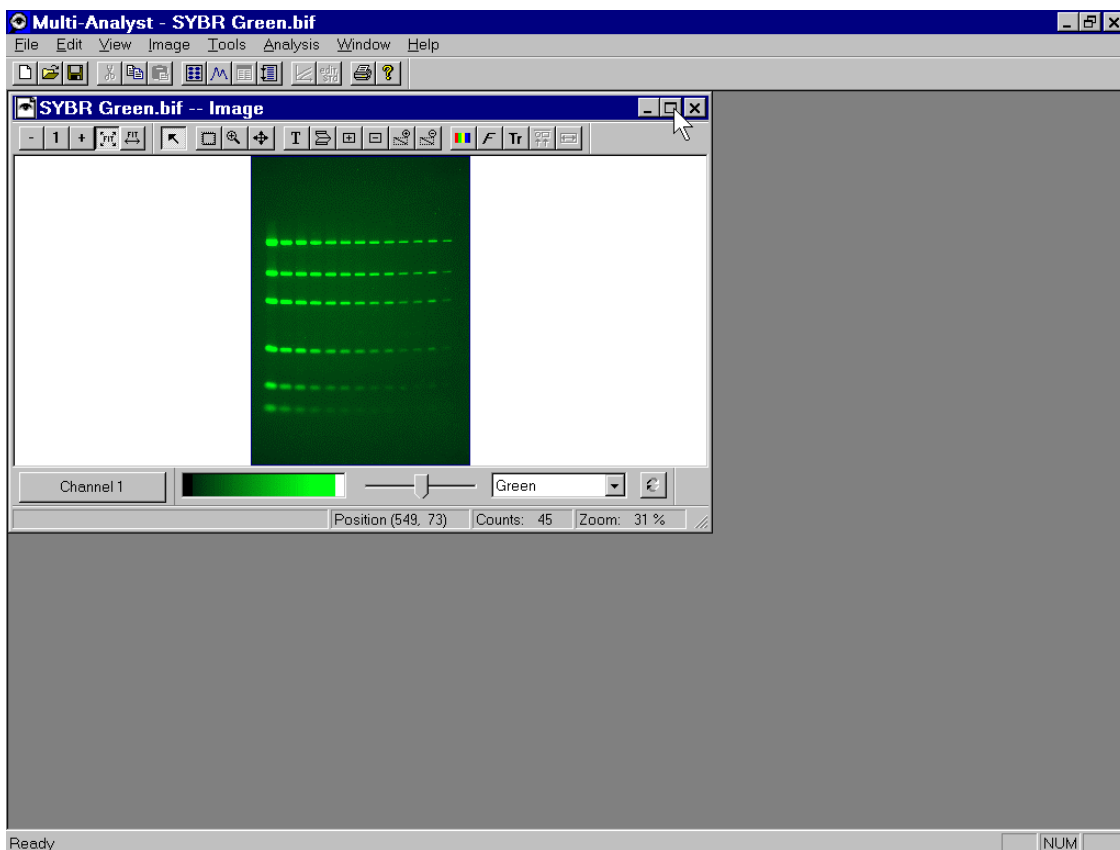
When you are finished using the Fluor-S Multi-Imager, it is a good ideal to close down the *Acquisition* window to reduce the clutter on your screen.

You have now successfully acquired an image using the Fluor-S MultiImager. You may now close down the acquisition window and exit the program or continue on with the tutorial.

13.8 Open your gel

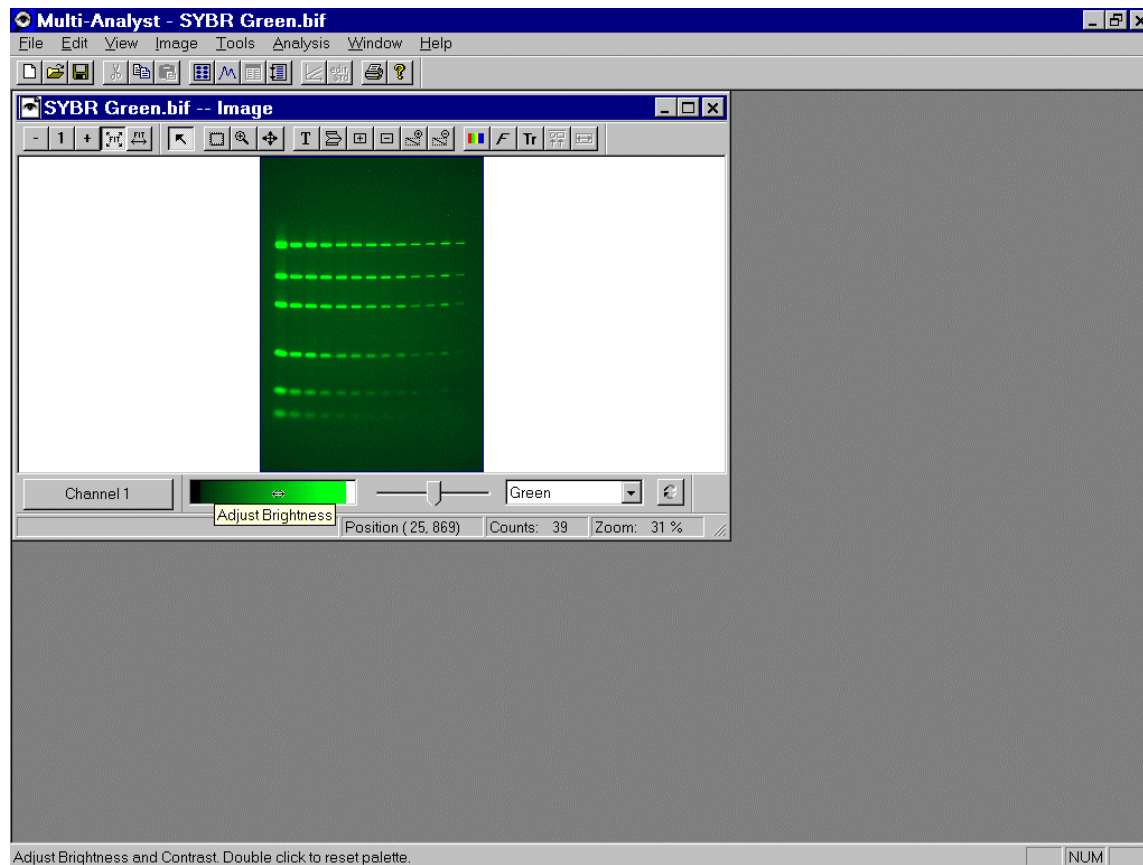


Go to *File, Open* or you can use the recently opened list to choose your file. For this tutorial, please select the SYBR Green.tif image.



You may want to enlarge your window by clicking on the window icon in the upper right corner.

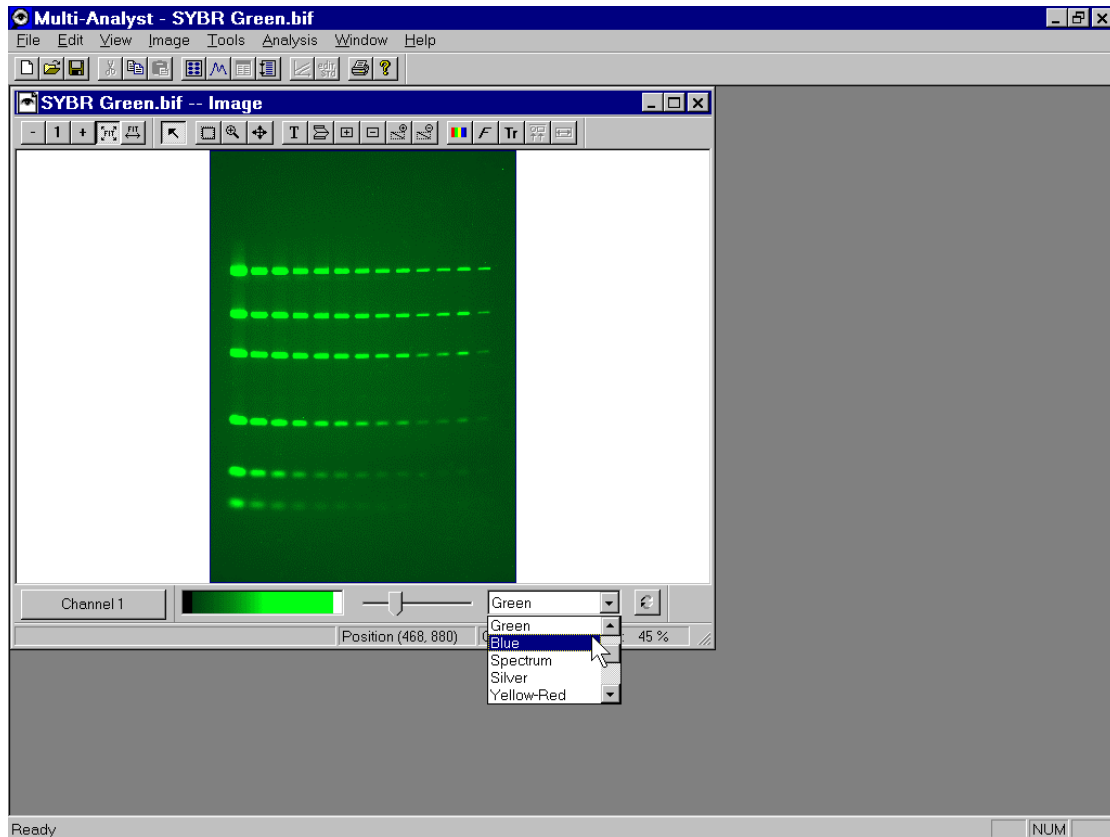
Another easy way to enlarge your image is by clicking and dragging the image window. Multi-Analyst will automatically adjust the size dimensions of your image to maintain the proper aspect ratio.



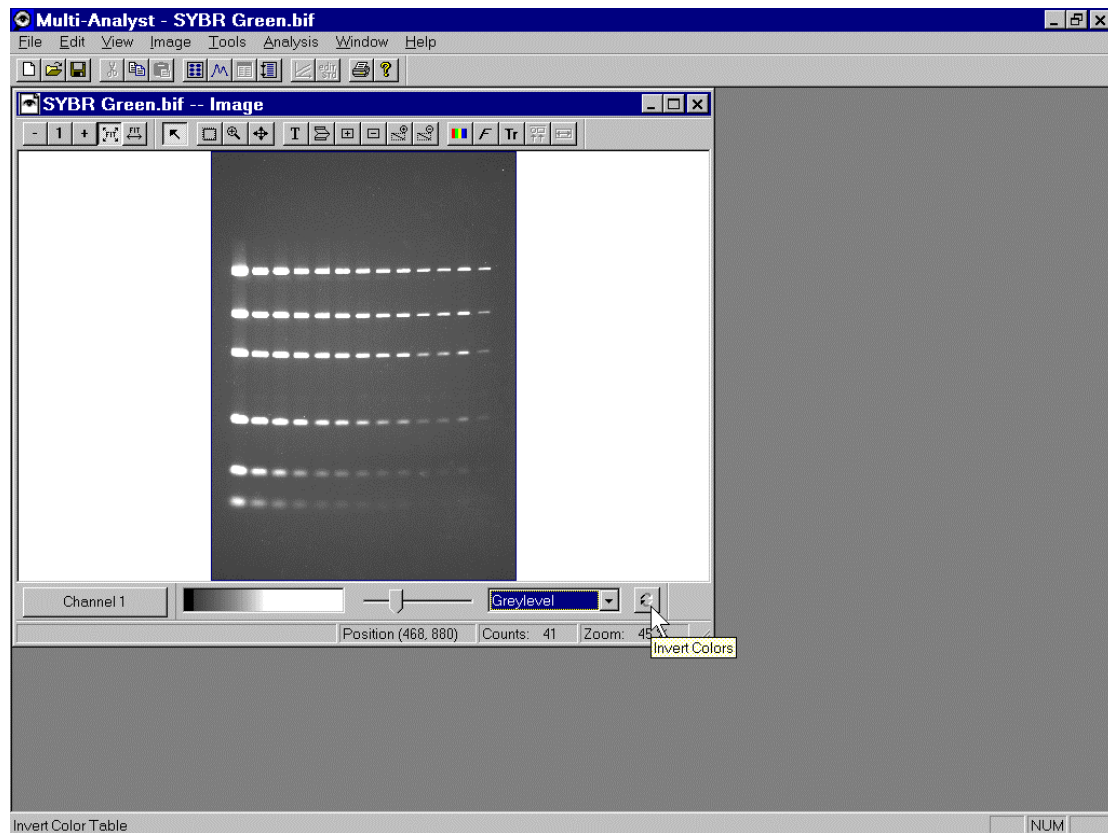
At the bottom of every image window is your color palette. Here you can change your brightness and contrast by simply dragging your cursor from the left or right. In addition, double clicking here will reset your pallet to it's default settings.

13.9 Optimizing your Gel Image

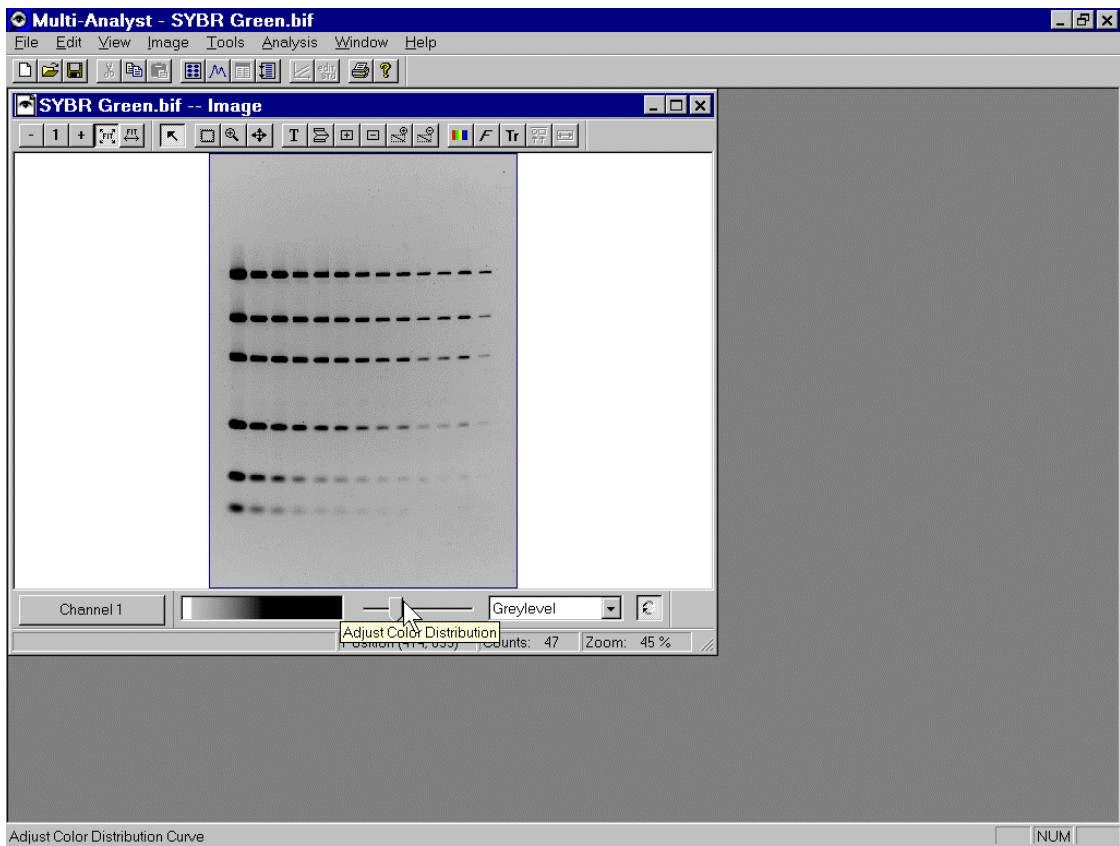
13.9.1 Image Enhancement Tools



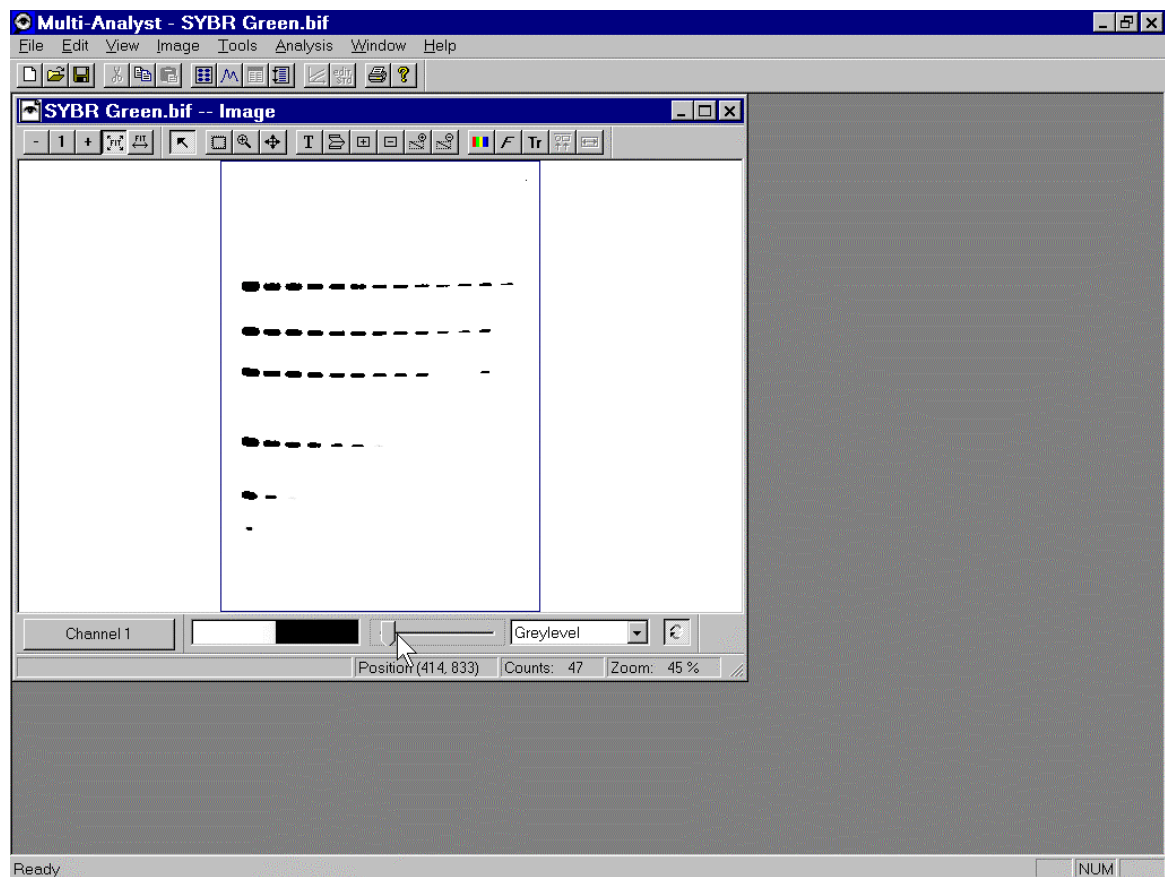
You may change the color of your image to one of the preset color palettes that most closely represents what you would see looking at your gel



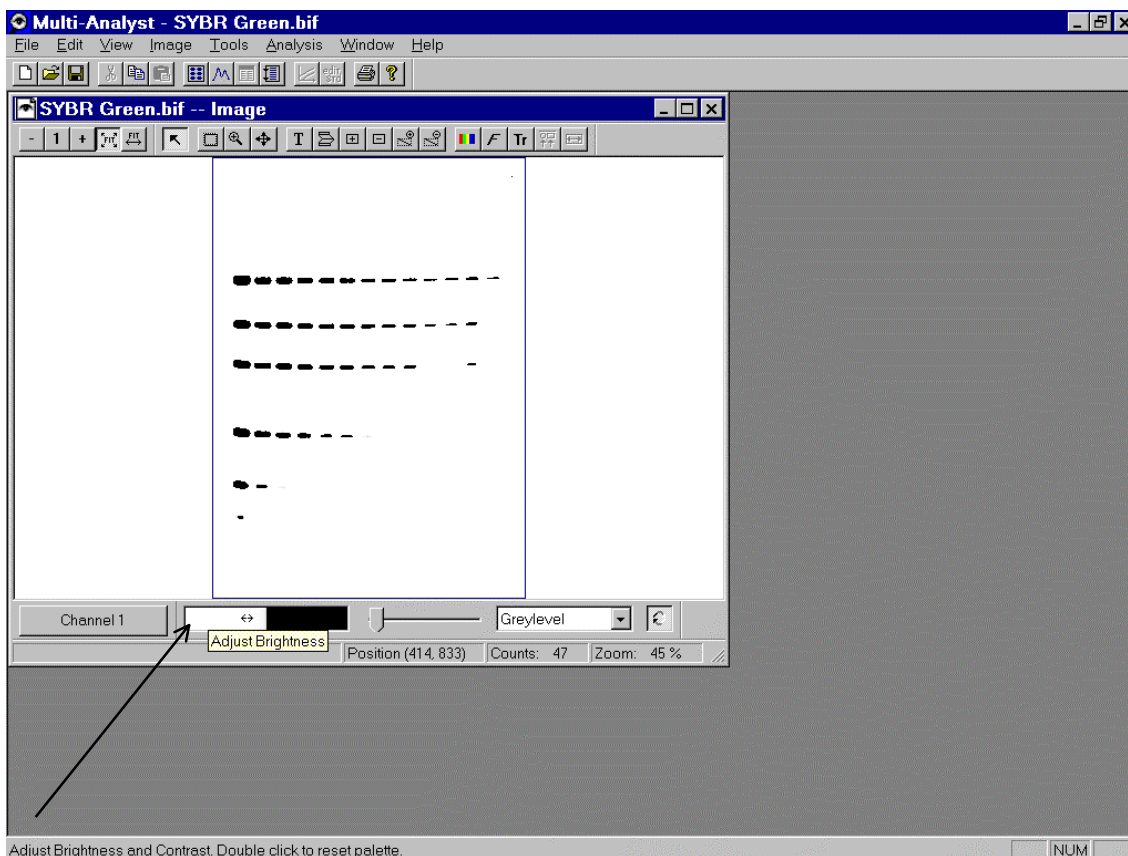
You can also invert your colors. You may prefer to choose black bands on a white background.



Use the slider bar to quickly visualize your bands

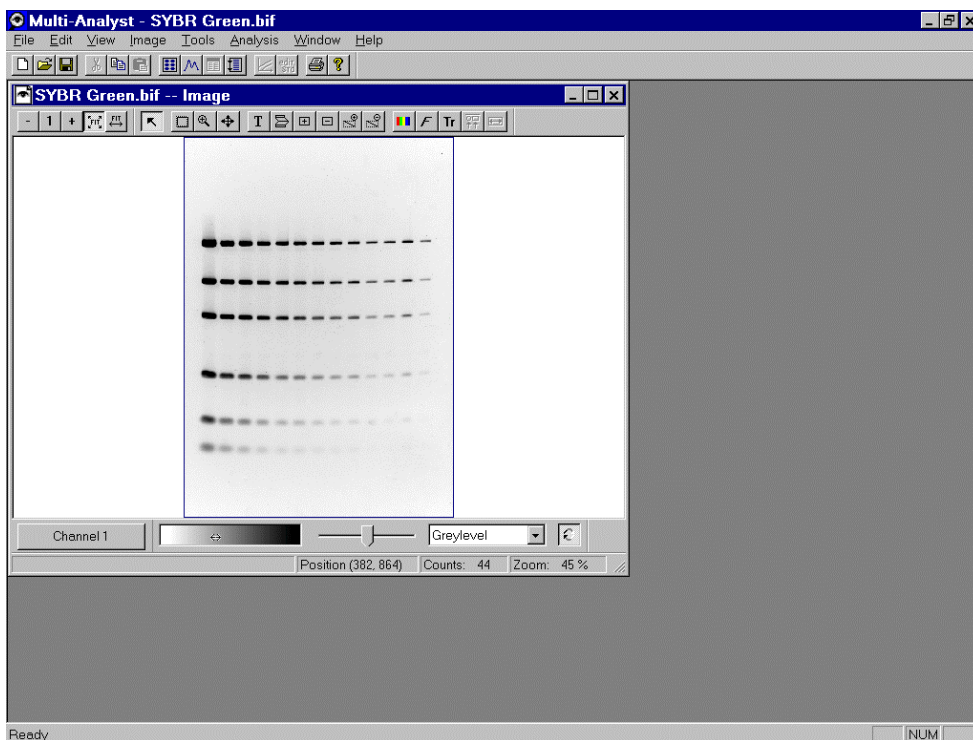


Don't worry about what you do to your view image. Multi-Analyst only uses the raw data to perform the numerical calculations. No matter how much you transform the appearance of your gel on the computer screen and your print outs, you will receive the same quantitative results each time.



Adjust Brightness and Contrast. Double click to reset palette.

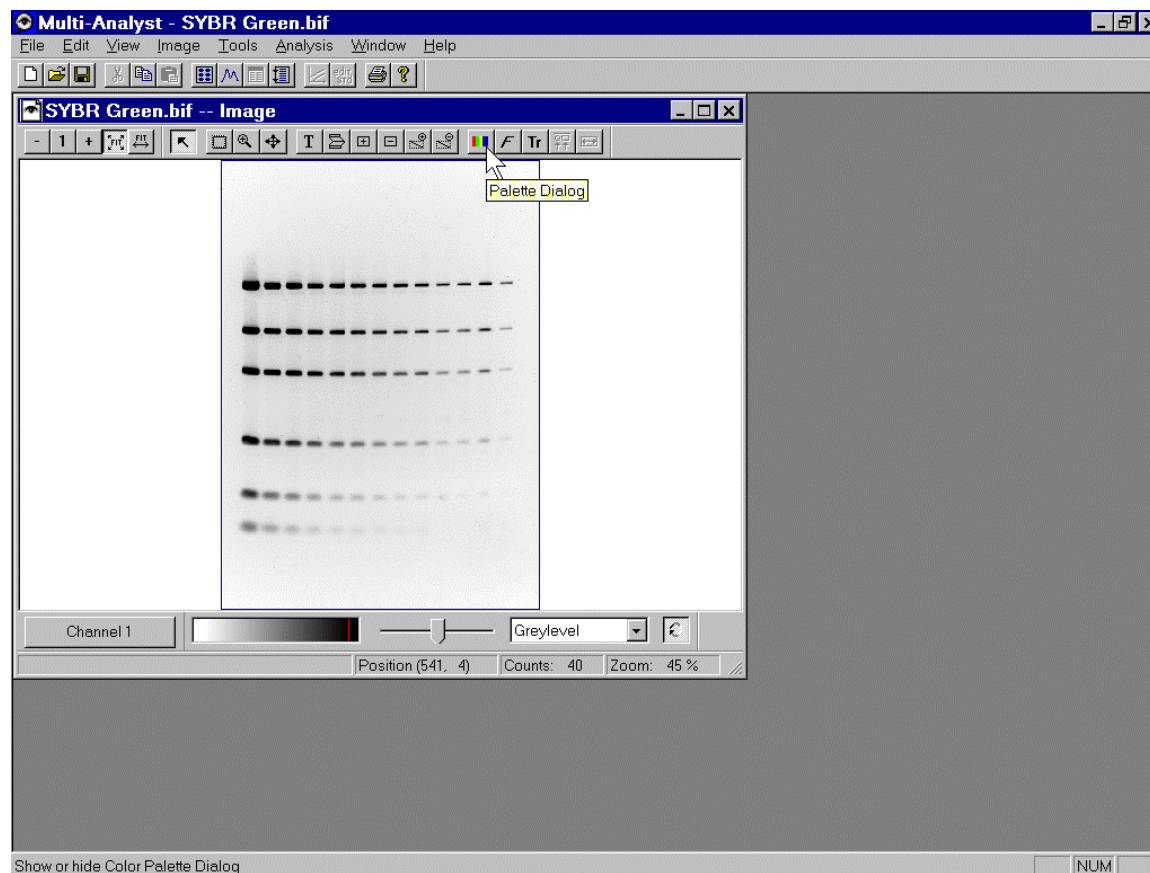
It's easy to reset your gel image to its original starting point. Just double click on color palette bar and Multi-Analyst resets the image back to the default settings.



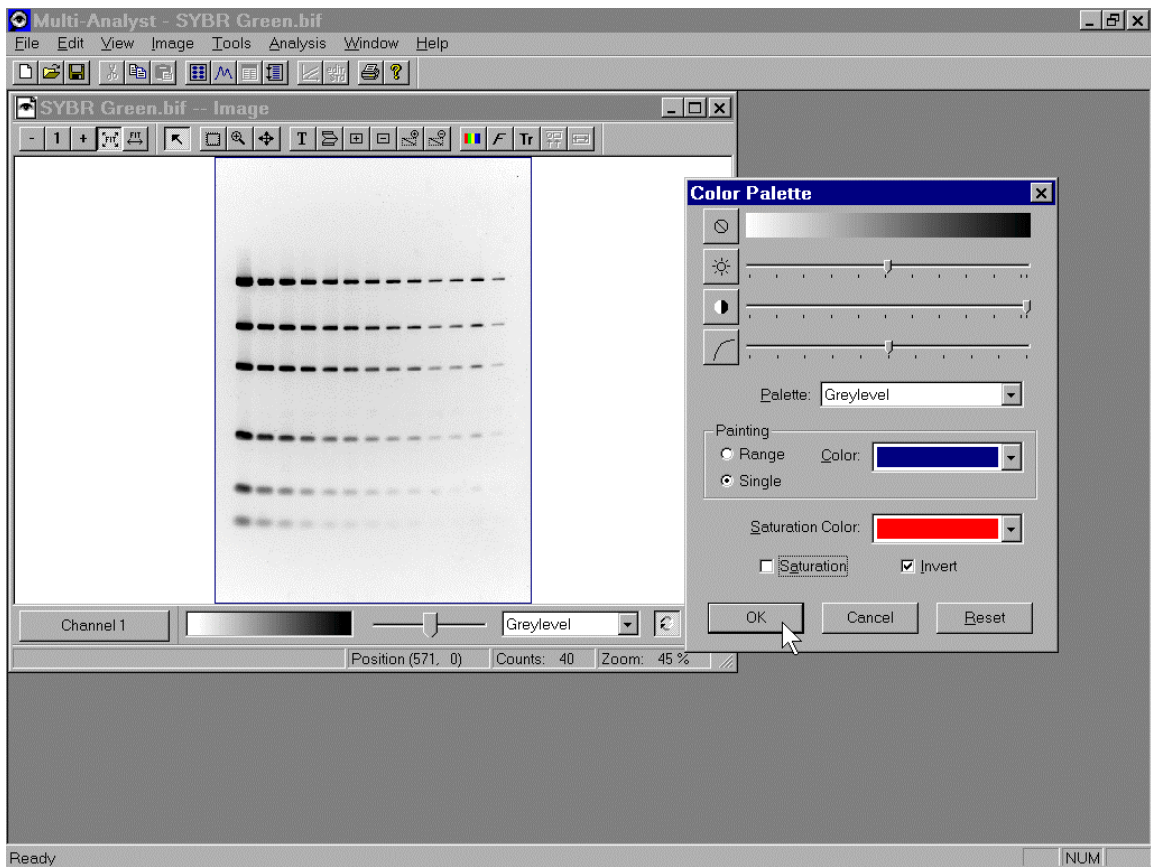
Ready

Image restored to original settings by double clicking on color palette bar.

13.9.2 Palette Dialog

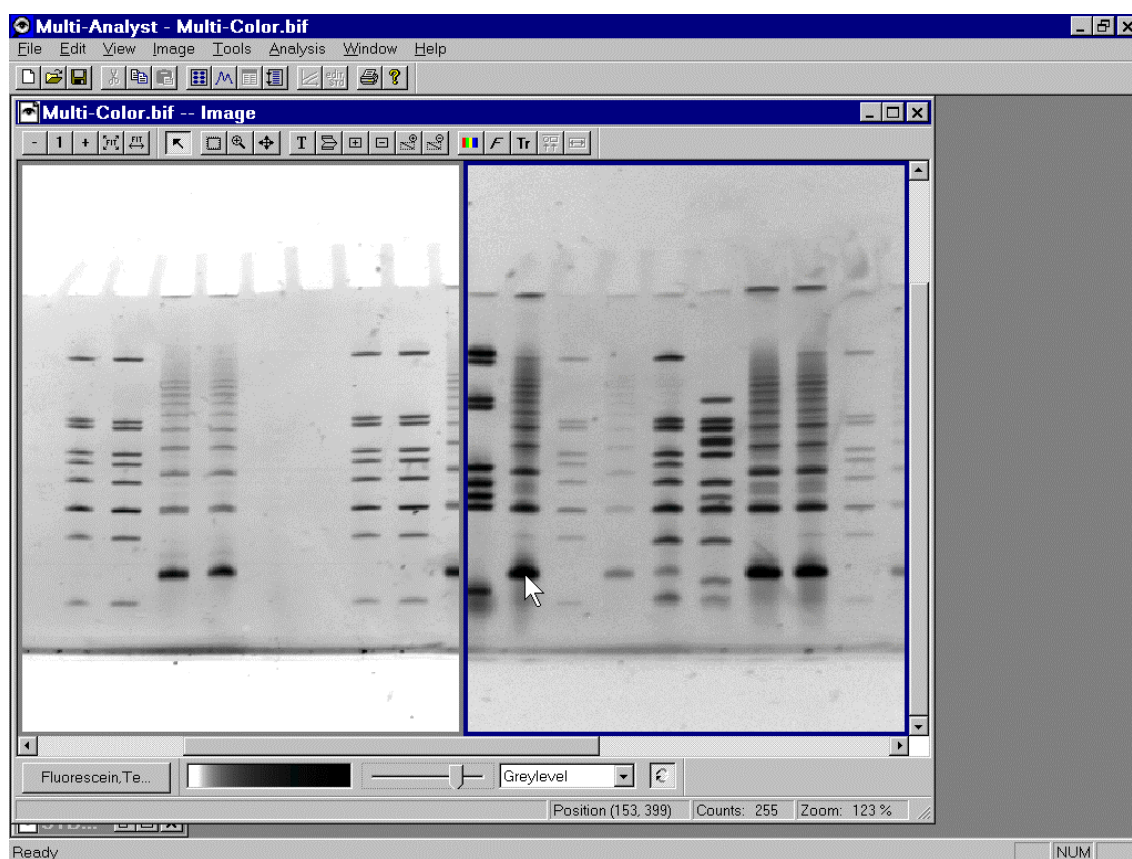


For more control over the image than what is presented at the bottom of the image window, select the *Palette Dialog* tool to access more palette commands to optimize the view image.

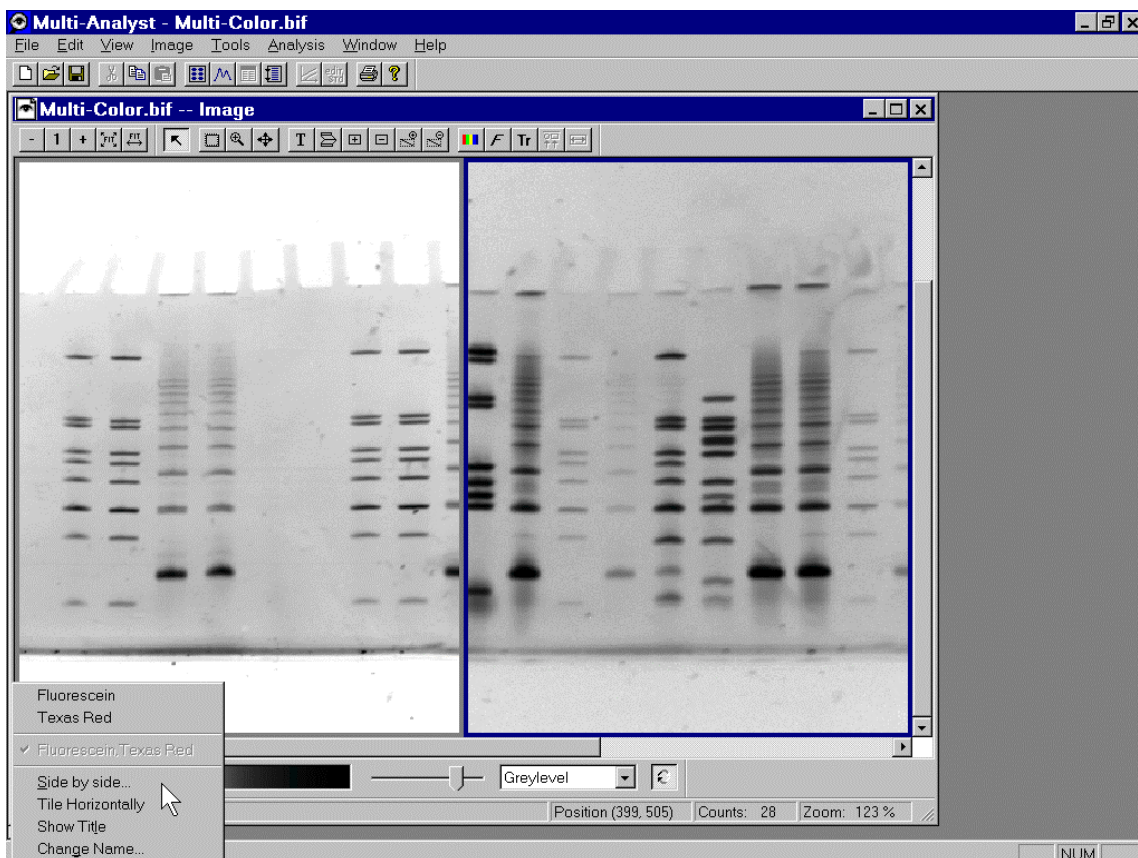


The Color palette window gives you many more options beyond the simple controls located in the slider bar.

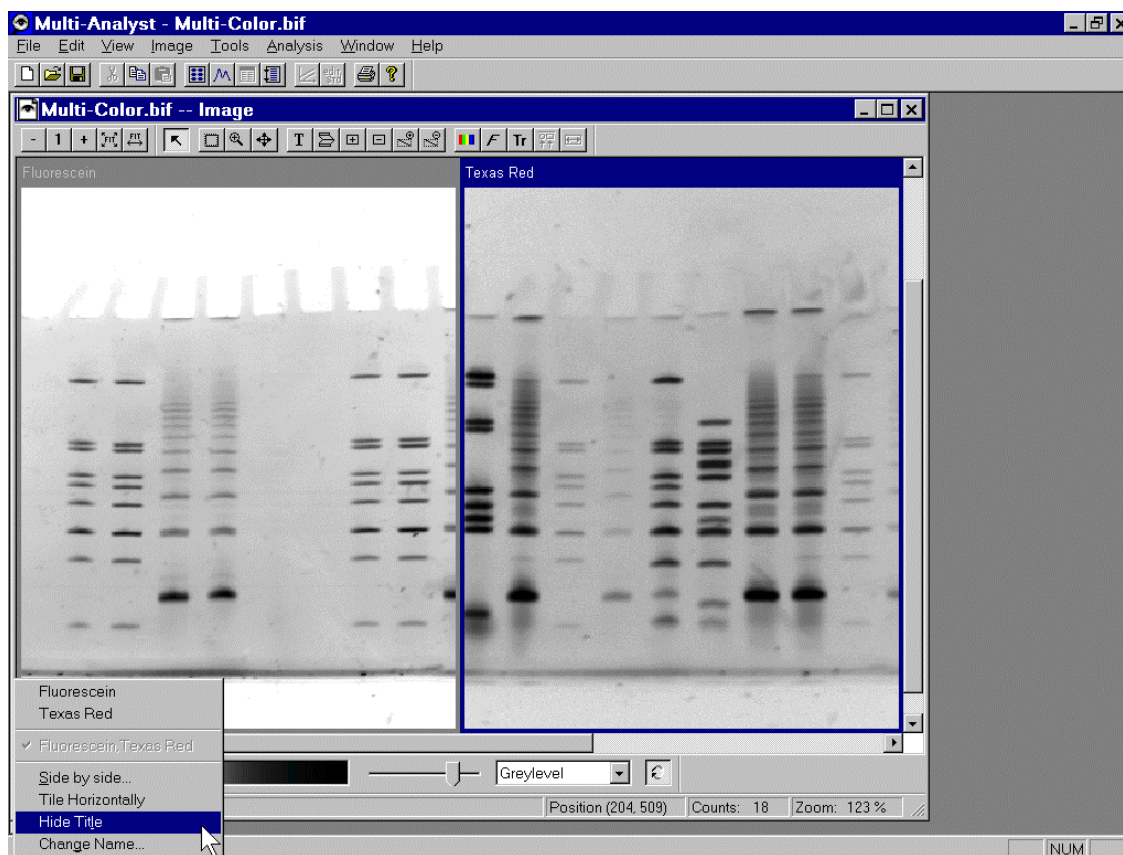
13.9.3 Multi-Channel Gels



Multi Channel gels are image linked. That means when you move the sliders to see different parts of the gel, the other channel moves in tandem.



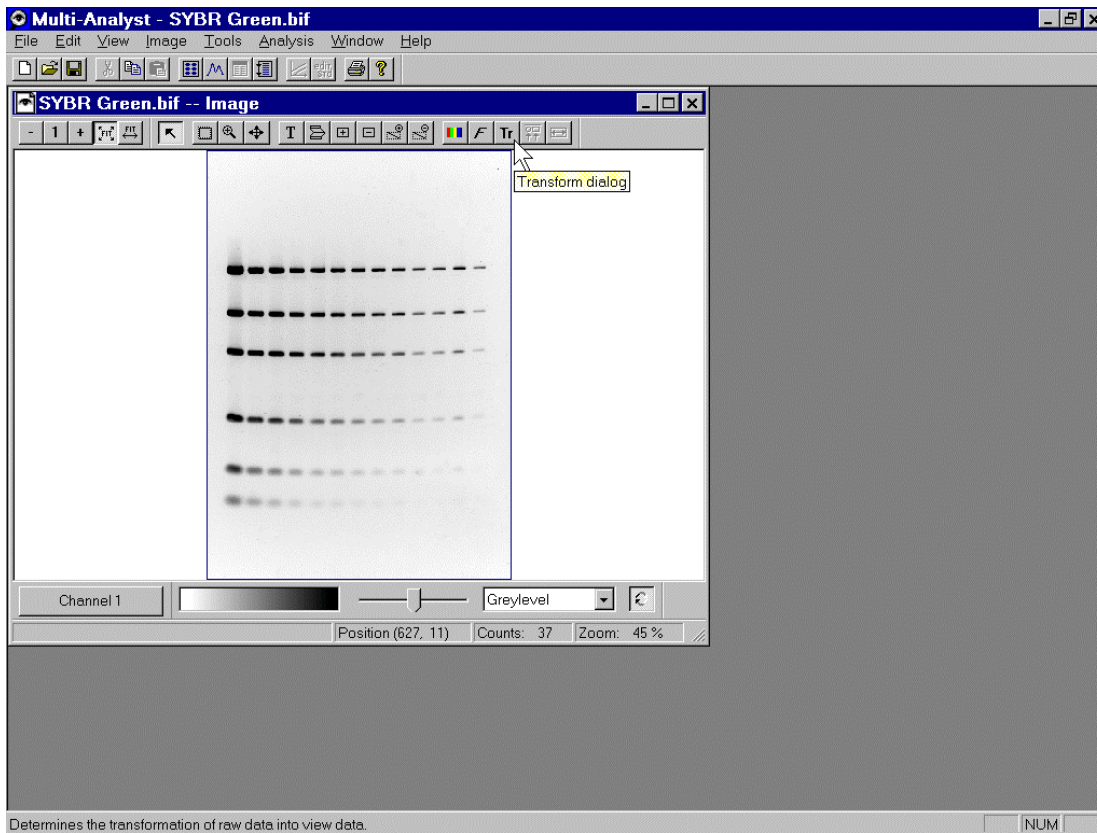
Clicking on the Channel button in the bottom left of your screen allows you to change how Multi-Analyst stacks the gel images and give names to each channel.



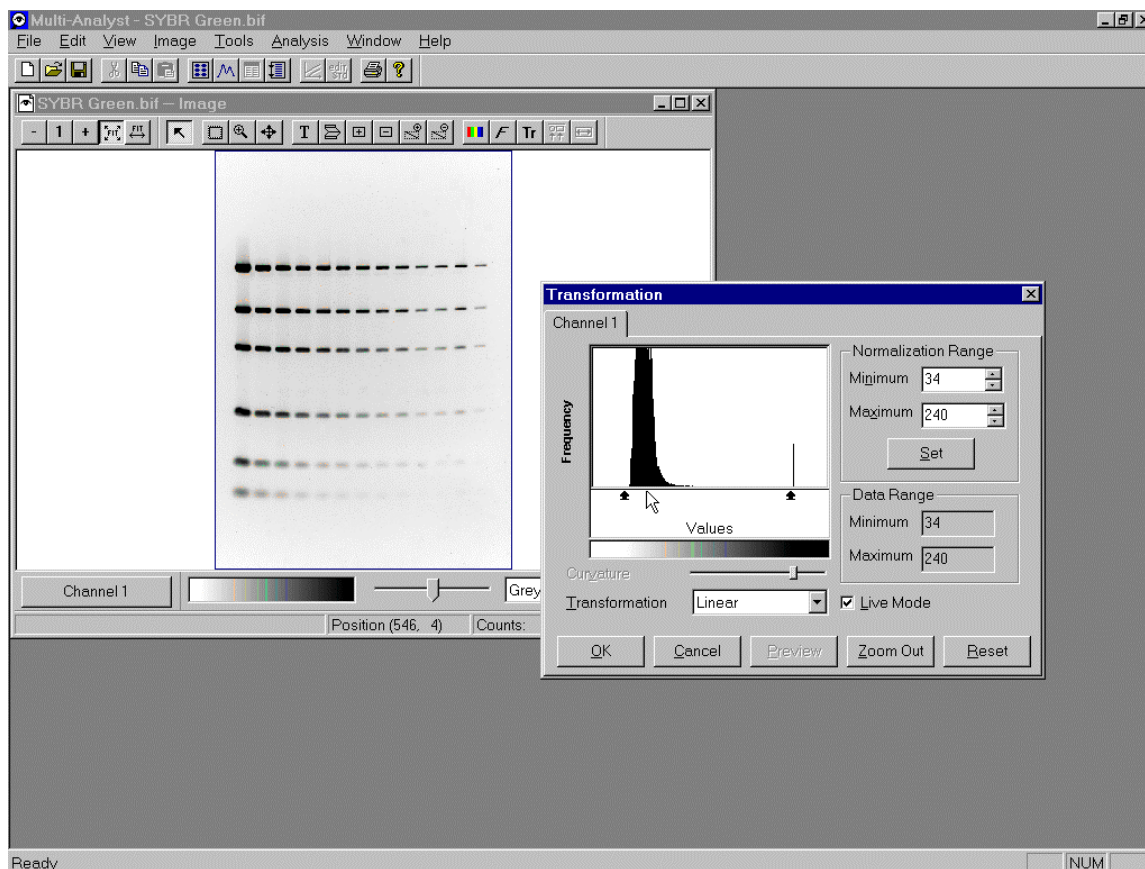
You can also show the Channel titles for easy stain identification.

One final note for Multi-Channel Images. The color palette and transform (see below) settings for each channel are independent of each other. This means that you are able to optimize each image separately and visualize the results simultaneously.

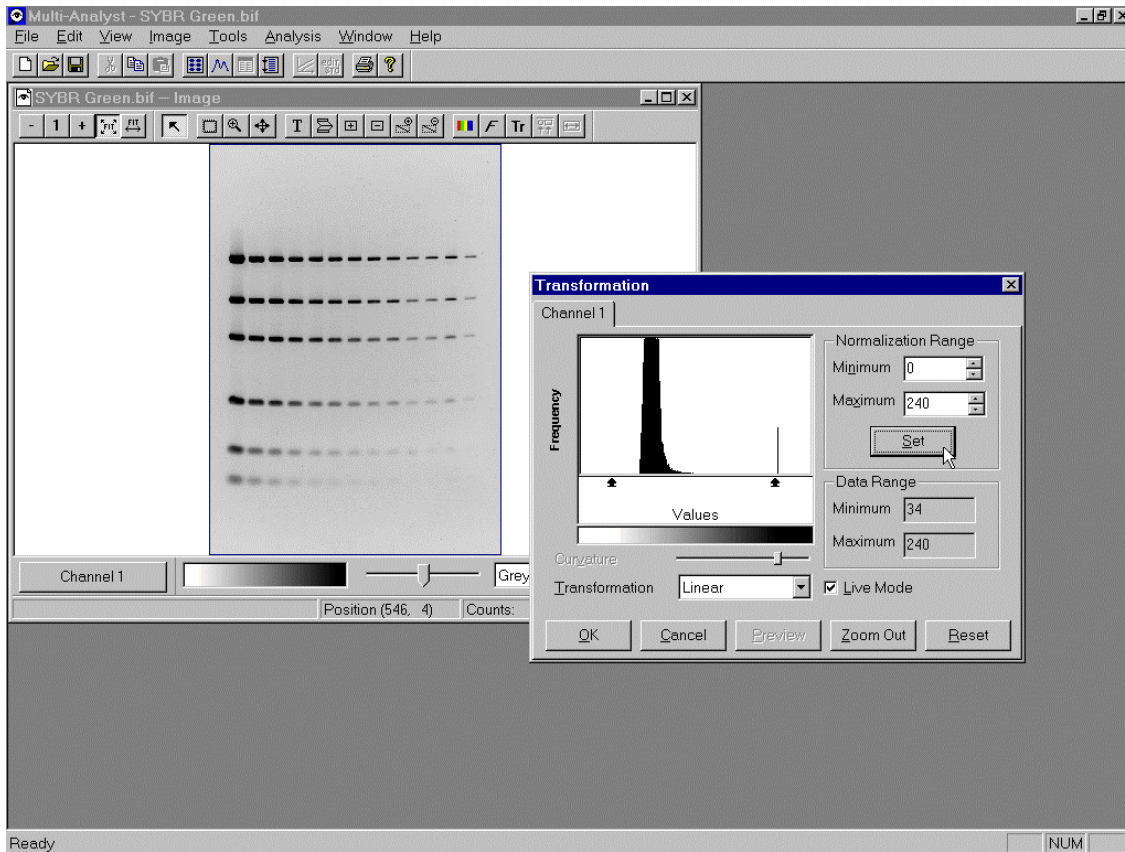
13.9.4 The Transformation window



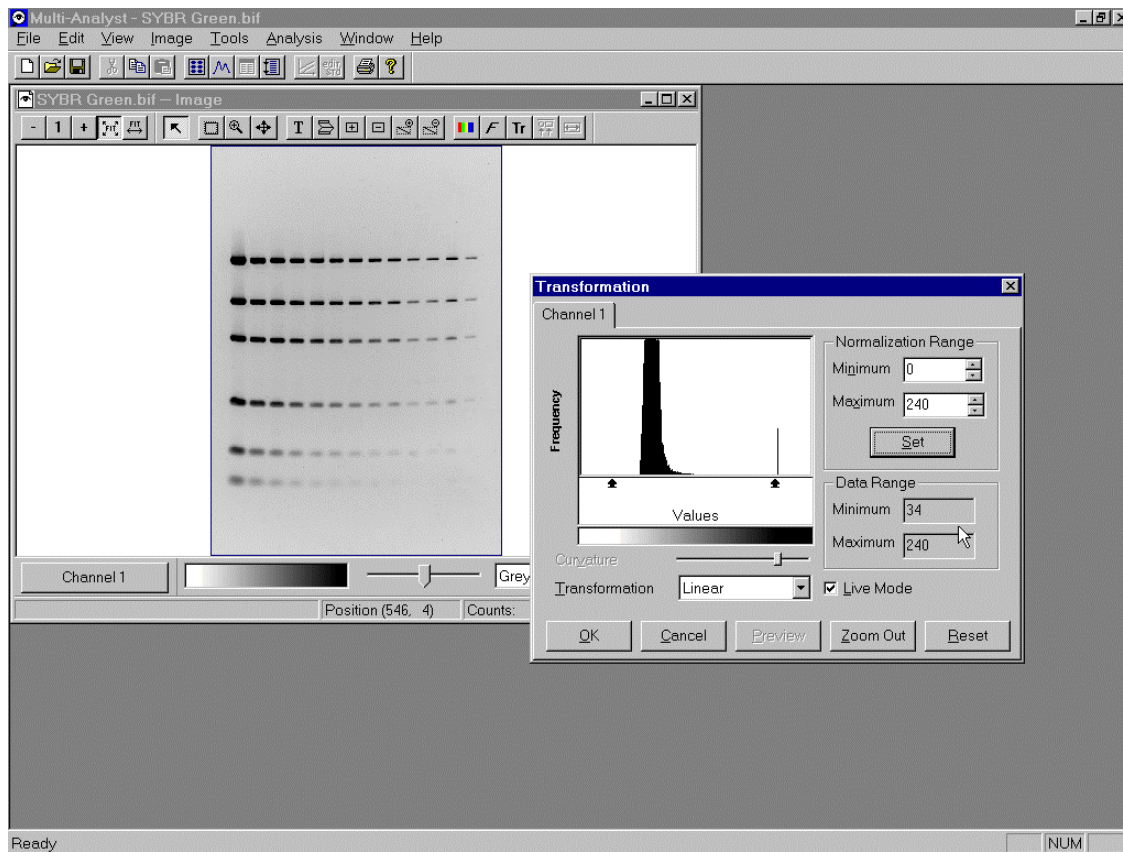
Many users find that the best way to quickly optimize their gel image is to use the *Transform Dialog*.



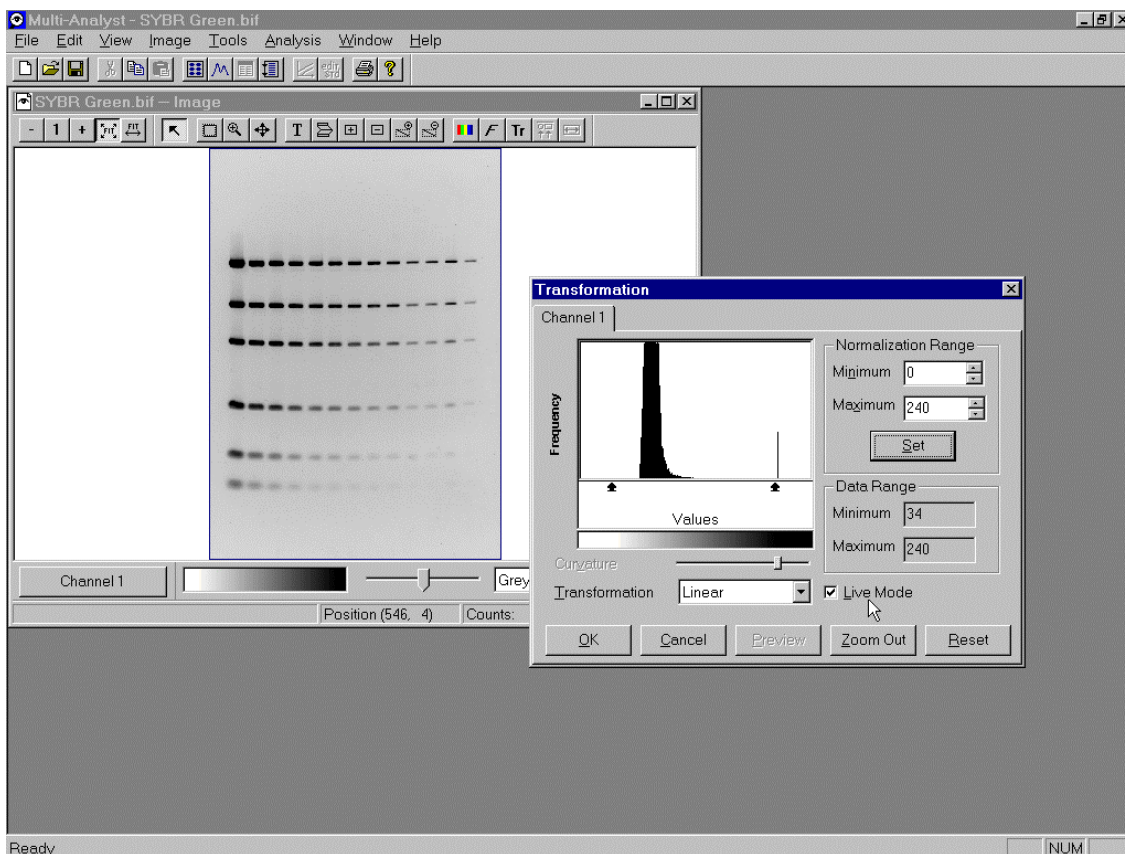
The *Transformation* window gives you a lot of useful data about their gel image in an easy to understand format. The histogram chart is a plot of the pixels values and the total quantity of pixels found at each value. In the above example image, we see a large peak of pixels in the gray range. This is to be expected because in many gels, background pixels account for the majority of the image.



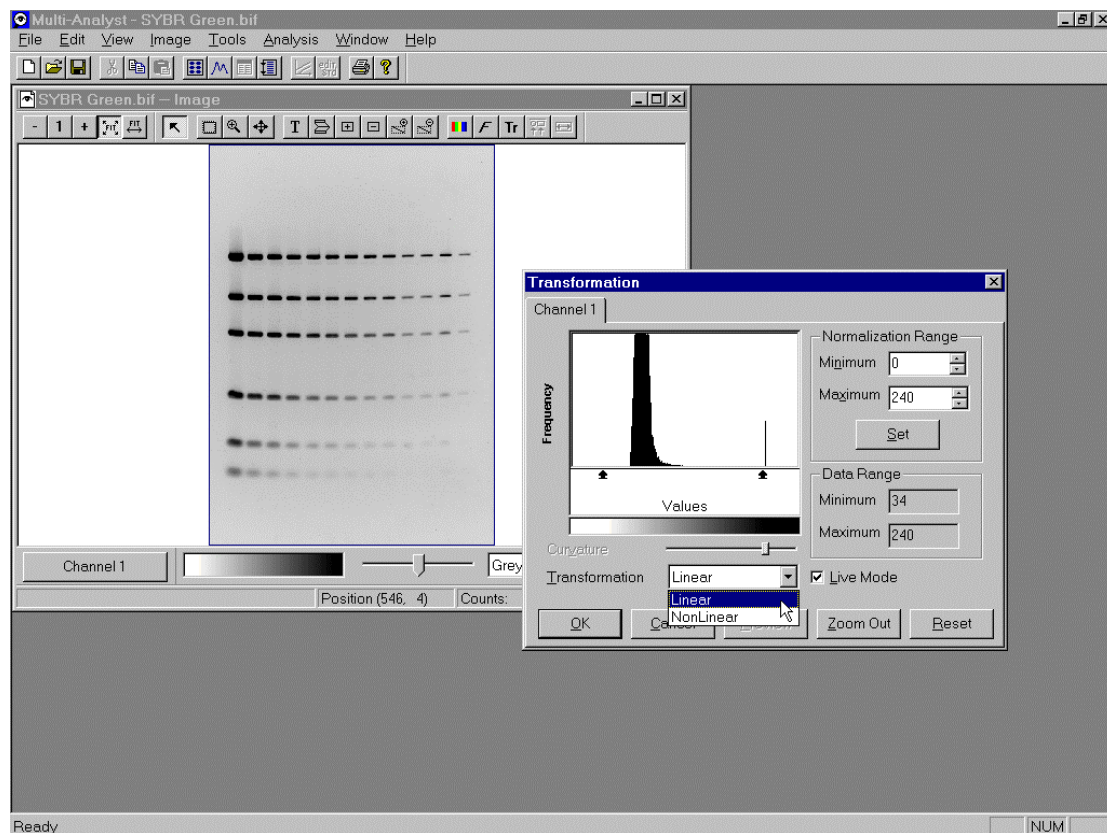
You can change the settings in the histogram window either by using the slider under the histogram or by manually typing in the values in the *Normalization Range* window. If you type them in, you **must** click on *Set* to enter the changes.



The Data Range window does not allow any changes to be made. It provides useful information about the high and low pixel values in the image.

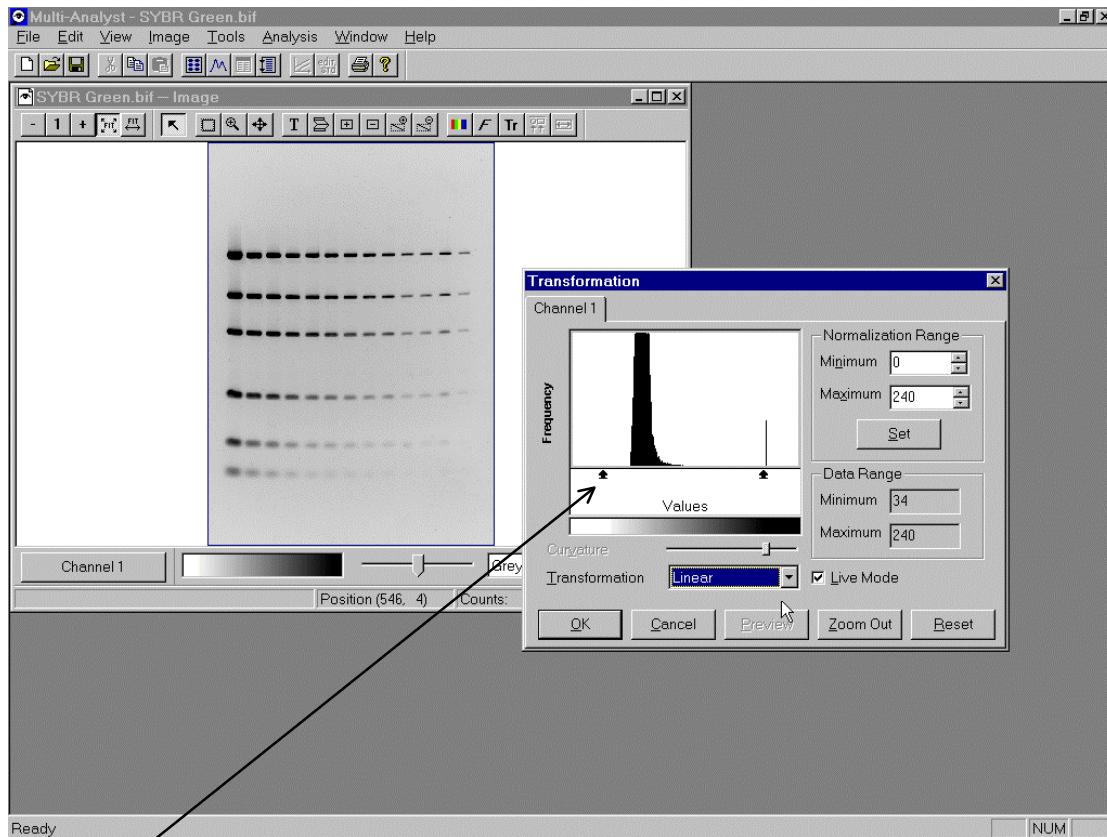


You may want to check the *Live Mode* box. This will cause any changes made in the *Transformation* window to immediately change the onscreen display of your gel. If you like the results, you must click on the OK button. Click on cancel or reset to start over.

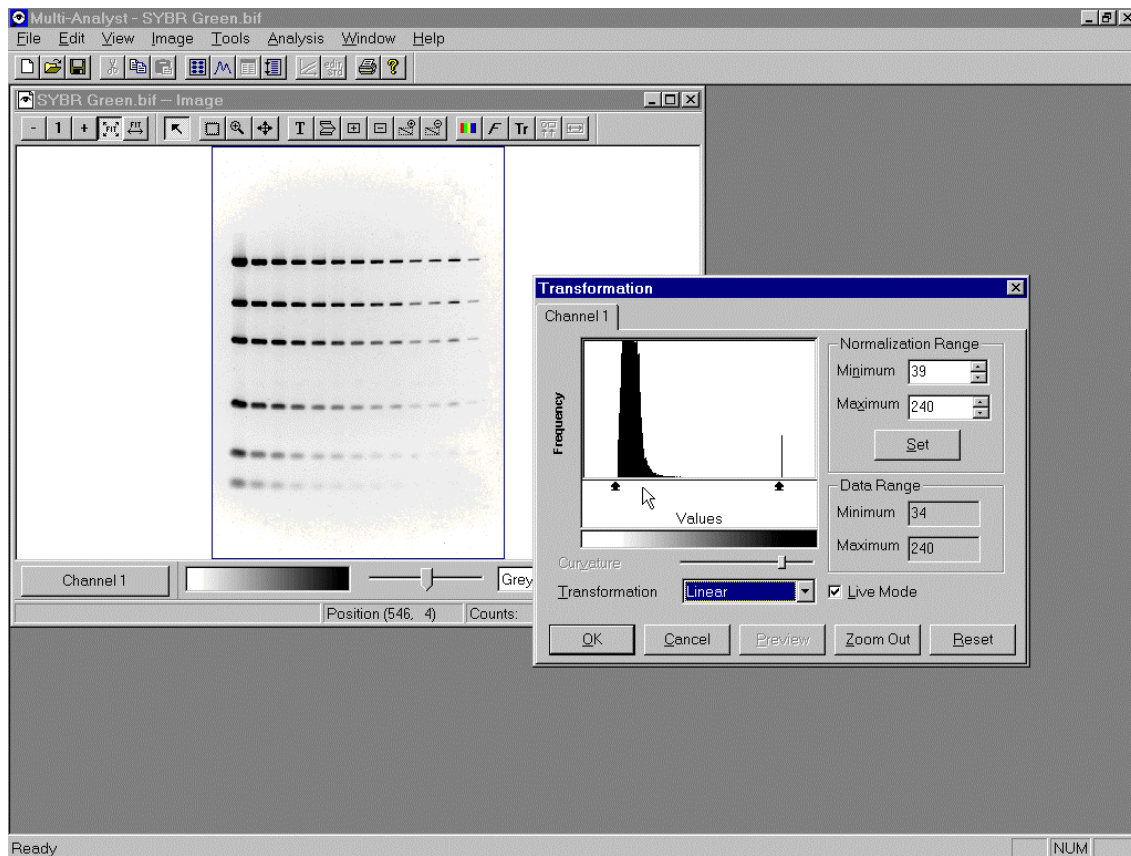


Some gels may be more efficiently visualized if their pixels are mapped in a non linear fashion. Non linear transformation activates the curvature slider.

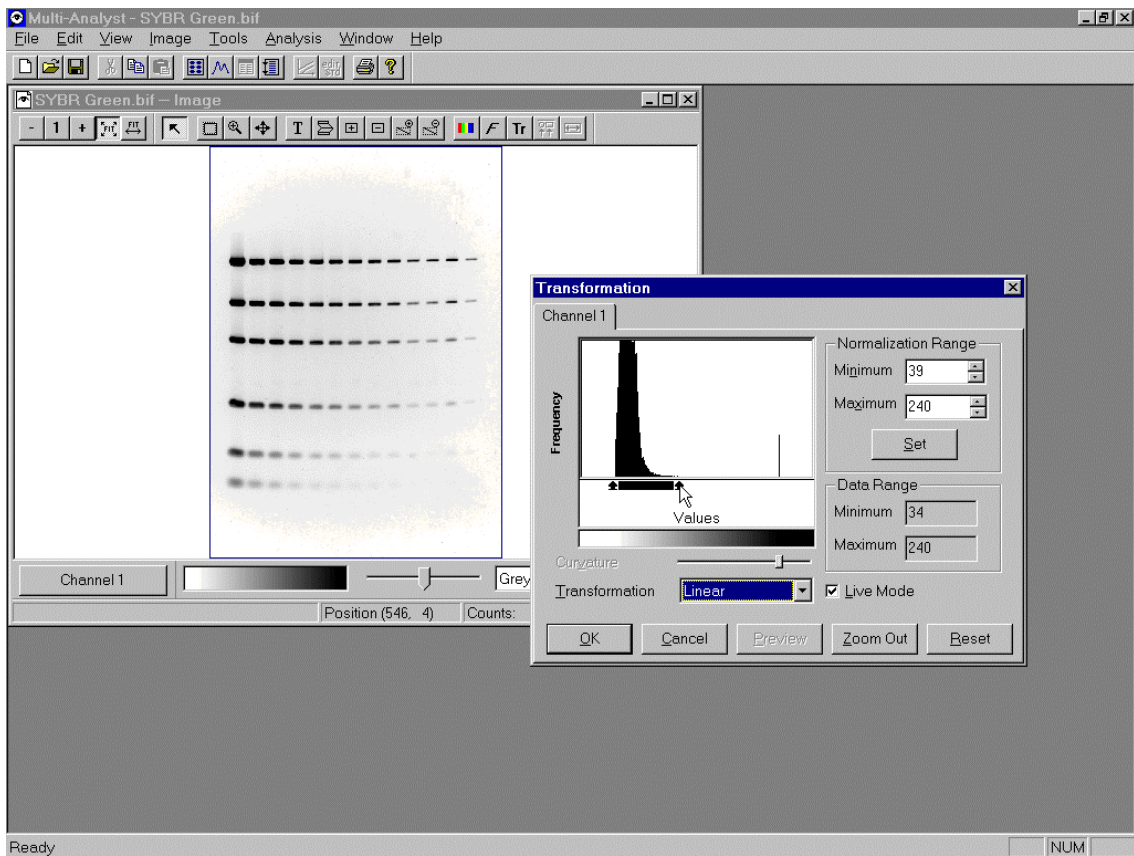
13.9.4.1 4.9.4.1 The Transform Histogram Controls



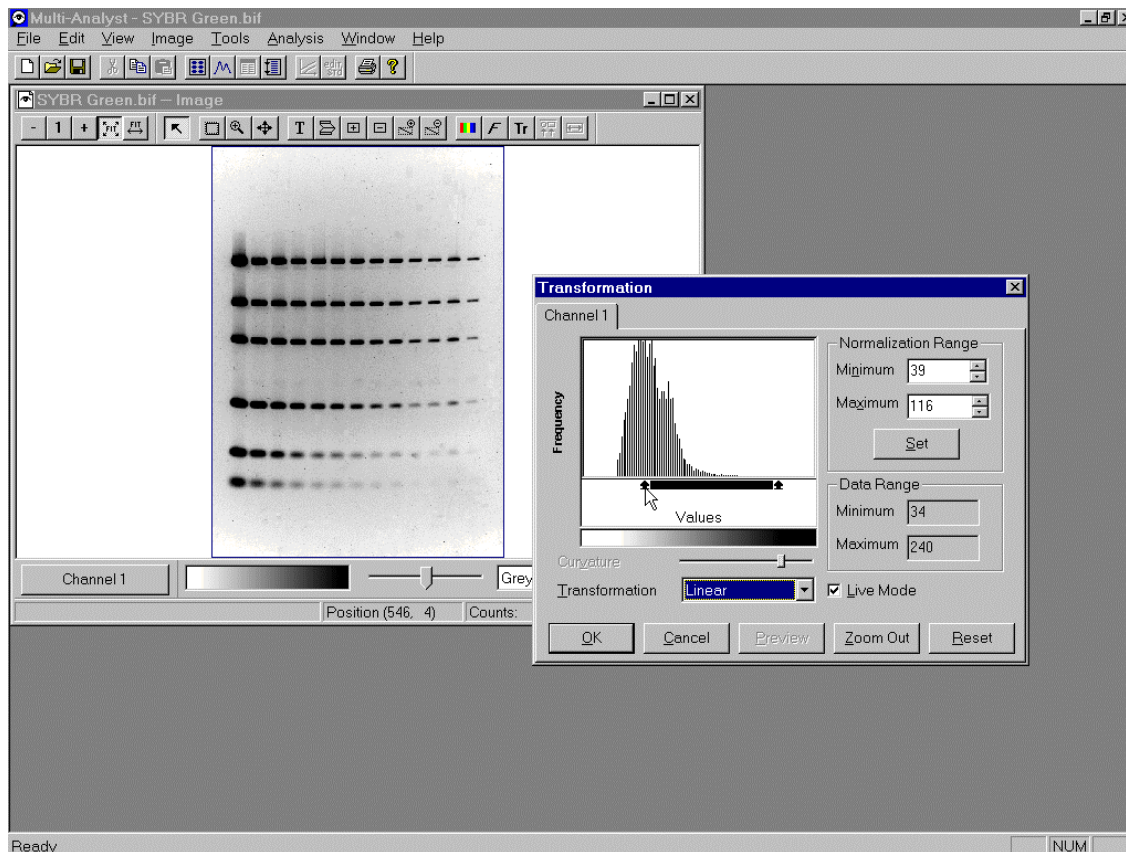
The easiest way to reduce the image background is to grab the left slider arrow under the histogram and slide it to the base of the curve.



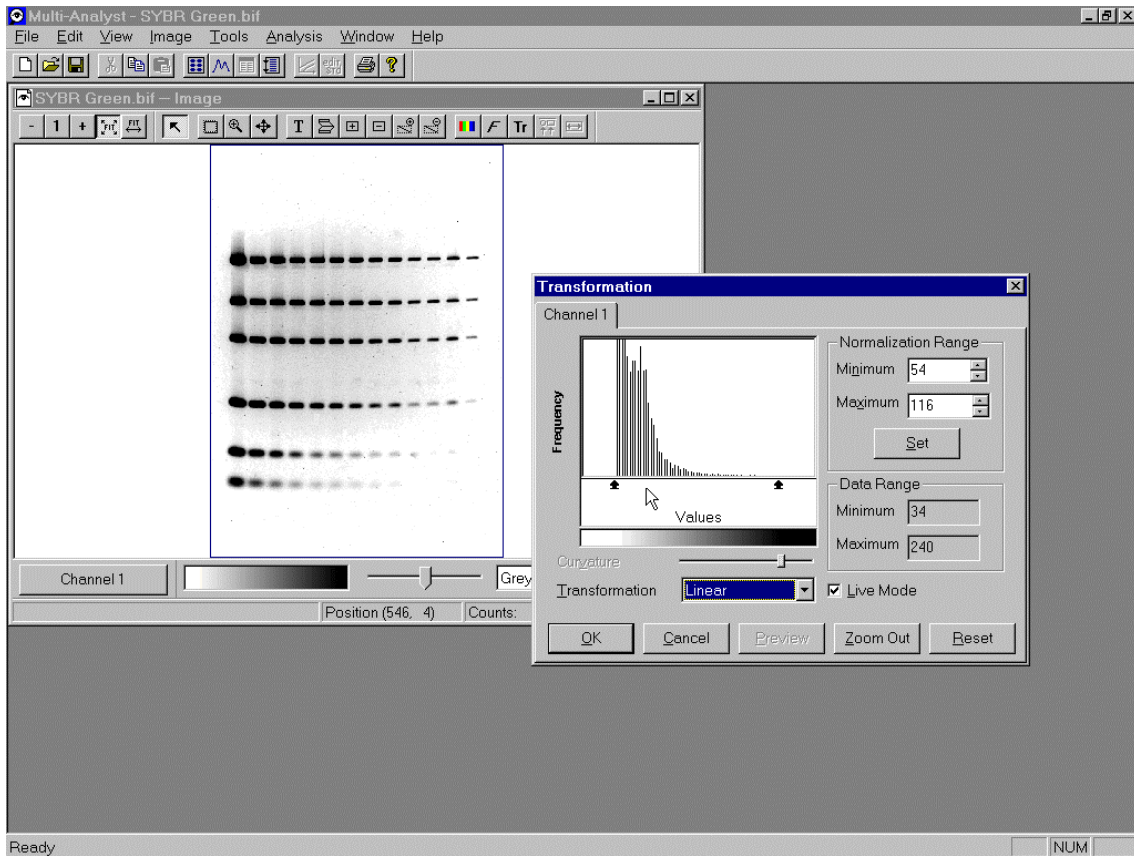
Most of the background is now remapped to white, while the rest of the data set is spread over the remaining 256 gray scales your monitor is capable of displaying.



Next, slide the right slider over to the bottom of the peak.



This causes the weak bands to be remapped to the darker pixels. Notice the numbers in the *Normalization Range* 39 -- 116. This means that any pixel that has a value of 38 or less is remapped to zero (absolute white), while any pixel that has a value of 117 or greater is remapped to black. The remaining 76 pixel values are stretched over the 256 gray scale of your monitor (an approximate jump of 3 gray values for every one increase in image pixel values).



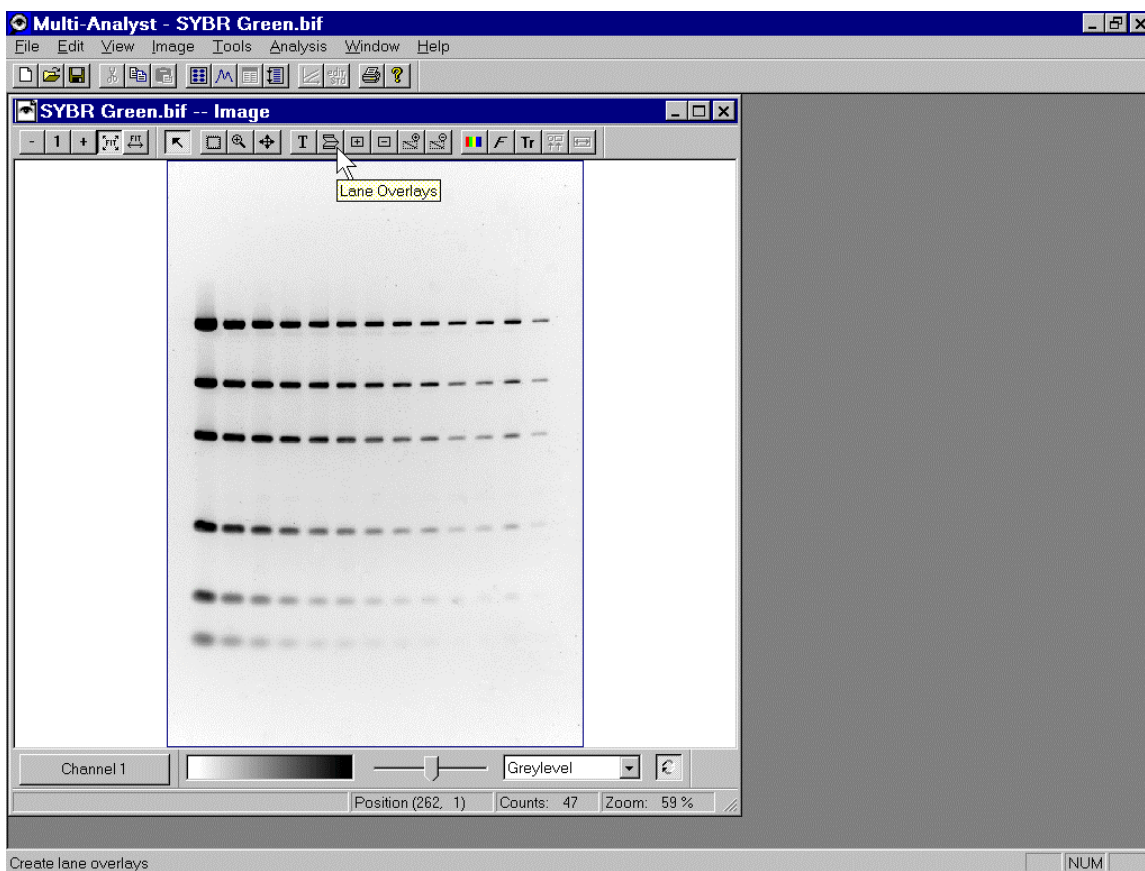
The end result of a successful transformation is the reduction of background while increasing the contrast of your weakest bands.

A good rule of thumb is that a high contrast image is easier to quantitate than a good looking image. This is because you can more easily see the beginnings of your bands for more accurate quantitation.

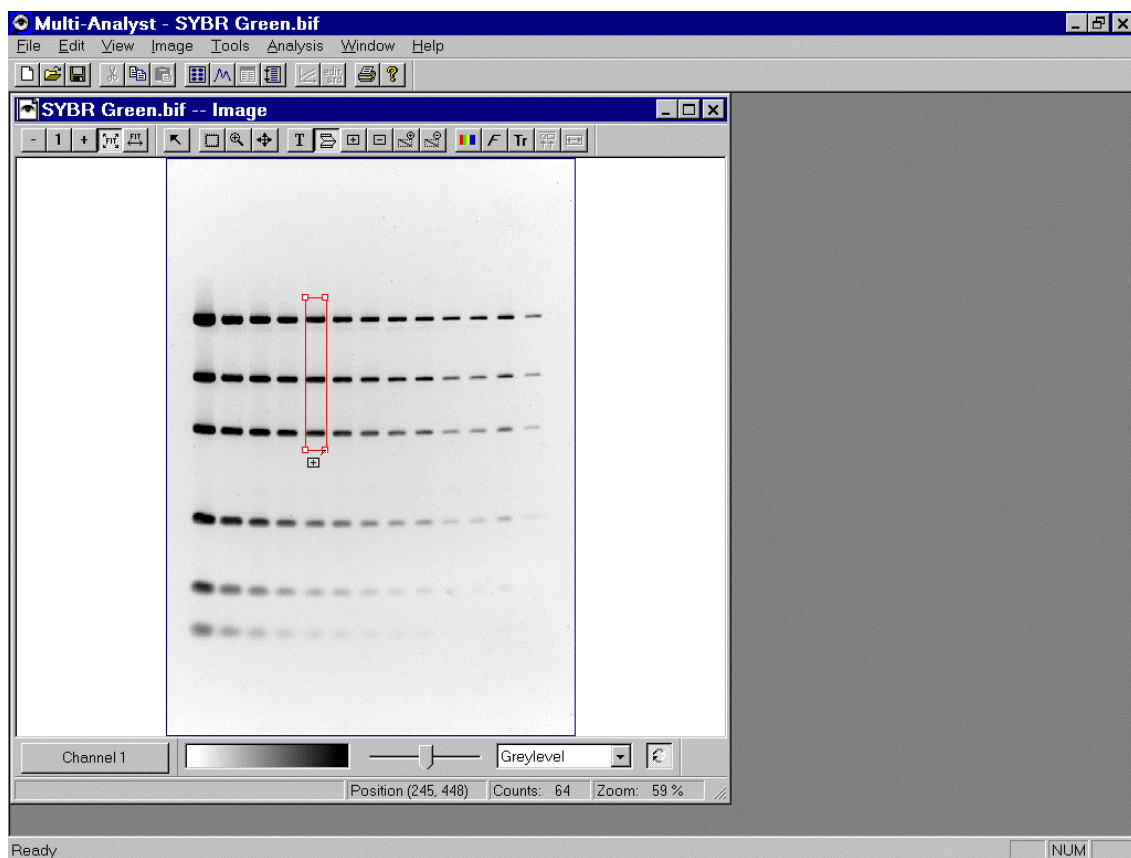
IMPORTANT: Remember, no matter how much you change the way your image looks, your raw data set is never changed.

14. Tutorial -- Quantitating your Gel

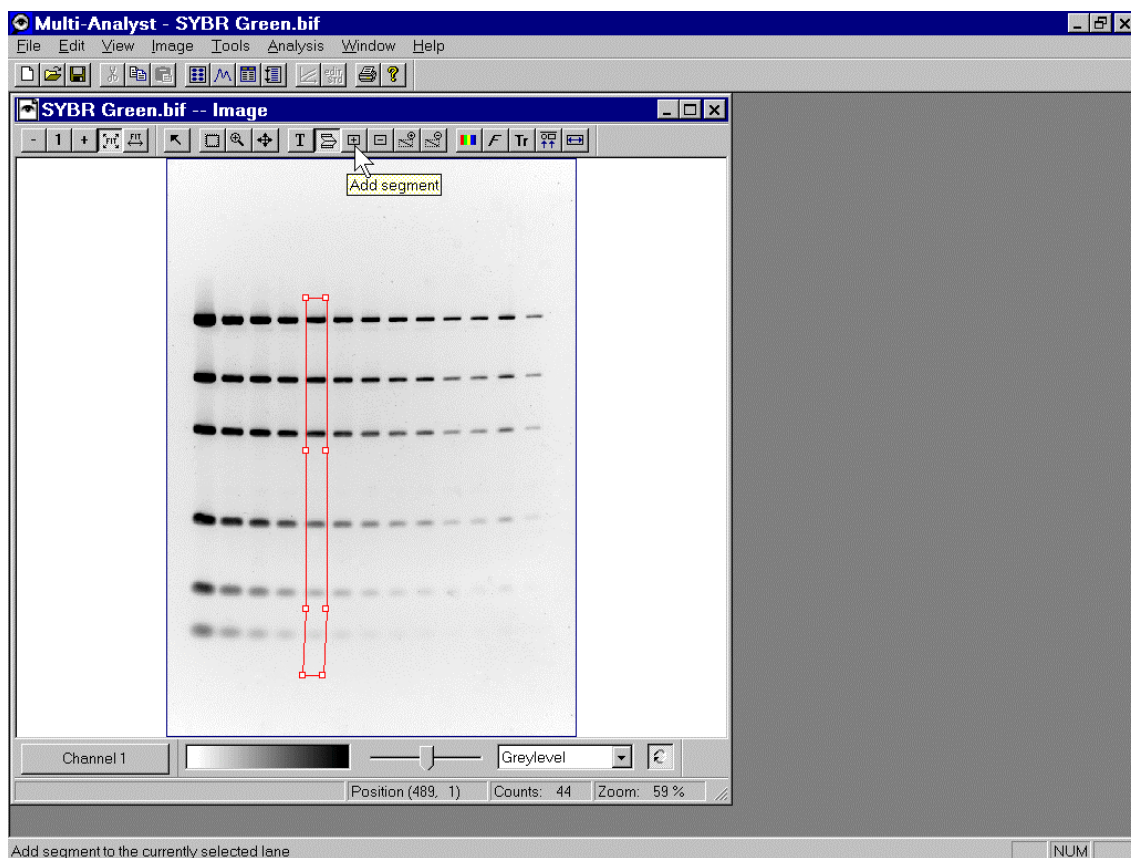
14.1.1 Manual Lane Identification



Use the *Lane Overlays* tool to manually draw your lane



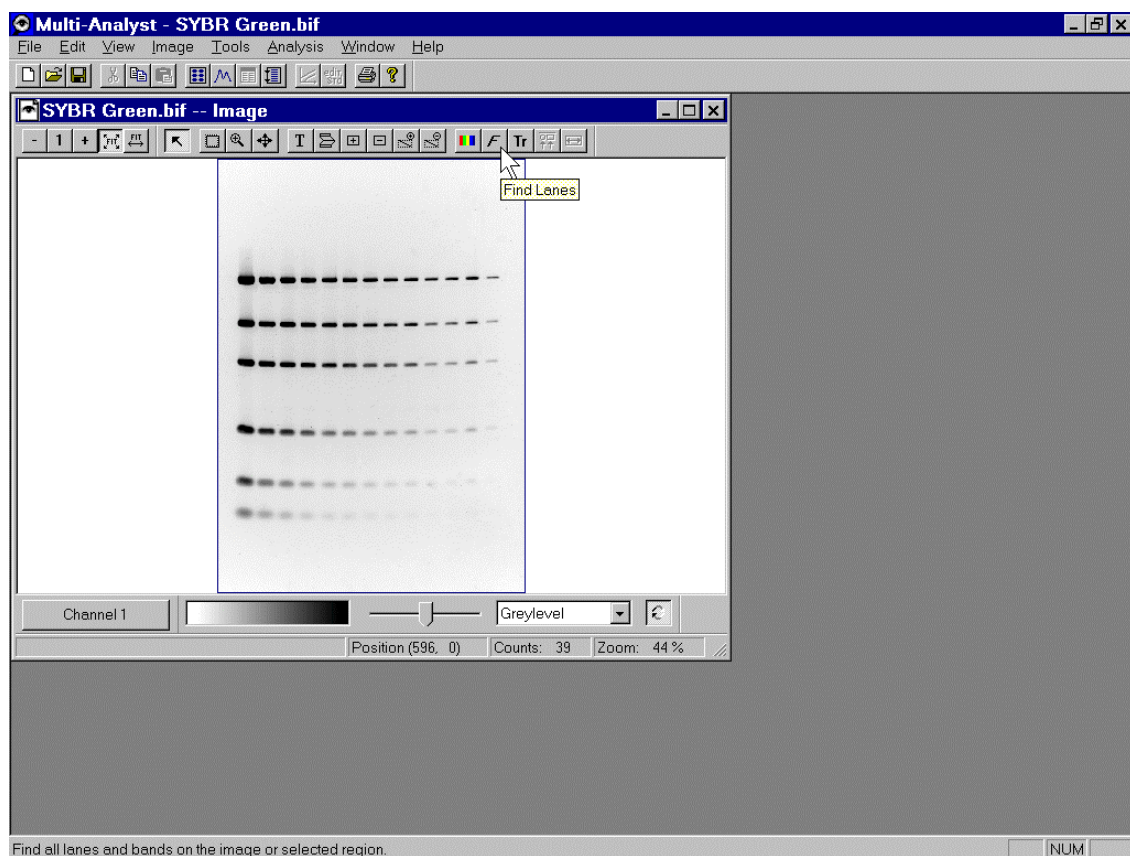
Click and drag the mouse over the lane of interest. You may let go of the mouse button at any point where you want to add a segment (node) to the lane.



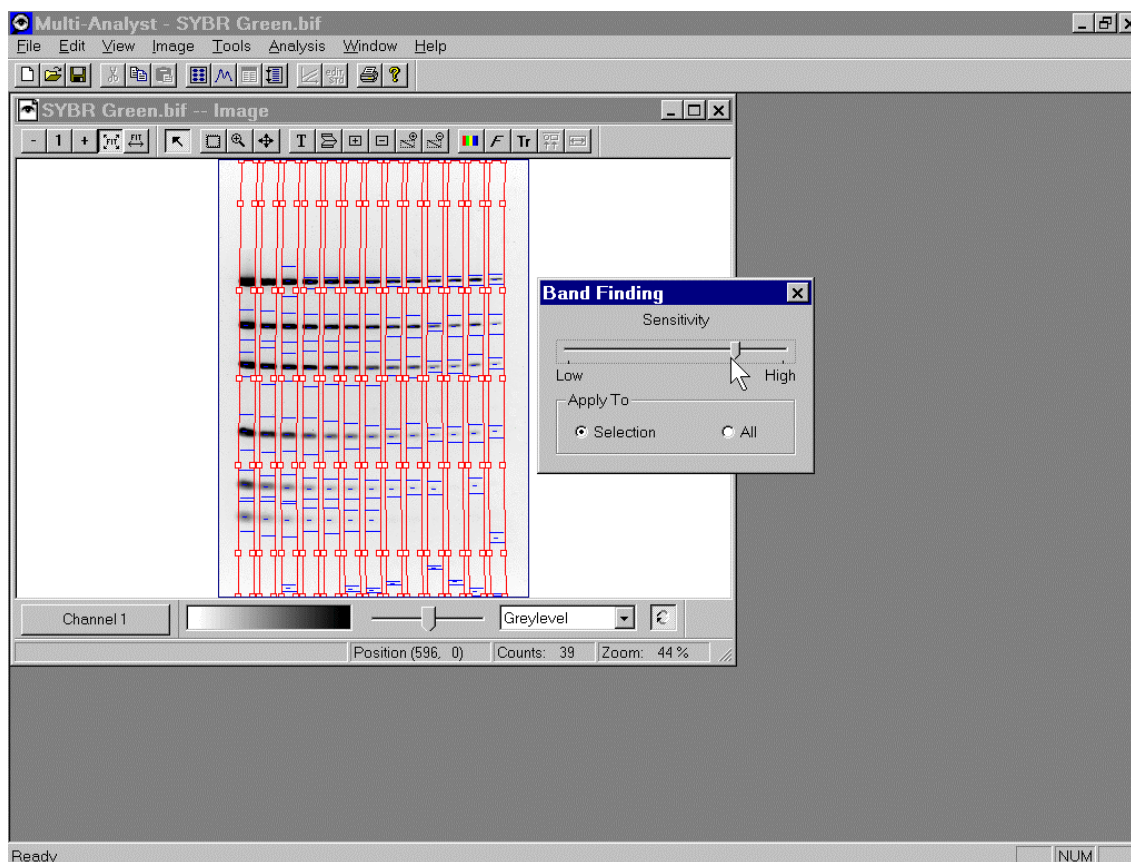
Click along the side of the lane to add segments. When you are finished drawing your lane, **double click** at the bottom of the last segment. You can manually add or remove nodes with the *Add/Remove node* tools.

You can use the *Delete* key on your keyboard to delete the active lane or the software will automatically delete the lane during automatic lane and band detection.

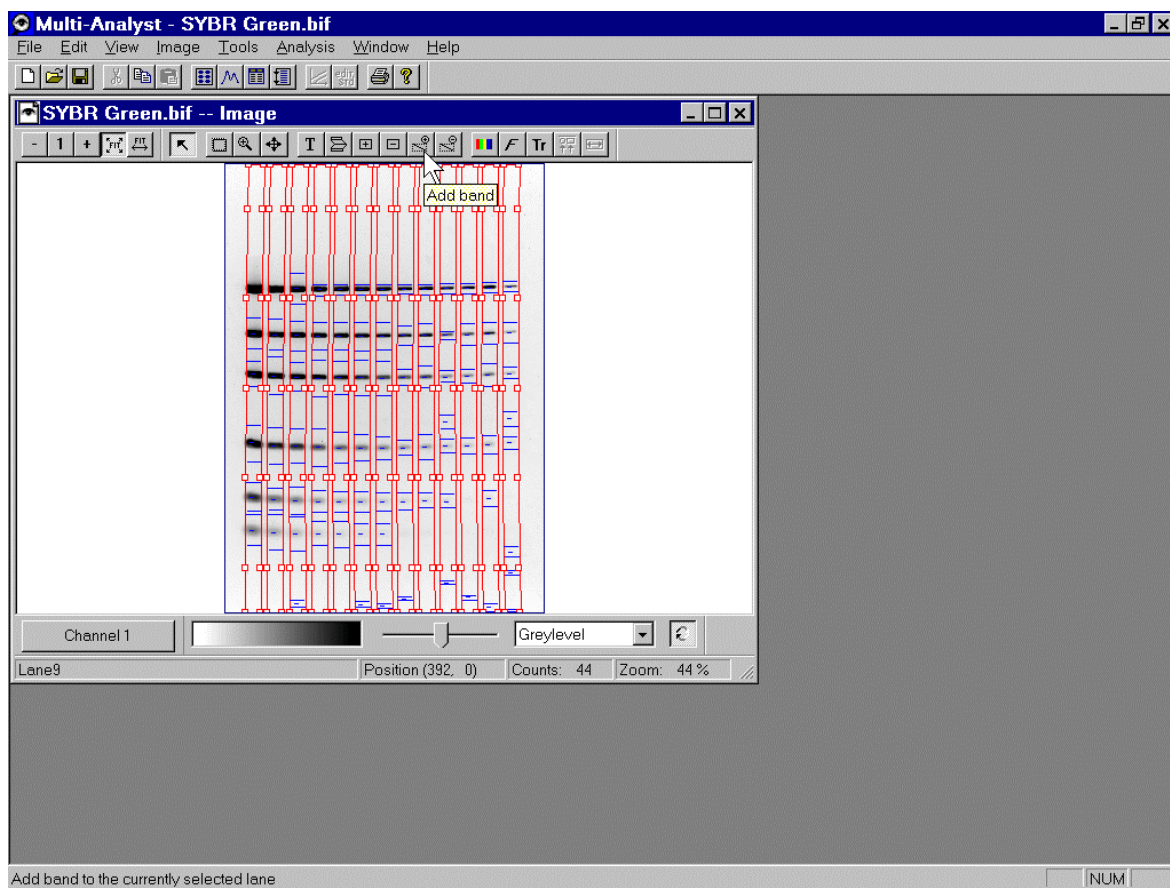
14.1.2 Automatic Lane and Band Identification



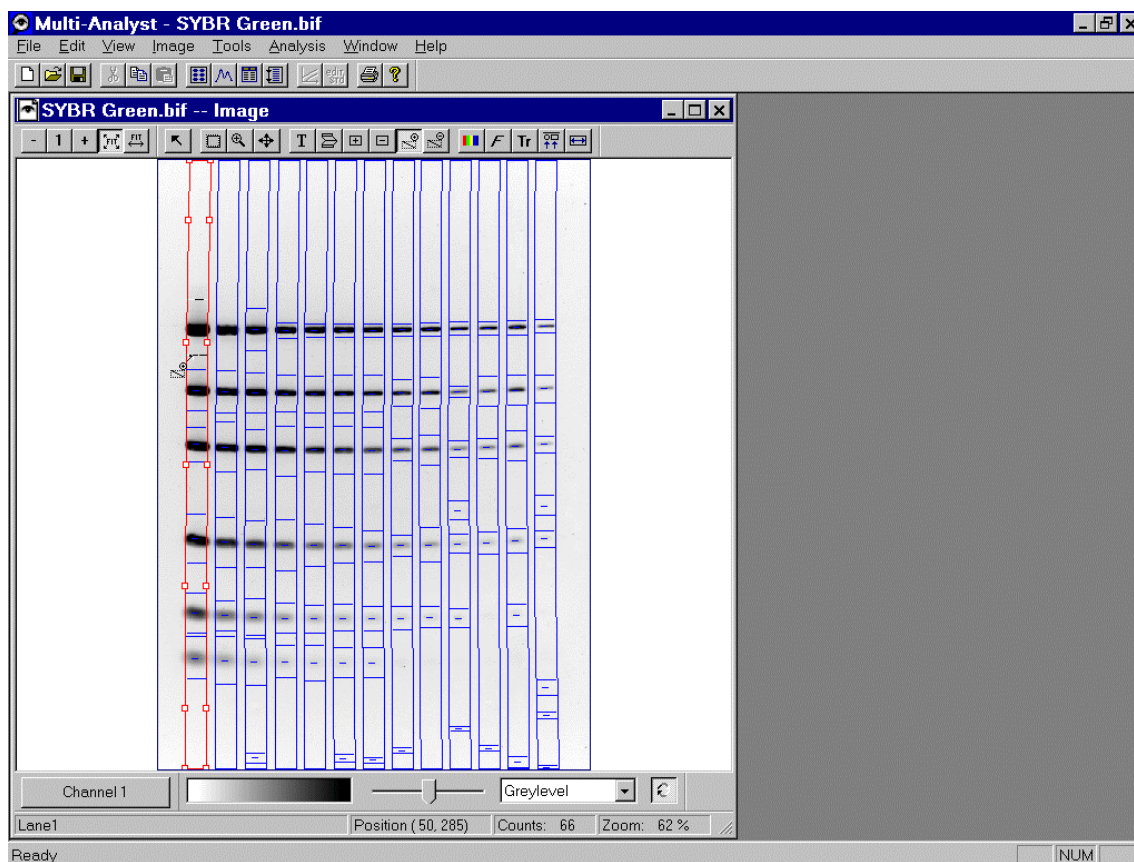
Click the *Find Lane* tool to automatically find your lanes.



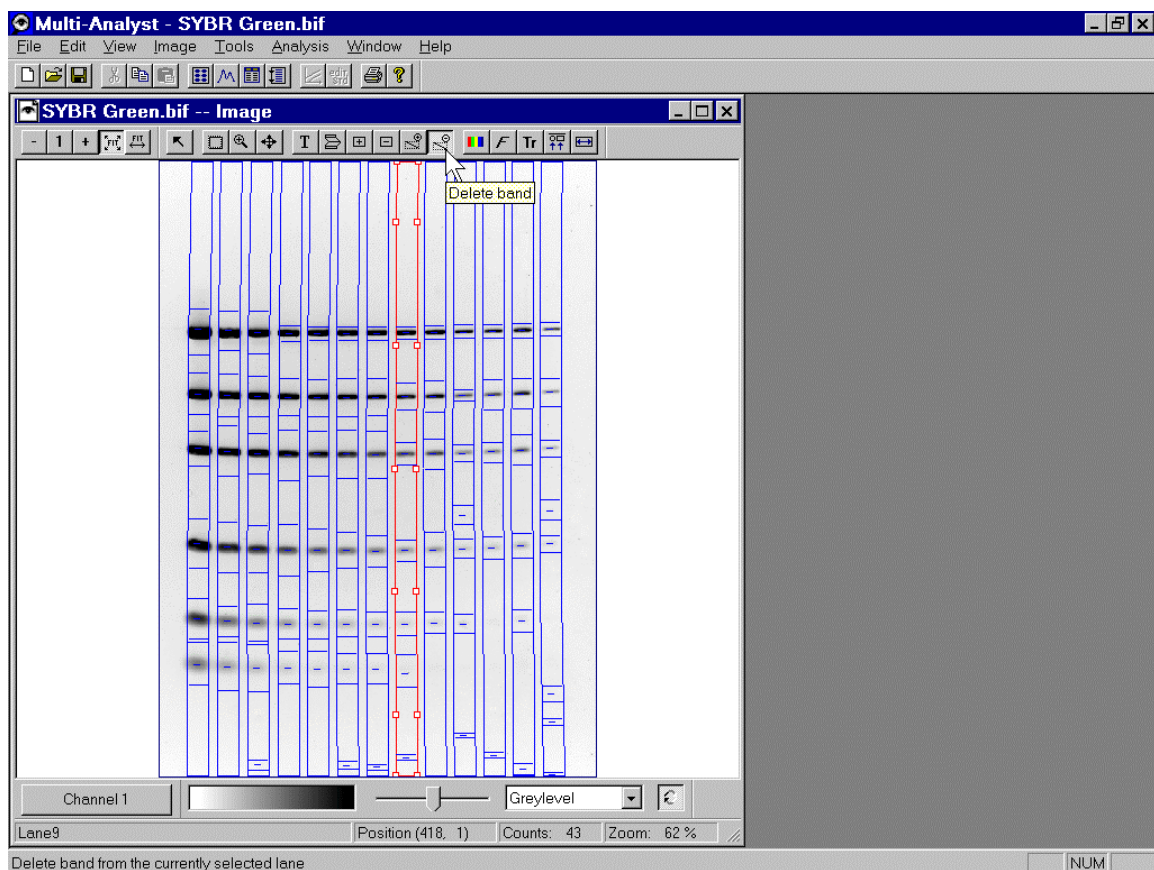
The *Band Sensitivity* window appears automatically. Adjust the slider to increase the sensitivity or decrease the sensitivity of the band finding algorithm. If you wish, you can apply the band sensitivity adjustment only to selected lanes. This is useful if only certain lanes need adjustment. To deselect the group, click outside of the lanes (on the image) and then click on the lane you want to select.



Once you have finished working with the band finding algorithm, you may use the *Add Band* tool to add or subtract those bands that were missed.

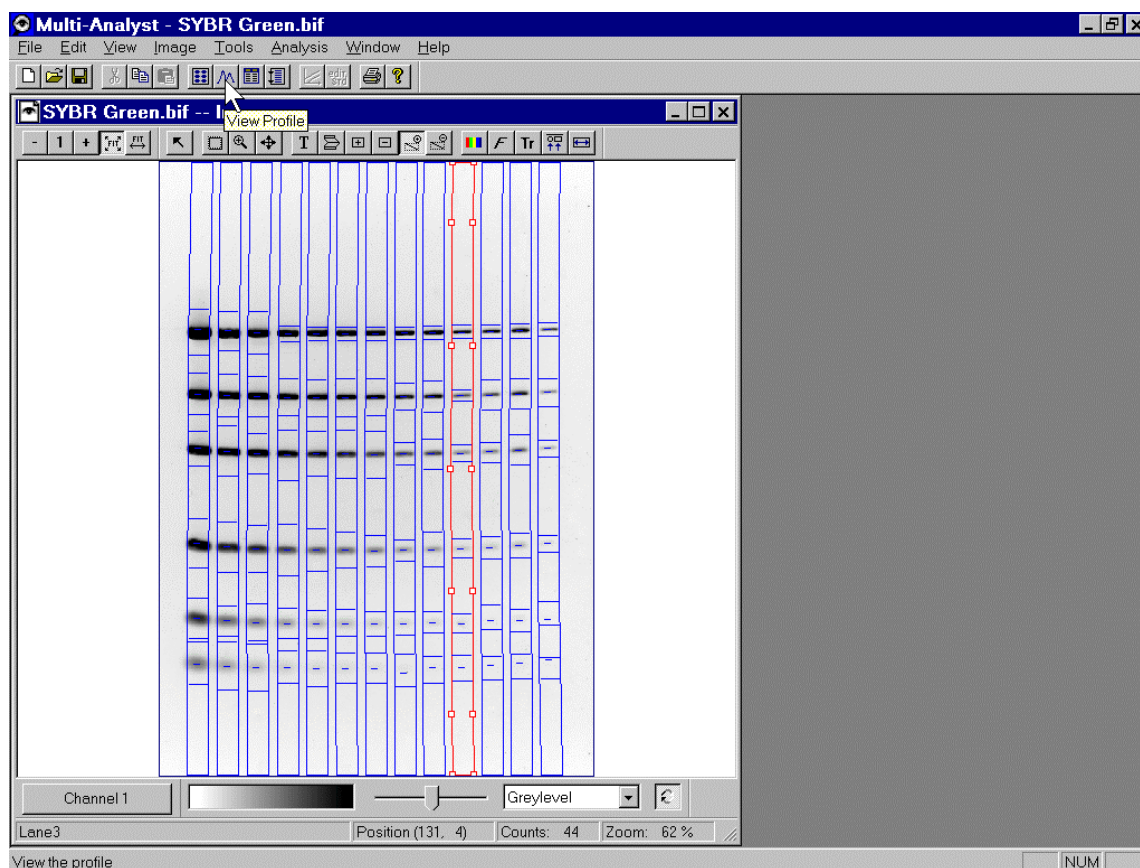


Just click and drag your mouse over the band you wish to identify. Multi-Analyst will automatically find the band peak and renumber the bands in the lane.

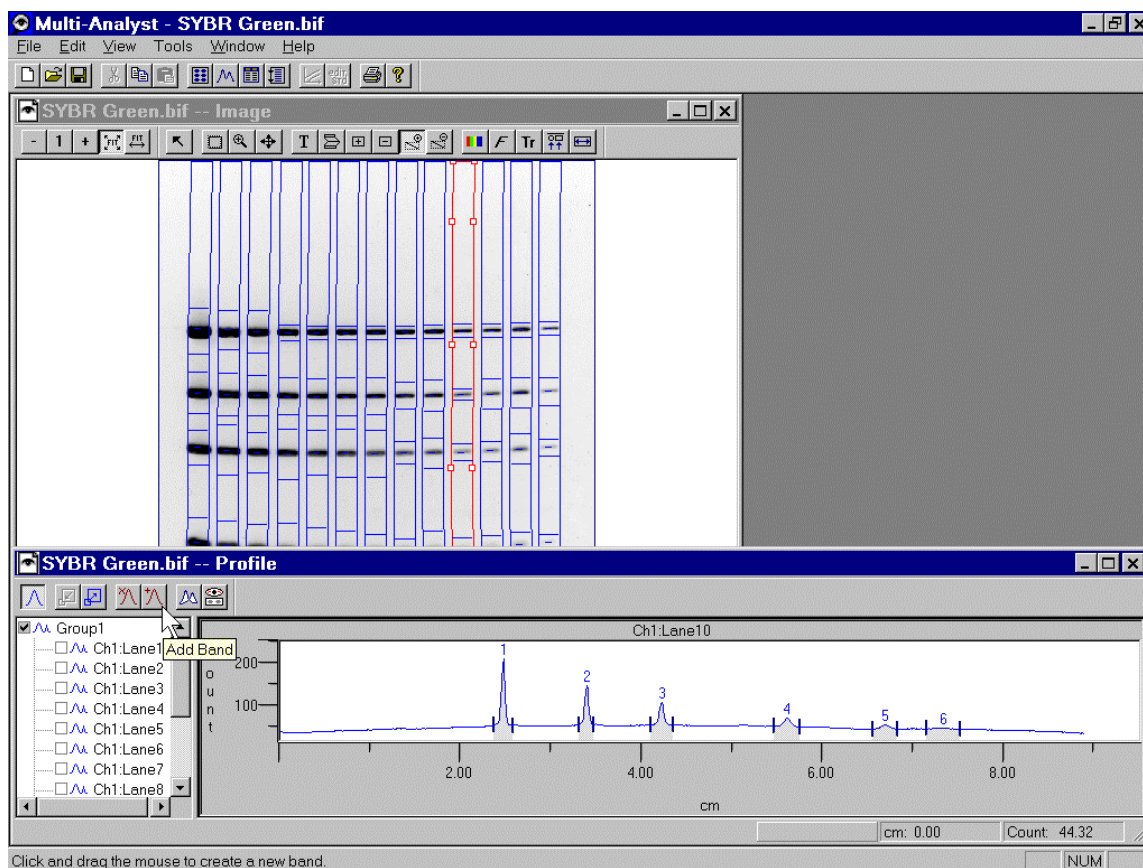


You can also use the *Delete Band* tool to automatically delete bands.
When you are satisfied with your lane and band selections, you can continue to the next step.

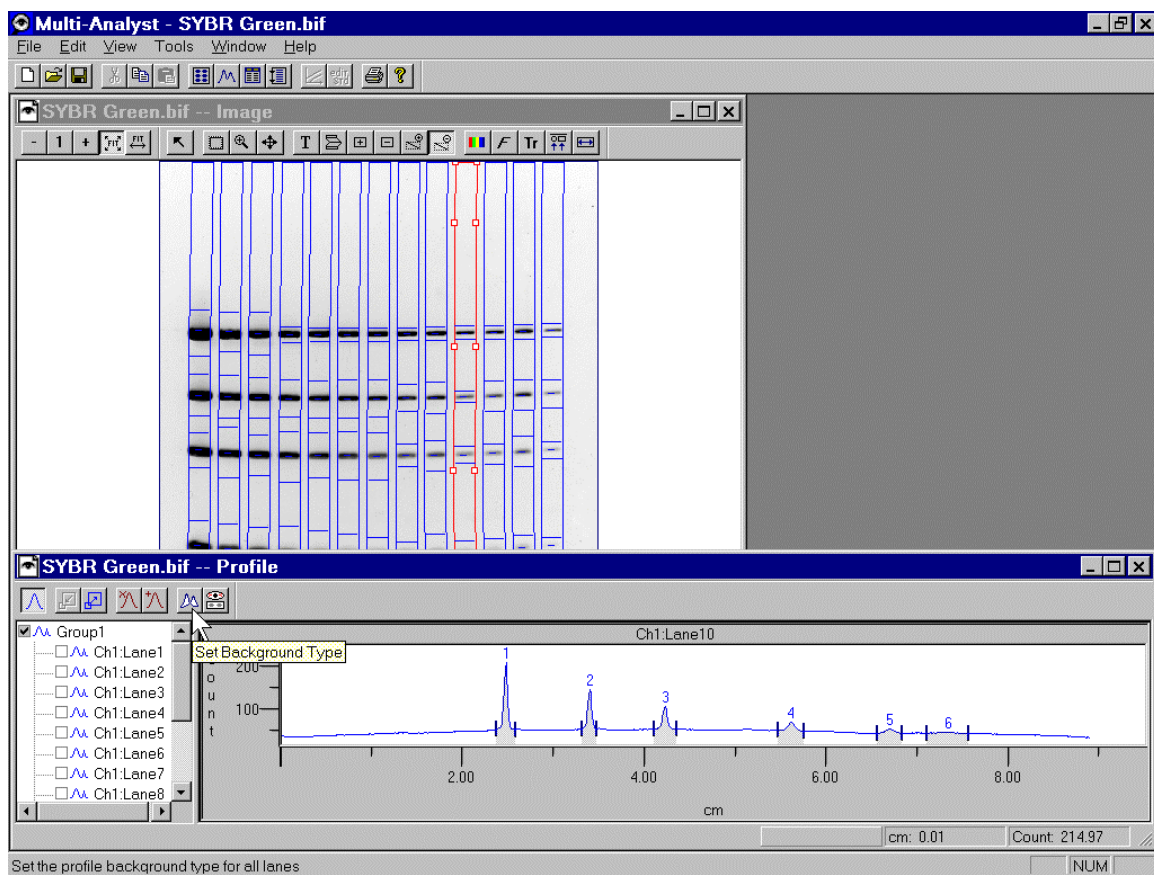
14.1.3 Profile Creation



Use the *View Profile* tool to open the *Profile Report*.

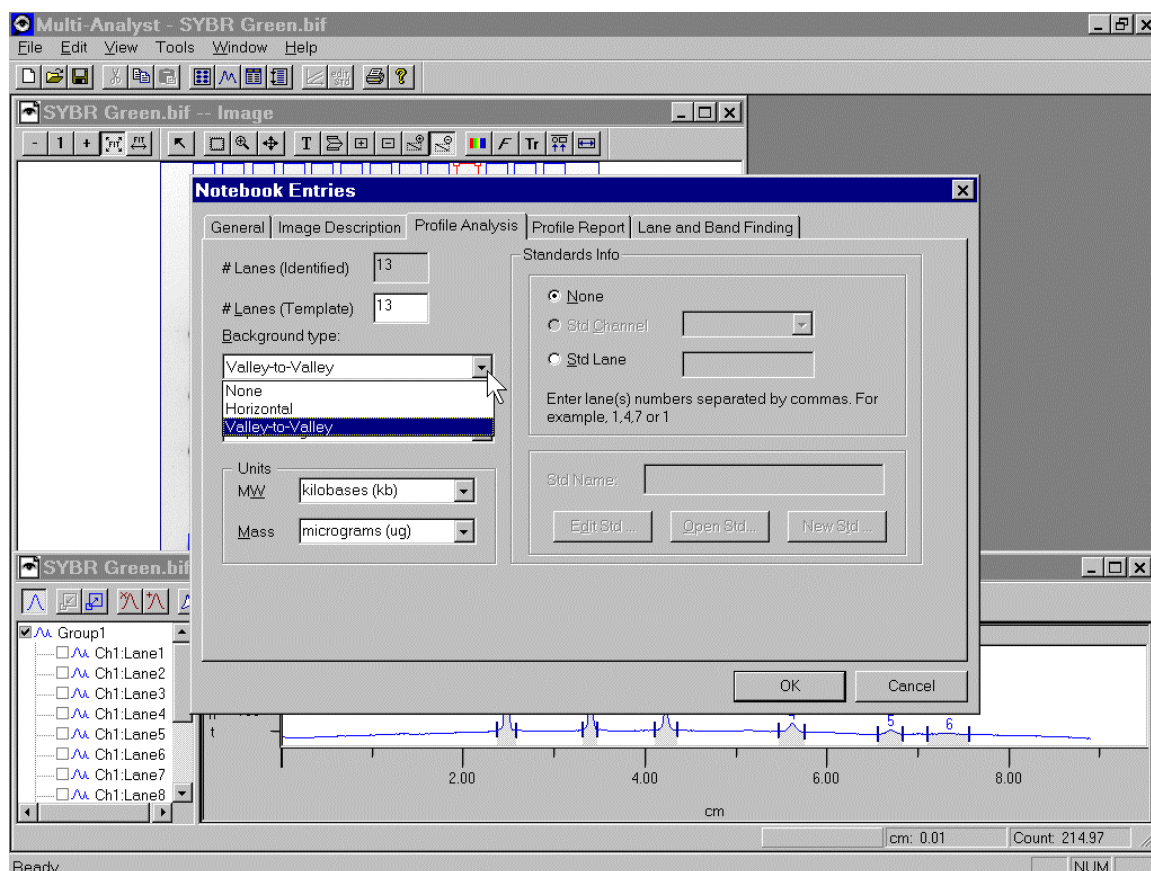


Multi-Analyst calculates the profiles of all the lanes and can show one, several or all at the same time. In addition, you can also use the *Add* and *Delete band* tools here too. Your work is dynamically linked so that any actions performed on the profile will be shown on the image window.



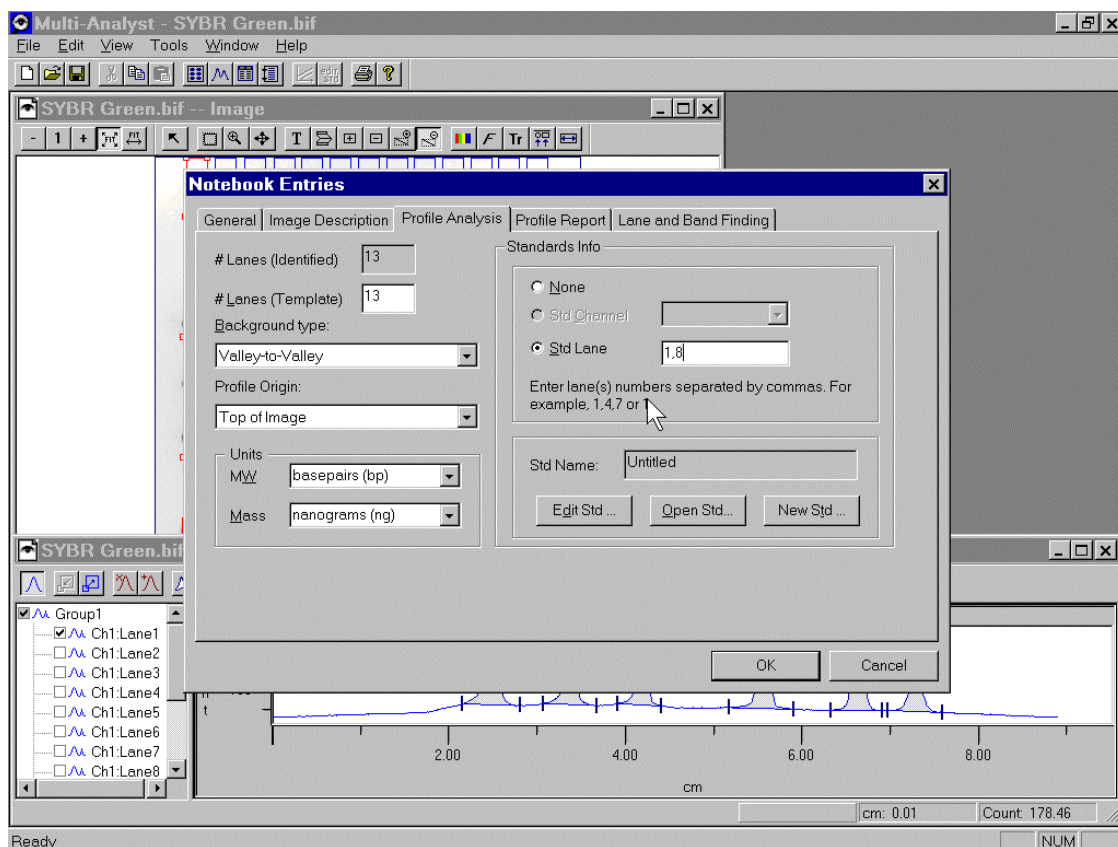
Use the *Set Background Type* tool to change the background subtraction algorithm.

14.1.3.1 Notebook Entries - Background Subtraction and MW standards



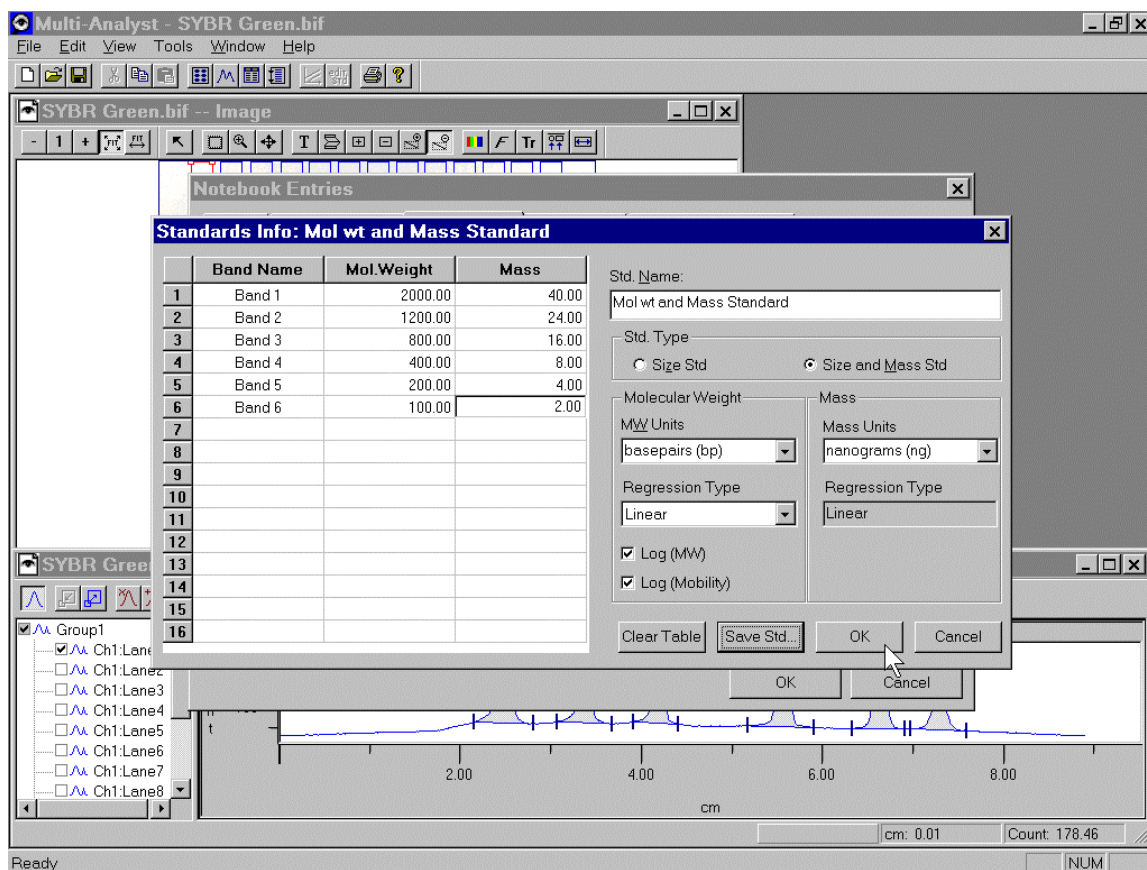
Change the *Background type* to *Valley to Valley*.

Several places in the program will take you to this screen, so now is a good time to review the *Notebook Entries* for your gel image. Every gel has associated with it a notebook. This *Notebook* is where the global settings are maintained for your gel as well as information gathered concerning your experimental settings, instrument used, notes, etc.



Set the lane that contains your standard. In this case we'll choose lane 8 since none of the bands are saturated in it. Notice that you can have multiple lanes identified as a standard. The software will normalize the standard curve across the gel, to account for variations in the gel based upon the runs in the standards lanes.

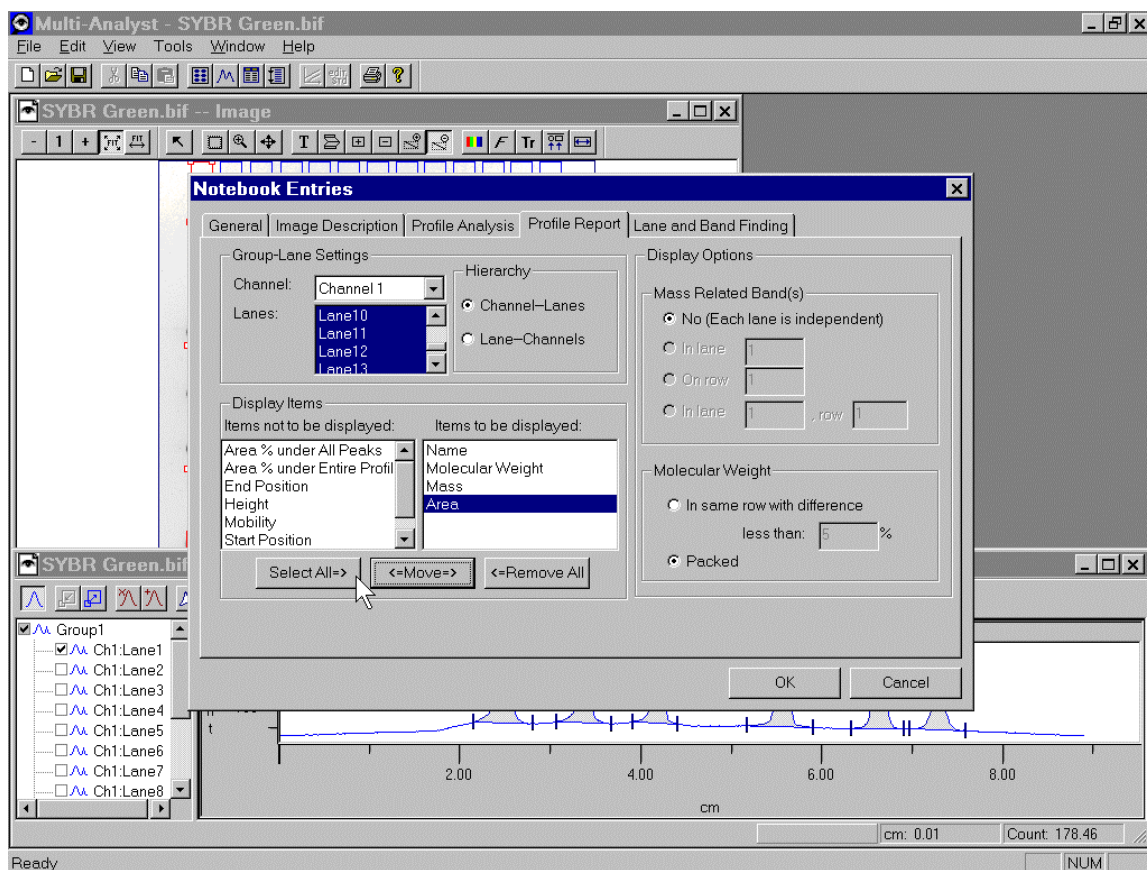
Click on *New Standard* to enter in the values for the SYBR Green gel.



Enter in the above values for your standard and give the standard a name such as the one above.

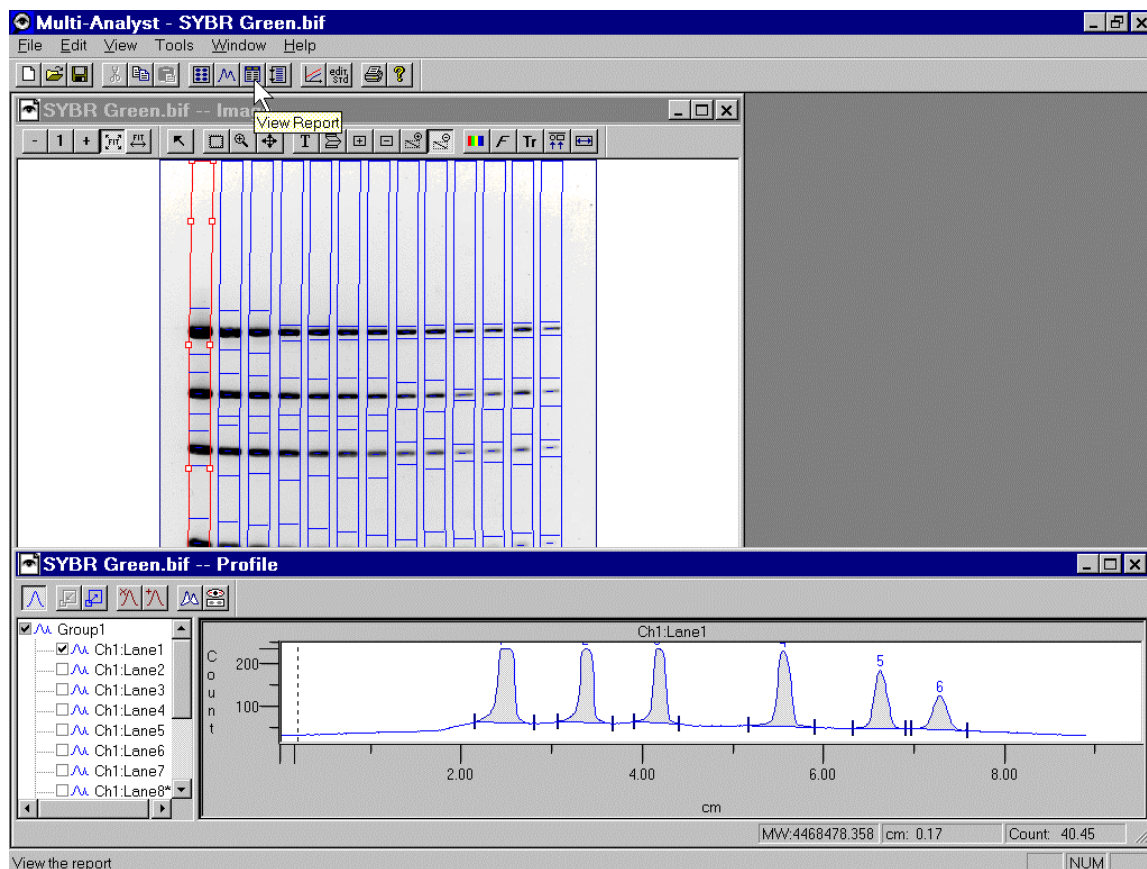
You can use and save any standard. Multi-Analyst allows for both Molecular Weight and Mass Standards to be used.

Once you save your standard, you can use it over and over.

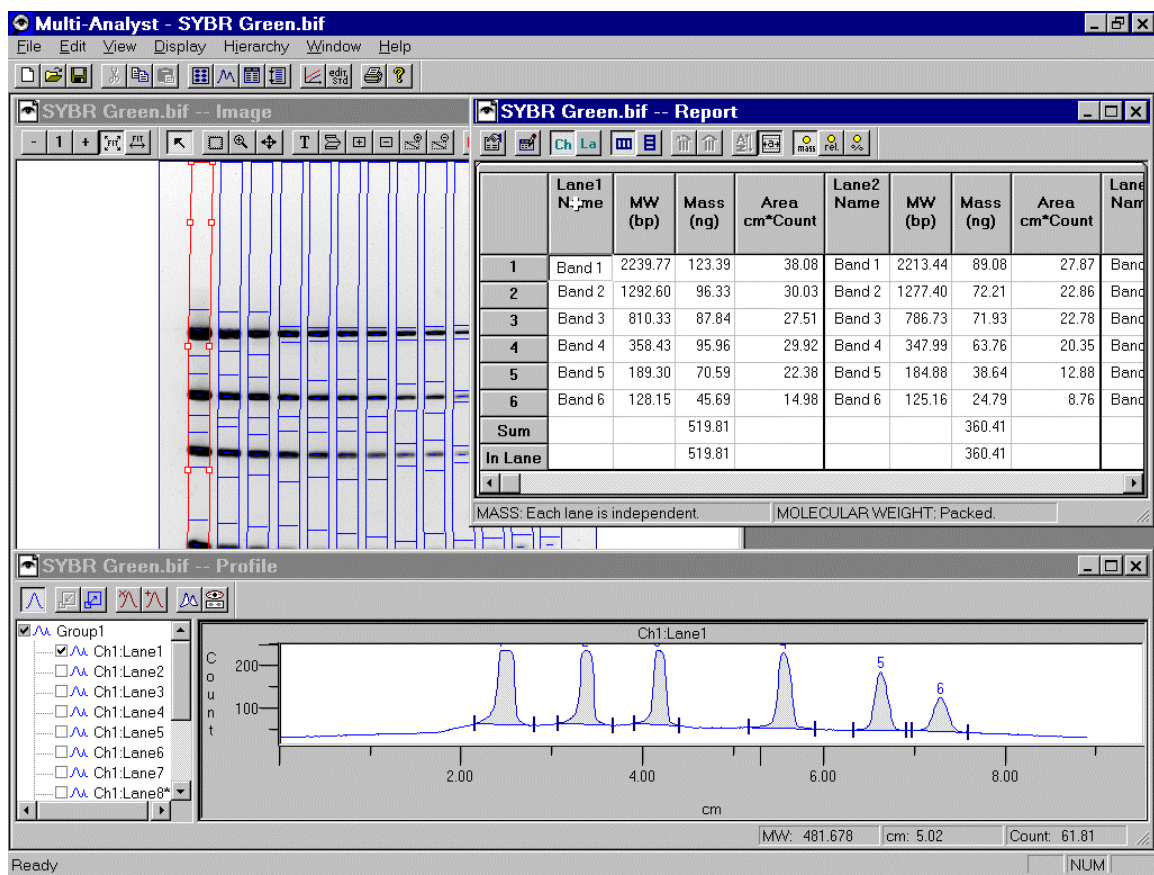


The *Profile Report* tab allows you to customize the data that you will see when you show the profile report.

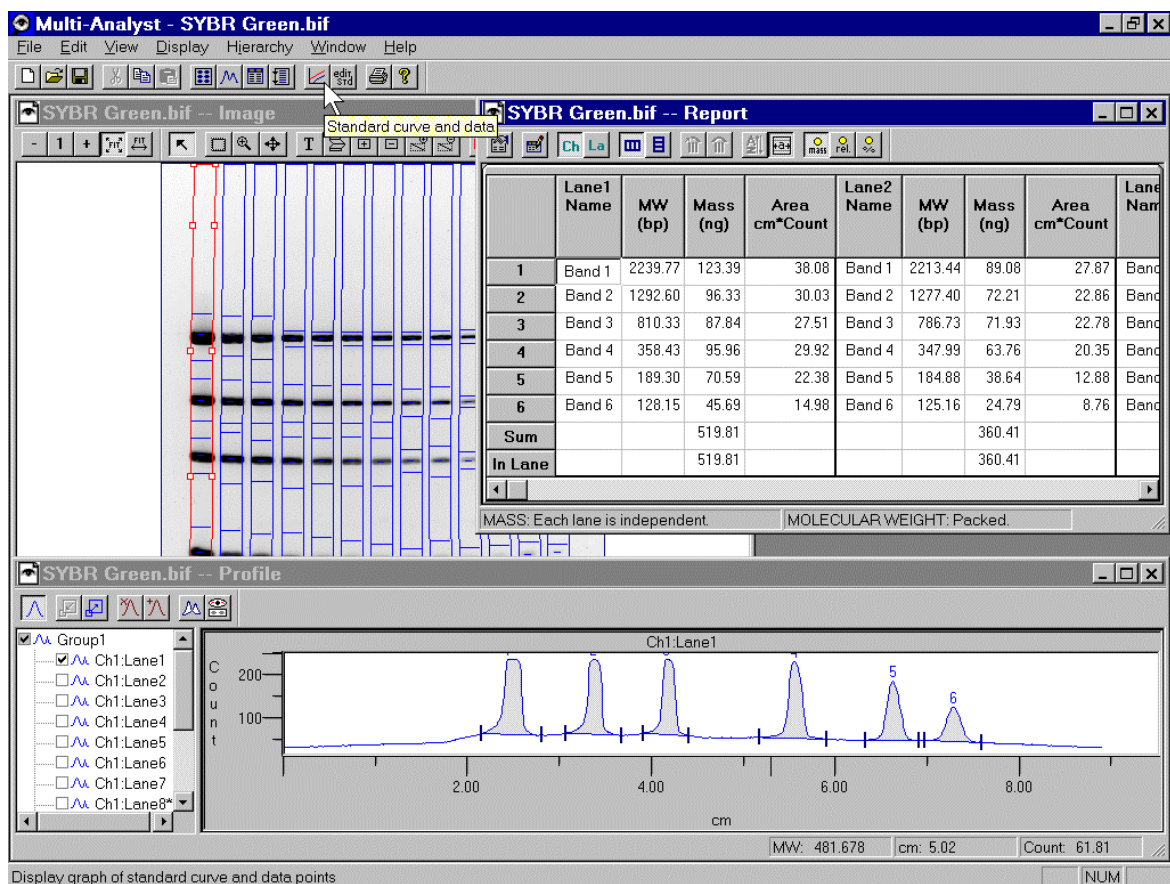
Obtaining your data results



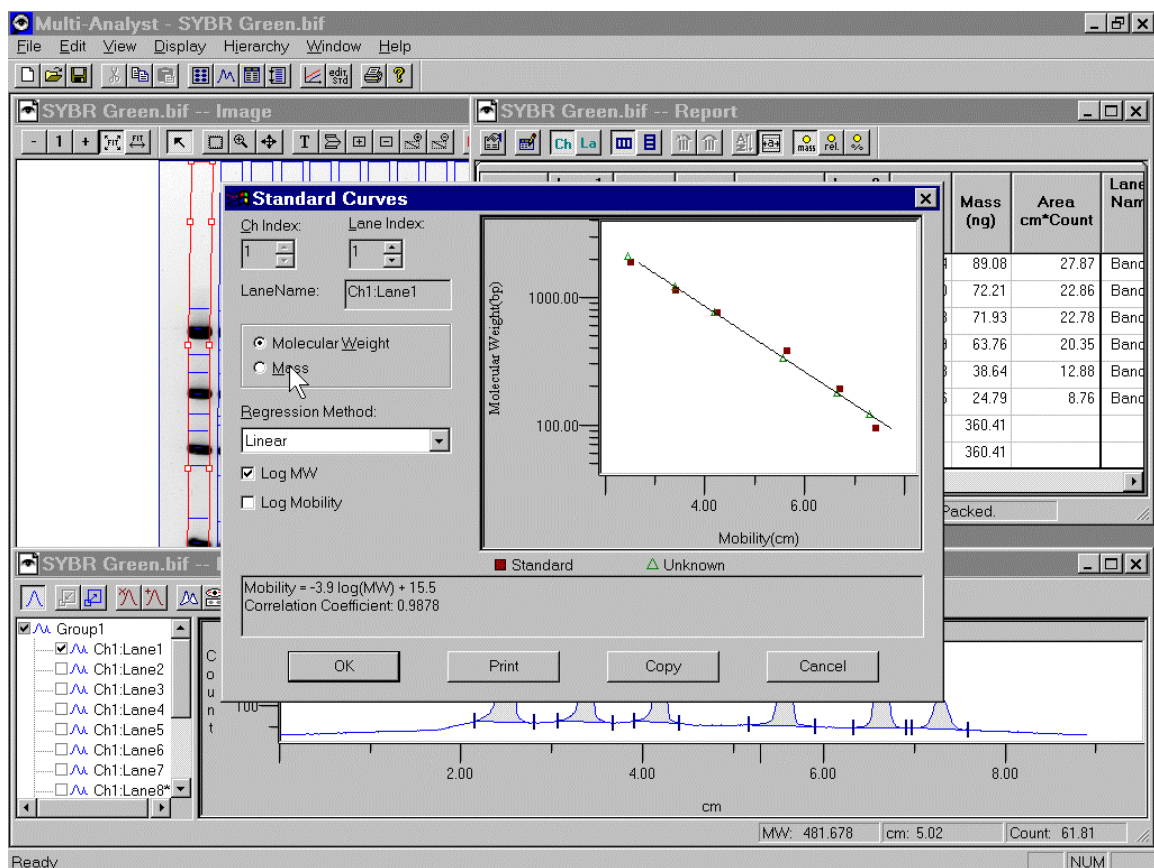
Click on the view report tool to bring up the report screen.



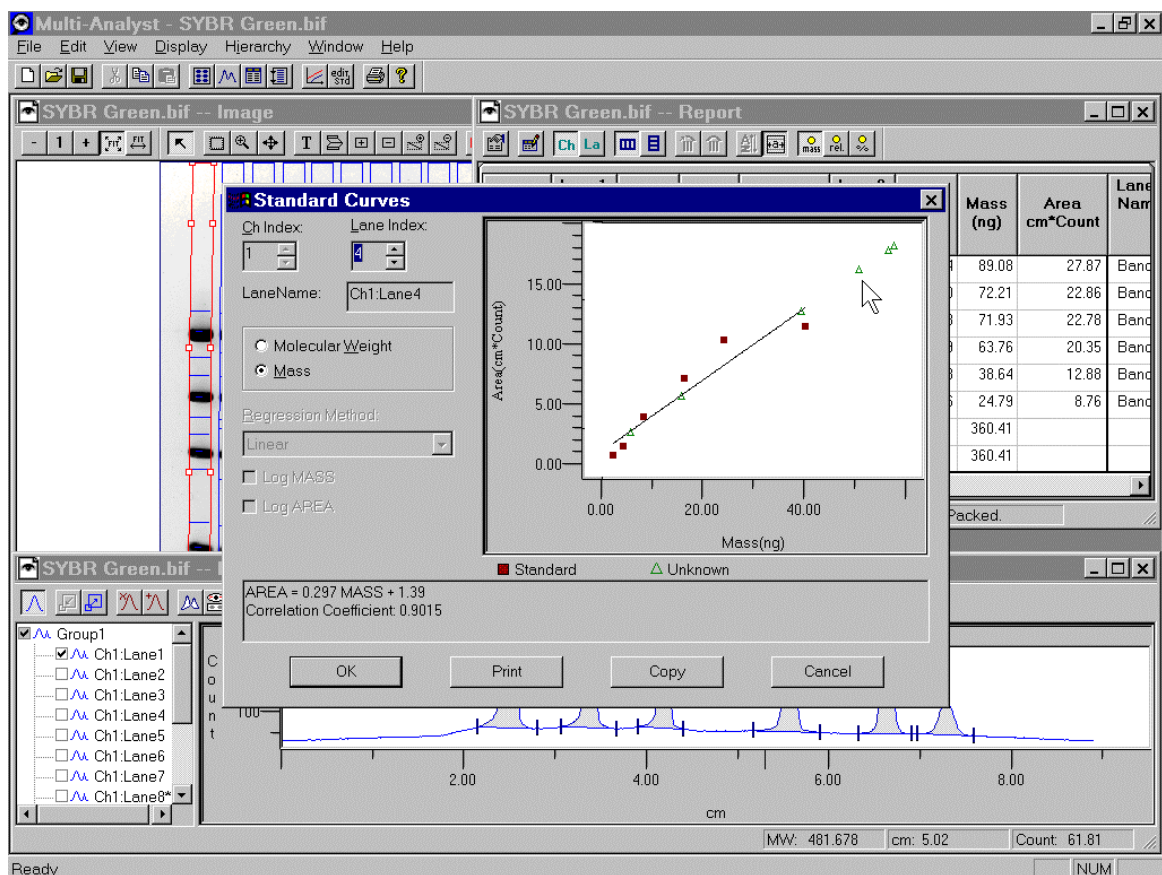
The profile report window will automatically show the finished calculations of your entire gel.



Click on the standard curve tool to bring up the standard curve window.



You can customize the standard curve calculation methodology to fit your experiments needs.



Your unknown data points as well as the Standard data points are shown in the graph.

14.1.4 Export your data

The screenshot displays the Multi-Analyst software interface. The 'File' menu is open, showing the 'Export to Excel' option. The 'SYBR Green.bif -- Report' window is visible, showing a table of data for two lanes. The 'SYBR Green.bif -- Profile' window is also visible, showing a chromatogram with peaks labeled 1 through 6.

SYBR Green.bif -- Report

	Lane1 Name	MW (bp)	Mass (ng)	Area cm*Count	Lane2 Name	MW (bp)	Mass (ng)	Area cm*Count	Lane Name
1	Band 6	128.15	45.69	14.98	Band 6	125.16	24.79	8.76	Band
2	Band 5	189.30	70.59	22.38	Band 5	184.88	38.64	12.88	Band
3	Band 4	358.43	95.96	29.92	Band 4	347.99	63.76	20.35	Band
4	Band 3	810.33	87.84	27.51	Band 3	786.73	71.93	22.78	Band
	Band 2	1292.60	96.33	30.03	Band 2	1277.40	72.21	22.86	Band
	Band 1	2239.77	123.39	38.08	Band 1	2213.44	89.08	27.87	Band
Sum			519.81				360.41		
In			519.81				360.41		

MASS: Each lane is independent. MOLECULAR WEIGHT: Packed.

SYBR Green.bif -- Profile

Ch1: Lane1

Count

2.00 4.00 6.00 8.00

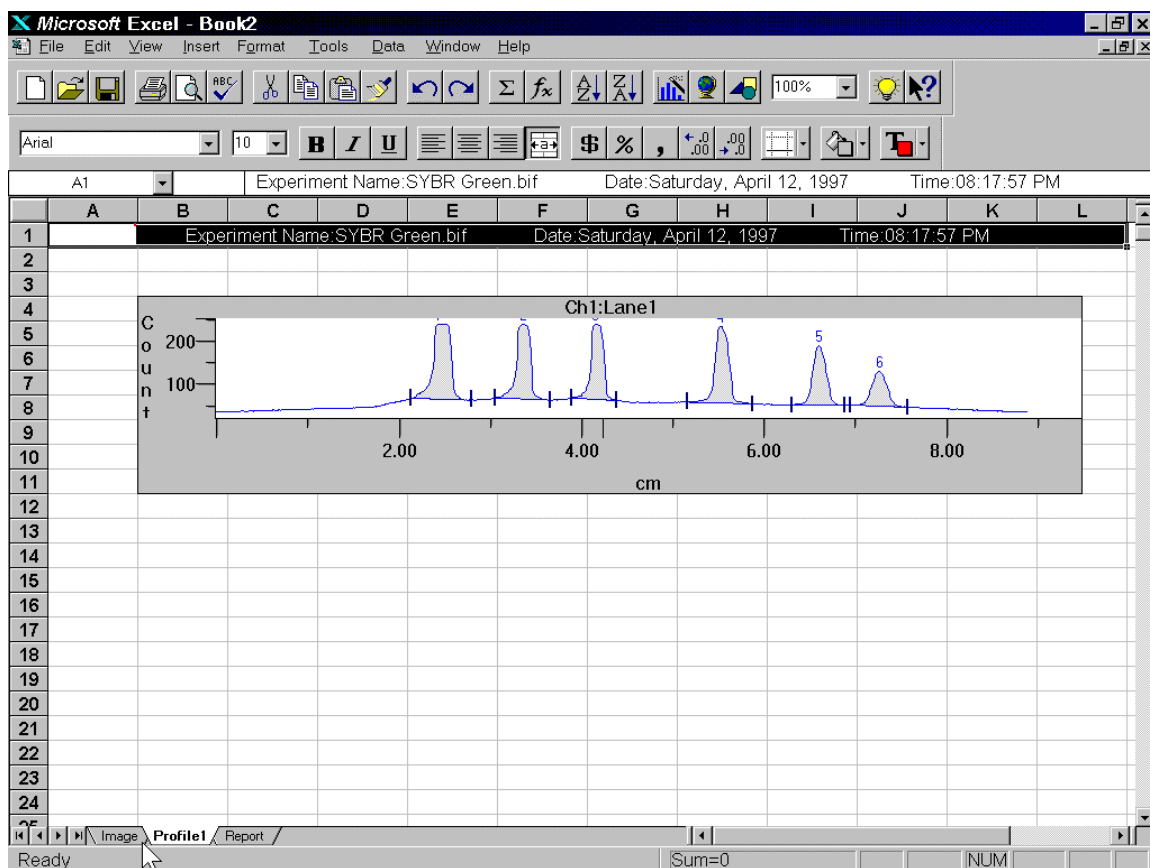
cm

MW: 9037.293 cm: 0.06 Count: 40.19

Export all open windows for the current document to Excel

NUM

Exporting your data is easy. Just use the *File* ,*Export to Excel* menu option and choose between exporting just the active window (header in blue) and the entire notebook, (image, data, and profile). If you have *Excel*, Multi-Analyst will open the program and automatically load and label everything for you!



Each report is listed as a separate page in your Excel notebook. Your image is on the first page, the profile is on the second page (along with the data in columnar format) and the report is on the last page.

Congratulations! You have finished the tutorial. We recommend that you spend time with your new software and instrument in order to become comfortable with the windows and the work flow.

15. Tutorial -- Molecular Weight and Mass Determination

Multi-Analyst software can automatically calculate molecular weight and mass values on your samples provided you are using a molecular weight and mass standard such as Bio-Rad's cat #170-8207 Mass Ruler.

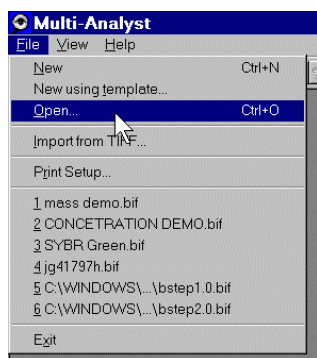
If you are only using a molecular weight standard, you may still perform a relative mass (or concentration) calculation of your samples. There are three ways to do this.

1. Obtain the relative mass % values of all your identified bands with one lane.
2. Obtain the relative mass % values of all your identified bands within one row of your gel.
3. Obtain the relative mass % values of all your identified bands based upon one specific band within the entire gel.
4. Performing your own calculations.

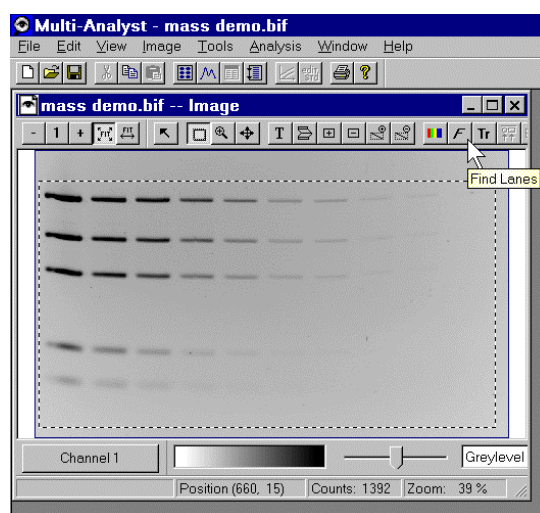
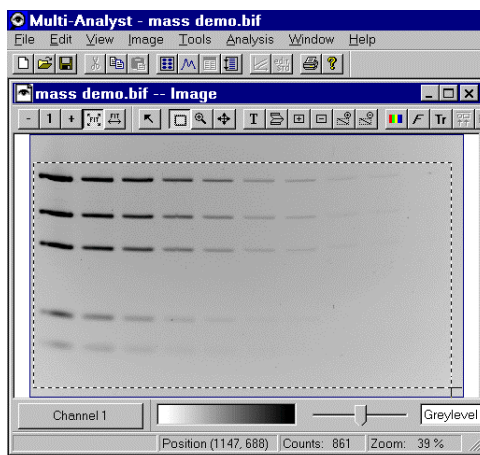
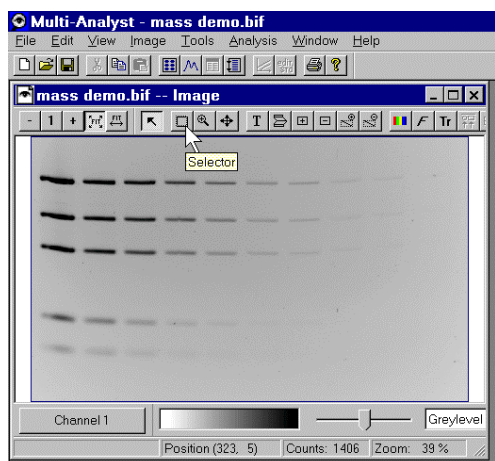
All of these calculations are based upon the area under the peaks (i.e. bands) that are identified in your gel.

15.1 Molecular Weight and Mass based upon a Molecular Weight and Mass standard.

1. Open up your gel that you are going to use. In this case it is *mass demo.bif*.

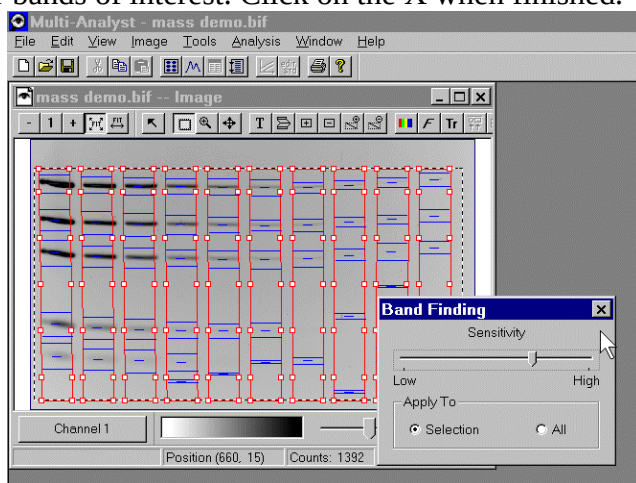


2. Select the area of the gel you want to find your lanes in with the selector tool. This is highly suggested as it will improve the automatic lane and band finding results.

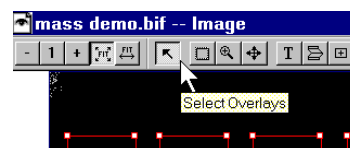
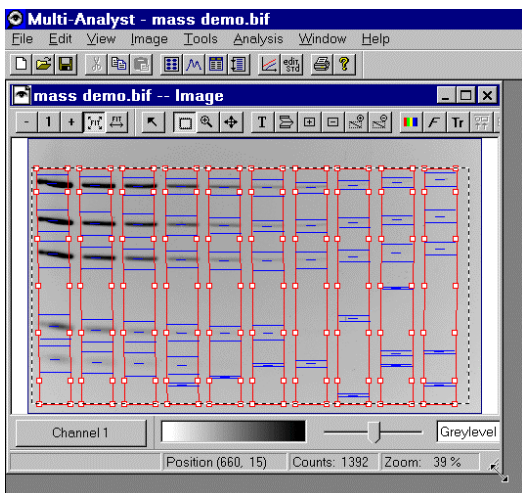


3. Find your lanes with the auto lane and band finding tool.

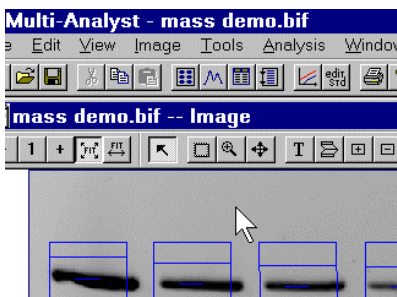
4. Adjust the band finding slider in the band finding window until you are satisfied that it picked up most of your bands of interest. Click on the X when finished.



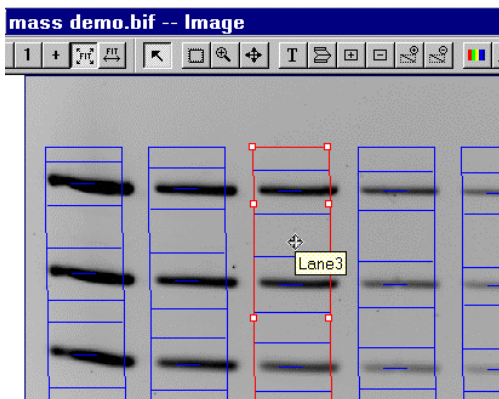
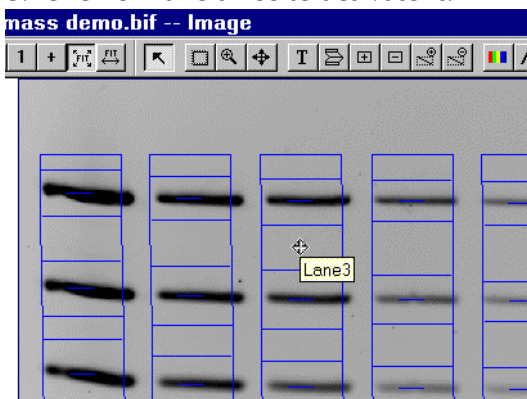
5. Grab the bottom corner and drag it down and to the left to make the image bigger. This is to help you when you are adjusting your lanes.



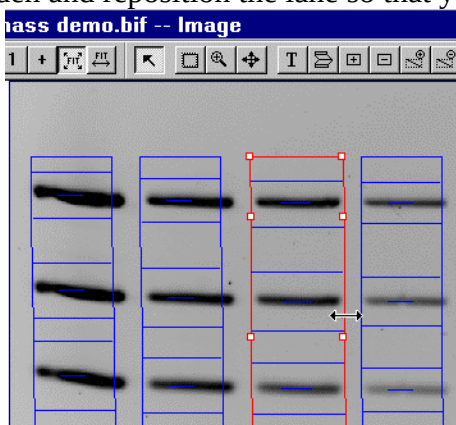
5. Click on the Select overlay tool to work with your lane overlays.
6. Adjust the lanes widths to include the entire width of the band.
 - a. click on the image to deselect everything.



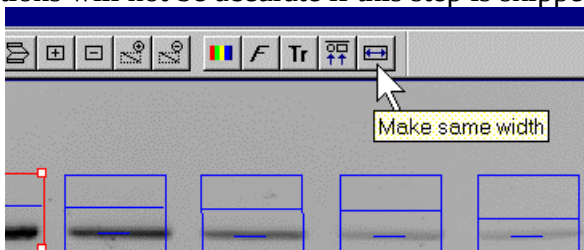
- b. click on lane three to activate it.



- c. Widen and reposition the lane so that you have the entire band within the lane.

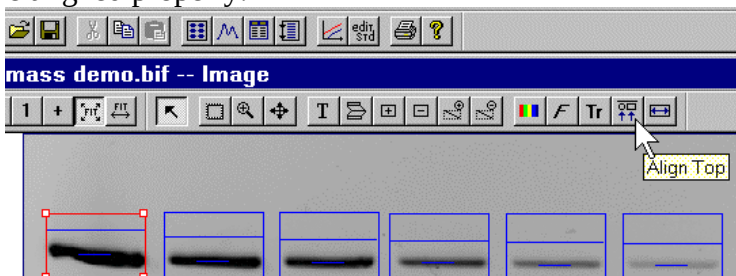


- d. Click on the *Make same width* button to resize all of the lanes .CRITICAL STEP. Your calculations will not be accurate if this step is skipped.

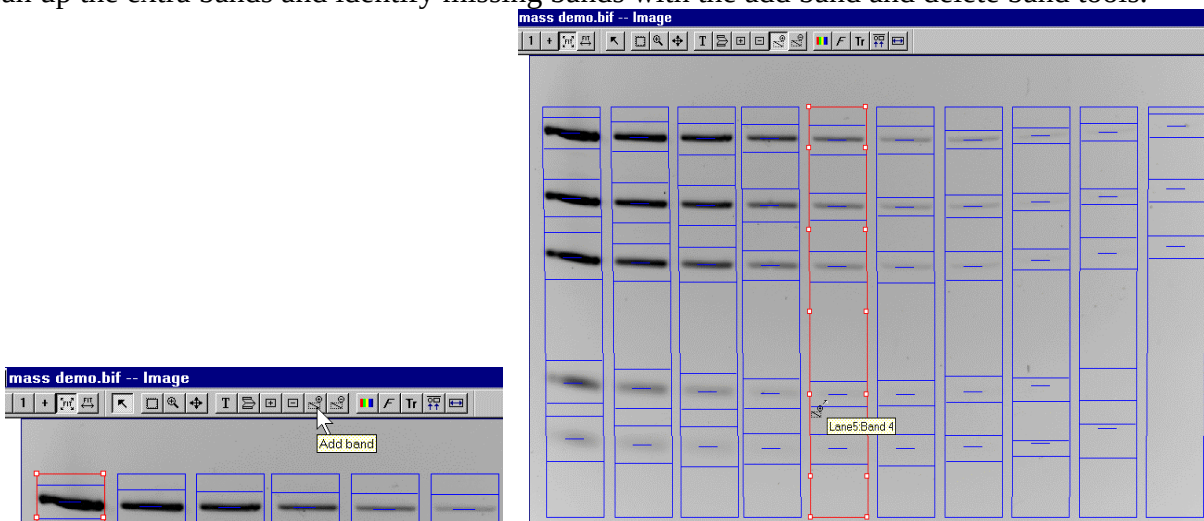


- e. Lanes 1 and 2 are overexposed but you may still want to reposition the lanes so that the entire bands are included in the lane.

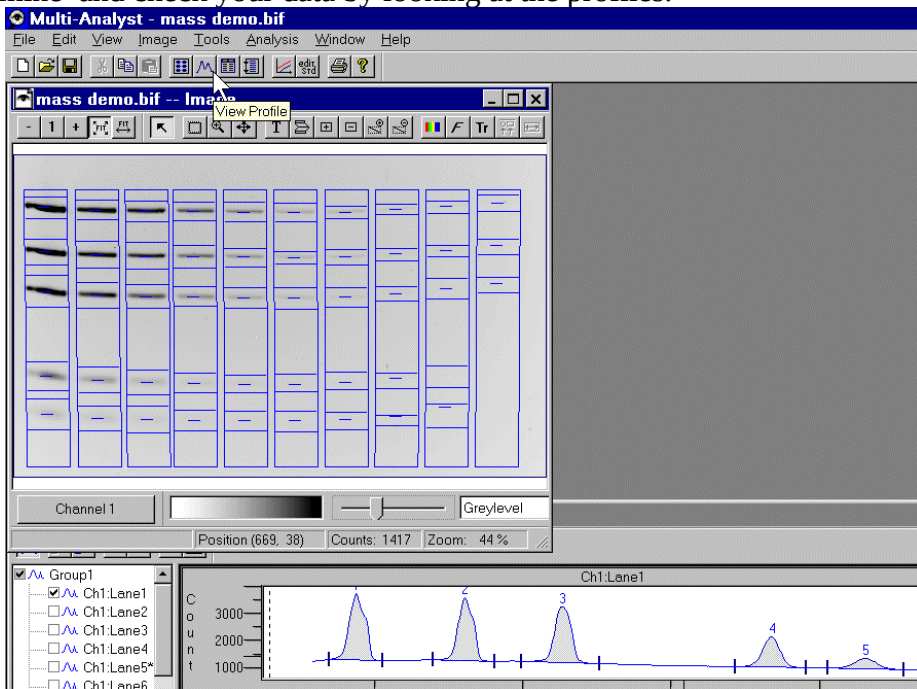
7. If you have moved the lanes around a lot you may want to click on Align Top to make sure all of your lanes are aligned properly.



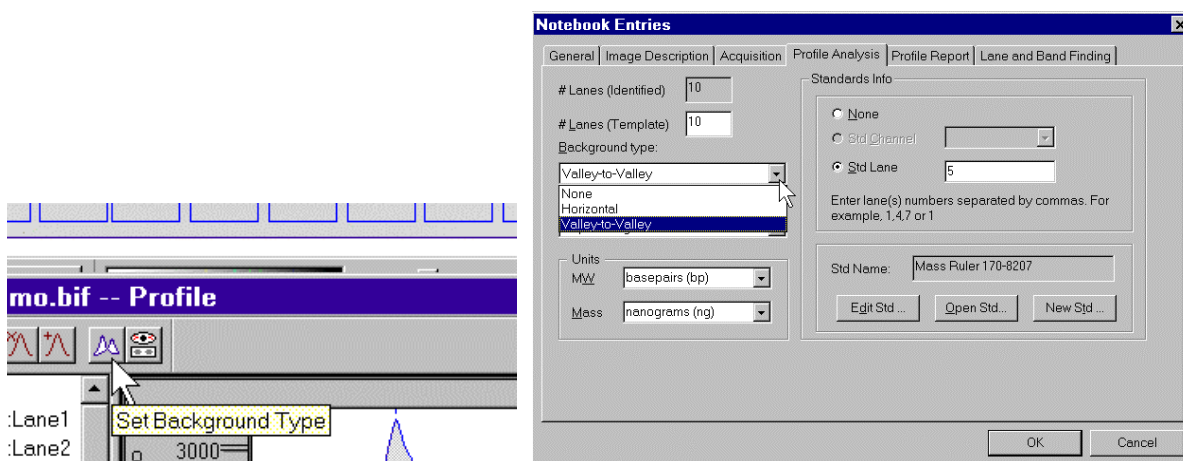
8. Clean up the extra bands and identify missing bands with the add band and delete band tools.



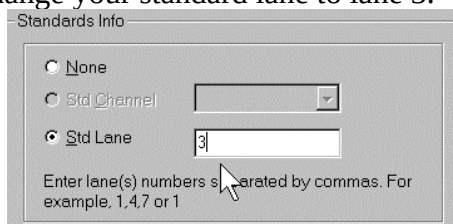
9. Examine and check your data by looking at the profiles.



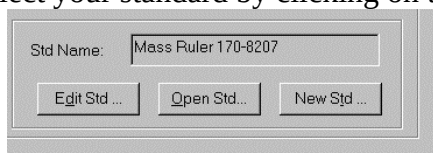
10. Set your background type to valley-to-valley by clicking on the background button and selecting valley to valley in the pull down menu.



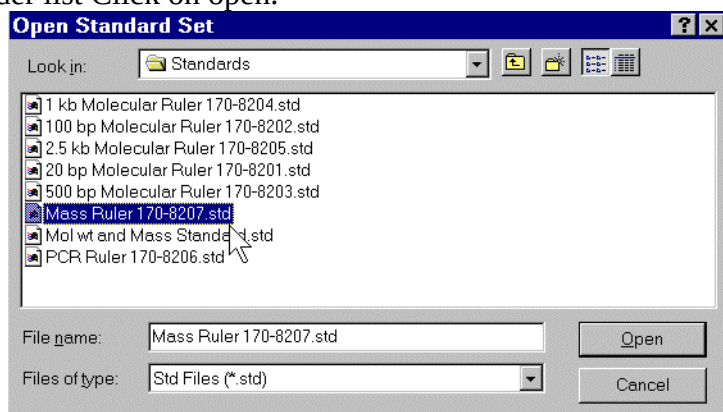
11. Change your standard lane to lane 3.



12. Select your standard by clicking on the open Standard button.



13. This example image used Bio-Rad's Mass Ruler cat # 170-8207, so select that one from the standard folder list Click on open.



14. The standards have already been entered correctly. Click on the OK button.

Standards Info: Mass Ruler 170-8207

	Band Name	Mol.Weight	Mass
1	StdBand 1	1000.00	100.00
2	StdBand 2	700.00	70.00
3	StdBand 3	500.00	50.00
4	StdBand 4	200.00	20.00
5	StdBand 5	100.00	10.00
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			

Std. Name: Mass Ruler 170-8207

Std. Type: ☐ Size Std ☒ Size and Mass Std

Molecular Weight: MW Units: basepairs (bp) Regression Type: Elder Southern
☐ Log (MW)
☐ Log (Mobility)

Mass: Mass Units: nanograms (ng) Regression Type: Linear

Clear Table Save Std... OK Cancel

15. Check to see that your Report settings are set to see the most important data fields by clicking on the Profile Report Tab.

Notebook Entries

General | Image Description | Acquisition | Profile Analysis | **Profile Report** | Lane and Band Finding

Group-Lane Settings: Channel: Channel 1 Lanes: Lane 7

Hierarchy: ☒ Channel-Lanes

Display Options: Mass Related Band(s): ☐ No (Each lane is independent)

16. In the Display Items, make sure Name, Molecular Weight, Area, and Mass are in the Items to be displayed column. You can have more, but having just these four make it easier to see what you are doing.

Display Items

Items not to be displayed: Area % under All Peaks, Area % under Entire Profil, End Position, Height, Mobility, Start Position

Items to be displayed: Name, Molecular Weight, Mass, Area

Select All=> <=Move=> <=Remove All

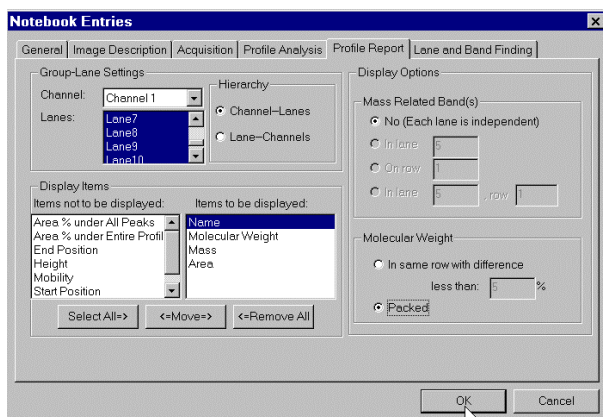
17. In the Molecular Weight Window, make sure the Packed button is clicked.

Molecular Weight

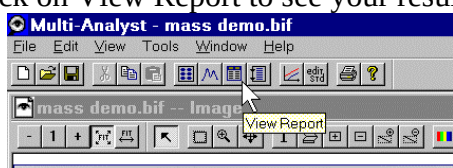
☐ In same row with difference less than: 5 %

☒ Packed

18. Click on OK to close the Notebook Entries Window.



19. Click on View Report to see your results.



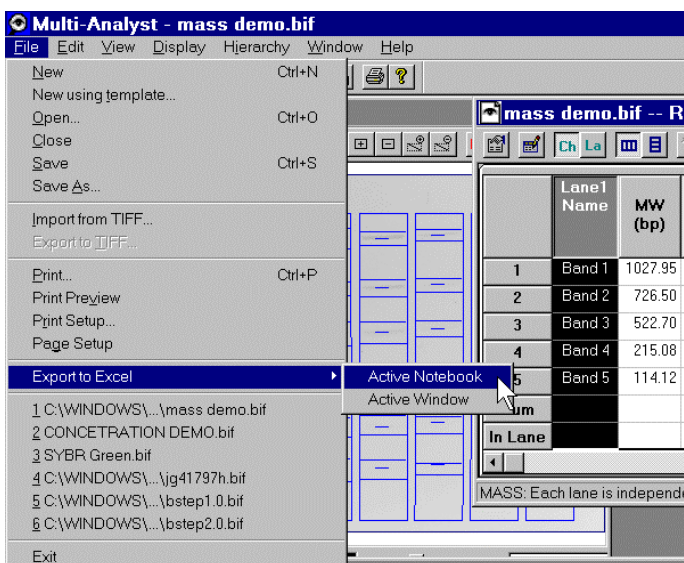
20. If your bands appear in reverse order, you can resort them by double clicking on the column name or by clicking on the sort button.

mass demo.bif -- Report

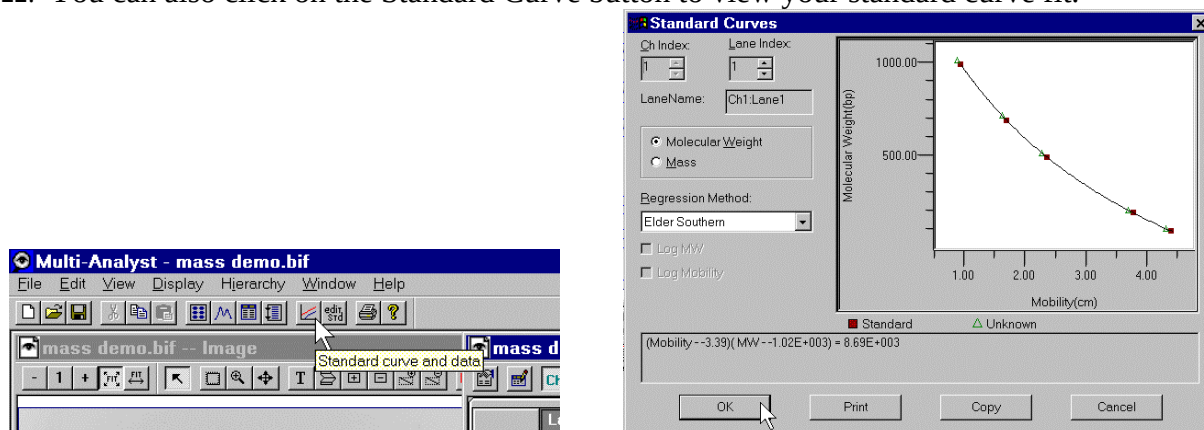
	Lane1 Name	MW (bp)	Mass (ng)	Sort Area cm*Count	Lane2 Name	MW (bp)	Mass (ng)	Area cm
1	Band 1	1027.95	160.29	302.06	Band 1	1006.53	110.71	
2	Band 2	726.50	144.59	274.43	Band 2	708.33	103.35	
3	Band 3	522.70	143.39	272.31	Band 3	506.52	94.94	
4	Band 4	215.08	76.68	154.88	Band 4	204.70	41.20	
5	Band 5	114.12	27.10	67.60	Band 5	104.28	12.69	
	Sum		552.06				362.90	
	In Lane		552.06				362.90	

MASS: Each lane is independent. MOLECULAR WEIGHT: Packed.

21. You have now calculated Molecular Weight and Mass for all of your identified bands. You can print the report or send it to Excel with the File, Export to Excel function.



22. You can also click on the Standard Curve button to view your standard curve fit.

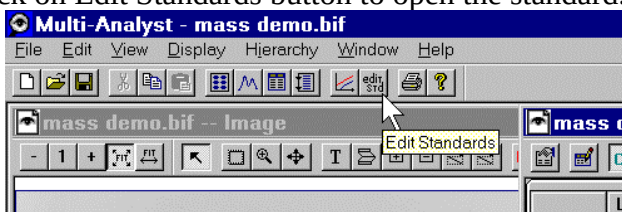


15.2 Calculating Relative Mass Values of all bands Relative to one Lane.

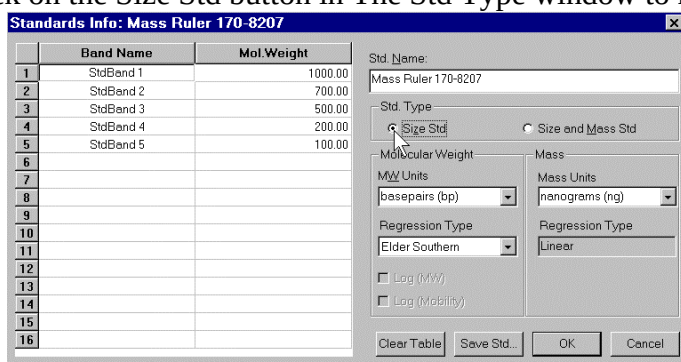
All of the following calculations must be performed with a Molecular Weight Standard and it is assumed you are not using a mass standard.

For ease of use we will use continue to use the same image and file, starting where we left off..

1. Click on Edit Standards button to open the standard.



- Click on the Size Std button in The Std Type window to remove the mass values column.



- DON'T CLICK ON SAVE STANDARD UNLESS YOU WANT TO PERMANENTLY DELETE THE MASS VALUES!

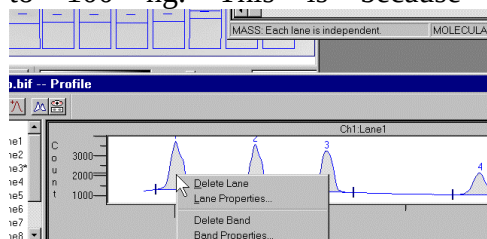
- Click on OK.

mass demo.tif -- Report

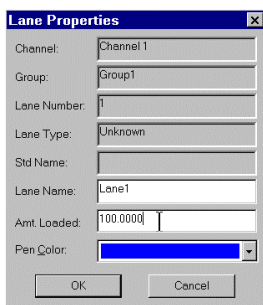
	Lane1 Name	MW (bp)	Mass (ng)	Area cm*Count	Lane2 Name	MW (bp)	Mass (ng)	Area cm*Count	La
1	Band 1	1027.95	28.20	302.06	Band 1	1006.53	29.09	214.79	€
2	Band 2	726.50	25.62	274.43	Band 2	708.33	27.34	201.83	€
3	Band 3	522.70	25.42	272.31	Band 3	506.52	25.33	187.02	€
4	Band 4	215.08	14.46	154.88	Band 4	204.70	12.52	92.41	€
5	Band 5	114.12	6.31	67.60	Band 5	104.28	5.72	42.24	€
Sum			100.00				100.00		
In Lane			100.00				100.00		

MASS: Each lane is independent. MOLECULAR WEIGHT: Packed.

- Scroll through the report window and notice that all of the Mass columns in the report window add up to 100 ng. This is because 100 ng is the default loading value for the lanes.



You can change this load value by right clicking on a lane in the image window or in the profile window and selecting lane properties.



Lane Properties

Channel: Channel 1

Group: Group1

Lane Number: 1

Lane Type: Unknown

Std Name:

Lane Name: Lane1

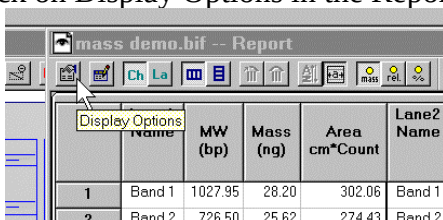
Amt. Loaded: 100.0000

Pen Color: [Blue]

OK Cancel

For this example we will assume you do not know how much you loaded in a lane. Click on OK to continue.

- Click on Display Options in the Report Window to set up how you will calculate relative masses.



	name	MW (bp)	Mass (ng)	Area cm*Count	Lane2 Name
1	Bend 1	1027.95	28.20	302.06	Bend 1
2	Bend 2	726.50	25.62	274.43	Bend 2

- Click on the Display Options Tab.

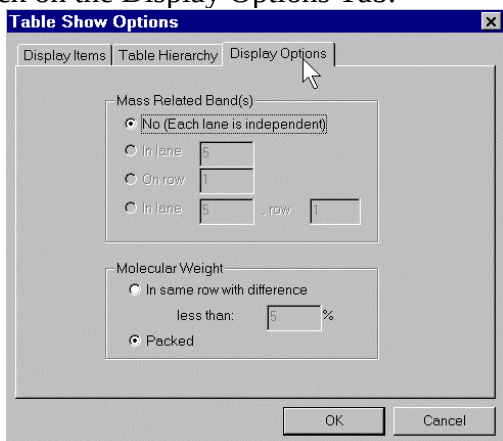


Table Show Options

Display Items | Table Hierarchy | Display Options

Mass Related Band(s)

☒ No (Each lane is independent)

☐ In lane: [5]

☐ On row: [1]

☐ In lane: [5] row: [1]

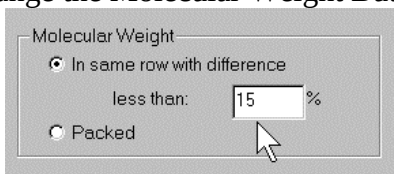
Molecular Weight

☐ In same row with difference less than: [5] %

☒ Packed

OK Cancel

- Change the Molecular Weight Button from packed to In same row with difference less than 15%.



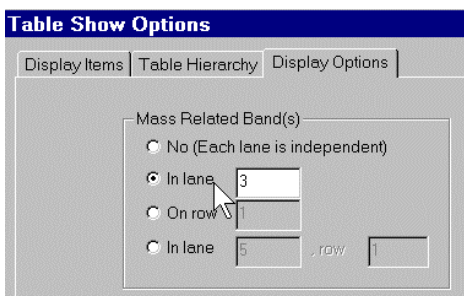
Molecular Weight

☒ In same row with difference less than: 15 %

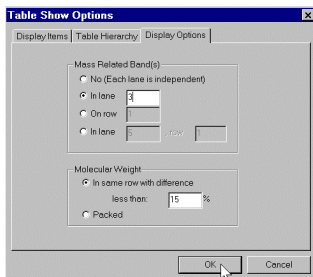
☐ Packed

This function is very useful for comparing related bands across a row. The only way “rows” are defined by Multi-Analyst is by molecular weight. If you use a lower difference number, you are tightening the variance allowed in a band’s molecular weight in order to group it in a row across the lanes of the gel. Lower difference numbers will therefore increase the number of rows, especially at the bottom of a lane where there is more variability in migration.

- Click on the In Lane button, and change the value to 3.



10. Click on OK.



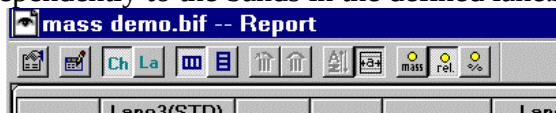
11. All mass values for each lane are now based upon the values done for lane 3. Notice that this is different from what you see in step 5 above because every lane assumes a 100 ng load and now the software is calculating loads based upon Lane 3.

mass demo.bif -- Report

	Mass (ng)	Area cm*Count	Lane3(STD) Name	MW (bp)	Mass (ng)	Area cm*Count	Lane4 Name	MW (bp)	Mass (ng)	Area cm*Count	Lane5 Name	MW (bp)	Mass (ng)	Area cm*Count	Lane6 Name
1	39.81	214.79	StdBand 1	1000.00	33.02	178.17	Band 1	999.48	20.22	109.08	Band 1	992.49	11.96	6455	Band 1
2	37.41	201.83	StdBand 2	700.00	28.95	156.21	Band 2	700.66	15.85	85.54	Band 2	695.58	9.07	48.95	Band 2
3	34.66	187.02	StdBand 3	500.00	23.93	129.10	Band 3	498.55	11.80	63.66	Band 3	494.60	6.84	36.90	Band 3
4	17.13	92.41	StdBand 4	200.00	9.55	51.55	Band 4	194.48	4.24	22.87	Band 4	193.22	2.20	11.87	Band 4
5	7.83	42.24	StdBand 5	100.00	4.54	24.51	Band 5	97.81	1.79	9.64	Band 5	97.81	1.08	5.84	Band 5
Sum	136.84				100.00				53.90				31.16		

MASS: Using lane 3 as a reference. MOLECULAR WEIGHT: In same row with difference less than 15.0%

12. Clicking on the 3 Mass buttons will change the mass calculations from ng, to relative mass, to % mass using $\text{RATIO OF (LANE X, BAND Y) / (LANE 3, BAND Y)}$ where each row is calculated independently to the bands in the defined lane



mass demo.bif -- Report

	Lane3(STD) Name	MW (bp)	Mass (rel)	Area cm*Count	Lane4 Name	MW (bp)	Mass (rel)	Area cm*Count	Lane5 Name	MW (bp)	Mass (rel)	Area cm*Count	Lane6 Name
1	StdBand 1	1000.00	1.00	178.17	Band 1	999.48	0.61	109.08	Band 1	992.49	0.36	6455	Band 1
2	StdBand 2	700.00	1.00	156.21	Band 2	700.66	0.55	85.54	Band 2	695.58	0.31	48.95	Band 2
3	StdBand 3	500.00	1.00	129.10	Band 3	498.55	0.49	63.66	Band 3	494.60	0.29	36.90	Band 3
4	StdBand 4	200.00	1.00	51.55	Band 4	194.48	0.44	22.87	Band 4	193.22	0.23	11.87	Band 4
5	StdBand 5	100.00	1.00	24.51	Band 5	97.81	0.39	9.64	Band 5	97.81	0.24	5.84	Band 5
Sum			1.00				0.54				0.31		
In Lane			1.00				0.54				0.31		

mass demo.bif -- Report

	Lane3(STD Name	MW (bp)	Mass %	Area cm*Count	Lane4 Name	MW (bp)	Mass %	Area cm*Count	Lane5 Name	MW (bp)	Mass %	Area cm*Count	Lane6 Name
1	StdBand 1	1000.00	100.00	178.17	Band 1	999.48	61.22	109.08	Band 1	992.49	36.23	64.55	Band 1
2	StdBand 2	700.00	100.00	156.21	Band 2	700.66	54.76	85.54	Band 2	695.58	31.34	48.95	Band 2
3	StdBand 3	500.00	100.00	129.10	Band 3	498.55	49.31	63.66	Band 3	494.60	28.58	36.90	Band 3
4	StdBand 4	200.00	100.00	51.55	Band 4	194.48	44.37	22.87	Band 4	193.22	23.03	11.87	Band 4
5	StdBand 5	100.00	100.00	24.51	Band 5	97.81	39.35	9.64	Band 5	97.81	23.84	5.84	Band 5
Sum			100.00				53.90				31.16		
In Lane			100.00				53.90				31.16		

13. These values are derived from the area under the peak (i.e. identified bands).

15.3 Obtain the relative mass %values of your identified bands within one row of your gel.

- Continuing on with your work so far.
- Click on Display Options again, and under Mass Related Bands click on row and enter 2.

mass demo.bif -- Report

	Lane2 Name	MW (bp)	Mass (ng)	Area cm*Count	Lane2 Name
1	Band 1	1027.95	28.20	302.06	Band 1
2	Band 2	726.50	25.62	274.43	Band 2

Table Show Options

Display Items | Table Hierarchy | Display Options

Mass Related Band(s)

☐ No (Each lane is independent)

☐ In lane

☒ On row 2

☐ In lane

Molecular Weight

☒ In same row with difference less than: 15 %

☐ Packed

OK Cancel

- Click on OK.
- Now all mass calculations are based on relative intensity of the bands within a lane based upon the 2nd band (i.e. row) within each lane.

mass demo.bif -- Report

	MW (bp)	Mass (rel)	Area cm*Count	Lane3(STD Name	MW (bp)	Mass (rel)	Area cm*Count	Lane4 Name	MW (bp)	Mass (rel)	Area cm*Count	Lane5 Name	MW (bp)	Mass (rel)
1	1006.53	1.06	214.79	StdBand 1	1000.00	1.14	178.17	Band 1	999.48	1.28	109.08	Band 1	992.49	1.
2	708.33	1.00	201.83	StdBand 2	700.00	1.00	156.21	Band 2	700.66	1.00	85.54	Band 2	695.58	1.
3	506.52	0.93	187.02	StdBand 3	500.00	0.83	129.10	Band 3	498.55	0.74	63.66	Band 3	494.60	0.
4	204.70	0.46	92.41	StdBand 4	200.00	0.33	51.55	Band 4	194.48	0.27	22.87	Band 4	193.22	0.
5	104.28	0.21	42.24	StdBand 5	100.00	0.16	24.51	Band 5	97.81	0.11	9.64	Band 5	97.81	0.
Sum		3.66				3.45				3.40				3.
In Lane		3.66				3.45				3.40				3.

5. You change how mass is calculated the same way as you did above by clicking on the Mass, Relative Mass, and % Mass buttons.

	Area cm*Count	Lane3(STD) Name	MW (bp)	Mass %	Area cm*Count	Lane4 Name	MW (bp)	Mass %	Area cm*Count	Lane5 Name	MW (bp)	Mass %	Area cm*Count
1	214.79	StdBand 1	1000.00	114.06	178.17	Band 1	999.48	127.51	109.08	Band 1	992.49	131.87	64.55
2	201.83	StdBand 2	700.00	100.00	156.21	Band 2	700.66	100.00	85.54	Band 2	695.58	100.00	48.95
3	187.02	StdBand 3	500.00	82.64	129.10	Band 3	498.55	74.41	63.66	Band 3	494.60	75.38	36.90
4	92.41	StdBand 4	200.00	33.00	51.55	Band 4	194.48	26.74	22.87	Band 4	193.22	24.25	11.87
5	42.24	StdBand 5	100.00	15.69	24.51	Band 5	97.81	11.27	9.64	Band 5	97.81	11.94	5.84
Sum				345.39				339.93				343.43	
In Lane				345.39				339.93				343.43	

15.4 Obtain the relative mass % values of all your identified bands based upon one specific band within the entire gel.

Continuing on...

1. Click on Display options once again.

	Name	MW (bp)	Mass (ng)	Area cm*Count	Lane2 Name
1	Band 1	1027.95	28.20	302.06	Band 1
2	Band 2	726.50	25.62	274.43	Band 2

2. Click on In Lane and Change values to Lane 3 Row 3.

Table Show Options

Display Items | Table Hierarchy | Display Options

Mass Related Band(s)

☐ No (Each lane is independent)

☐ In lane 3

☐ On row 2

☒ In lane 3, row 3

Molecular Weight

☒ In same row with difference

less than: 15 %

☐ Packed

OK Cancel

3. Click on OK.
4. All mass values are now based upon the Band found at position Lane 3, Row 3.

mass demo.bif -- Report

	Lane3(STD) Name	MW (bp)	Mass %	Area cm*Count	Lane4 Name	MW (bp)	Mass %	Area cm*Count	Lane5 Name	MW (bp)	Mass %	Area cm*Count	Lane6 Name	MW (bp)	Mass %
1	StdBand 1	1000.00	138.01	178.17	Band 1	999.48	84.49	109.08	Band 1	992.49	50.00	64.55	Band 1	992.49	24.00
2	StdBand 2	700.00	121.00	156.21	Band 2	700.66	66.26	85.54	Band 2	695.58	37.92	48.95	Band 2	693.06	16.00
3	StdBand 3	500.00	100.00	129.10	Band 3	498.55	49.31	63.66	Band 3	494.60	28.58	36.90	Band 3	492.63	13.00
4	StdBand 4	200.00	39.93	51.55	Band 4	194.48	17.72	22.87	Band 4	193.22	9.19	11.87	Band 4	193.22	4.00
5	StdBand 5	100.00	18.98	24.51	Band 5	97.81	7.47	9.64	Band 5	97.81	4.53	5.84	Band 5	97.81	2.00
Sum			417.92				225.25				130.22				60.00
In Lane			417.92				225.25				130.22				60.00

5. You can change how mass is calculated the same way as you did above by clicking on the Mass, Relative Mass, and % Mass buttons.

mass demo.bif -- Report

	Area cm*Count	Lane3(STD) Name	MW (bp)	Mass (rel)	Area cm*Count	Lane4 Name	MW (bp)	Mass (rel)	Area cm*Count	Lane5 Name	MW (bp)	Mass (rel)	Area cm*Count
1	214.79	StdBand 1	1000.00	1.38	178.17	Band 1	999.48	0.84	109.08	Band 1	992.49	0.50	64.55
2	201.83	StdBand 2	700.00	1.21	156.21	Band 2	700.66	0.66	85.54	Band 2	695.58	0.38	48.95
3	187.02	StdBand 3	500.00	1.00	129.10	Band 3	498.55	0.49	63.66	Band 3	494.60	0.29	36.90
4	92.41	StdBand 4	200.00	0.40	51.55	Band 4	194.48	0.18	22.87	Band 4	193.22	0.09	11.87
5	42.24	StdBand 5	100.00	0.19	24.51	Band 5	97.81	0.07	9.64	Band 5	97.81	0.05	5.84
Sum				4.18				2.25				1.30	
In Lane				4.18				2.25				1.30	

15.5 Performing your own calculations

Continuing on...

1. Click on Display Options.

2. Click on the Display Items Tab.

Table Show Options

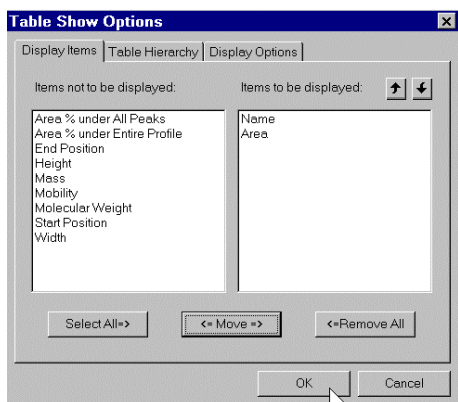
Display Items | Table Hierarchy | Display Options

Items not to be displayed: Items to be displayed:

Area % under All Peaks Name

Area % under Profile Molecular Weight

3. Move Molecular Weight and Mass to the Items not to be displayed column. You should now only have Name and Area showing in the items to be displayed column.

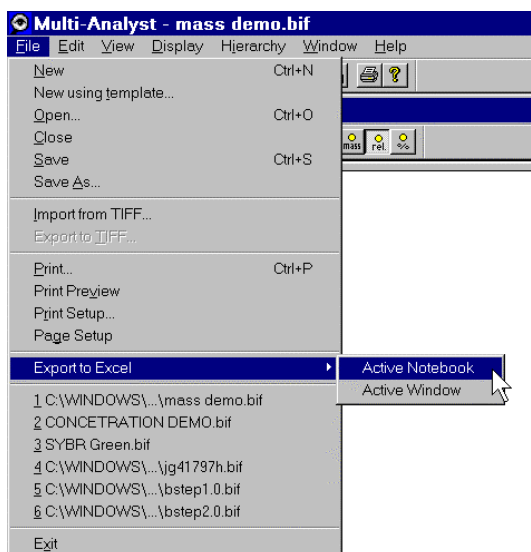


C lick on OK.

5. Click on the Portrait button to redraw the report in portrait mode.

	Name	Area cm*Count
Lane1 1	Band 5	67.60
2	Band 4	154.88
3	Band 3	272.31
4	Band 2	274.43
5	Band 1	302.06
Sum		
In Lane		
Lane2 1	Band 5	42.24
2	Band 4	92.41
3	Band 3	187.02
4	Band 2	201.83
5	Band 1	214.79
Sum		
In Lane		
Lane3(STD) 1	StdBand 5	24.51
2	StdBand 4	51.55
3	StdBand 3	129.10
4	StdBand 2	156.21
5	StdBand 1	178.17
Sum		
In Lane		

6. Click on File, Export to Excel, Active Notebook.



7. Your data is now in an easy to modify format for creating your own data calculations based upon your needs.

Microsoft Excel - Book1

File Edit View Insert Format Tools Data Window Help

Experiment Name mass demo.bif Date: Friday, May 16, 1997 Time: 08:31:41 AM

	A	B	C	D	E	F	G	H	I	J	K	L
1												
2												
3			Name	Area								
4				cm*Count								
5												
6	Lane1	1	Band 5	67.6								
7		2	Band 4	154.88								
8		3	Band 3	272.31								
9		4	Band 2	274.43								
10		5	Band 1	302.06								
11	Sum											
12	In Lane											
13	Lane2	1	Band 5	42.24								
14		2	Band 4	92.41								
15		3	Band 3	187.02								
16		4	Band 2	201.83								
17		5	Band 1	214.79								
18	Sum											
19	In Lane											
20	Lane3(Std)	StdBand 5		24.51								
21		2	StdBand 4	51.55								
22		3	StdBand 3	129.1								
23		4	StdBand 2	156.21								
24		5	StdBand 1	178.17								

Ready Sum=0 NUM

Good Luck!

16. Appendices

16.1 Appendix A Using Concentration Standards

The following cautions should be kept in mind when using any set of standards: The range of O.D.s of standards should always bracket expected O.D. levels in the sample. The greater the number of points in the standard data, the greater the conversion accuracy will be. The standards should be made in the same type of matrix as the sample.

16.2 Appendix B Technical Support

Bio-Rad provides complete technical support on all features of the Multi-Analyst/PC software. If you have any problems or any questions on the use of the features, contact your local Bio-Rad office, or in the U.S. call Bio-Rad Technical Service at 1-800-4BIORAD (1-800-424-6723). Please report any reproducible software problems to Bio-Rad so we can eliminate them from future versions.

16.3 Appendix C Adjusting The CCD Camera Focus When Zooming

This procedure should only be done when focus is completely off when zooming from high to low magnification.

Tools Required: Standard small screwdriver, resolution target (provided with camera), and .060" hex tool (provided with camera).

Purpose of Procedure: Installation of the lens on camera and adjustment of the flange back so that correct focus is held when zooming from high to low magnification.

1. Loosen lens mount set screw on camera body with hex tool.
2. Install zoom lens by screwing lens into camera body.
3. When lens is completely seated in the camera body, continue to rotate lens clockwise to line up zoom, focus, and iris index line with the face of the camera body. (This will be the face with the two (2) 1/4-20 tapped holes). Resistance will be felt as the lens is rotated to line up the index line. The lens will slip through a maximum of 350 degrees.
4. Tighten the lens mount setscrew with the .060" hex tool
5. Screw the lens diopter and other filters onto the zoom lens.
6. Mount the camera/bracket assembly onto the hood. Place the resolution target into the hood. Connect the camera-lens assembly so that a live image can be viewed on the computer monitor.

7. Remove the tracking adjustment hole cover. Loosen the tracking adjustment locking screw with the screwdriver. CAUTION: Do not allow dust or other particulates to accumulate in the camera body during this adjustment procedure.
8. Open up the iris on the lens until maximum illumination (completely open). Source light may need to be reduced in order to prevent saturation of CCD.
9. Rotate the zoom ring to one extreme so that the highest magnification of the resolution target is achieved. Adjust the focus ring for best focus. Do not adjust the tracking adjustment screw at this zoom.
10. Rotate the zoom ring to the other extreme so that the lowest magnification of the resolution target is achieved. Adjust the tracking adjustment screw for best focus.
11. repeat steps 9 and 10 above until rotation of zoom ring requires no adjustment for focus (Generally, this is an iterative procedure and will converge in no more that two steps).
12. Tighten the tracking adjustment screw and replace the tracking adjustment access hole cover.
13. The zoom lens should now be properly installed onto the CCD camera with the focus adjusted properly for high and low magnification.

16.4 Appendix D Adjusting The Mitsubishi P68 Thermal Printer for PC

Rear of printer:

- 1) Mode 1 DSQ pin configuration: All pins are OFF except pin # 2, #7, #8
- 2) Mode 2 DSQ pin configuration: All pins are OFF except pin #5

Front panel of printer:

- 1) Direction: Normal
- 2) Scan: Frame
- 3) Contrast: Normal

Recommended function settings

1. Connect the BNC cable (that came with the printer) from "Video" on the camera to "Video-In" on the rear of the printer. Turn on the CCD camera.
2. Connect the Gel Doc integrating cable to the frame grabber card in your PC.
3. Connect the "Camera Video" cable (1 of the 4 cables of the integrating cable) to "Video-Out" on the rear of the printer.
4. Connect the "Printer Remote" cable (1 of the 4 cables of the integrating cable) to "Remote" on the rear of the printer.
5. Turn on the printer (both rear and front power switches).
6. Start Multi-Analyst/PC/Mac and view a live display using the Acquire command from the FILE menu.
7. Open the front panel of the Mitsubishi printer, press the "Function" button and adjust the settings as seen below. Press the "Set" button to accept the settings.

RS-170, NTSC Systems

USER		USER	
HWidth	794	B-Lev	-3
HTRC	0	W-Lev	+4 or +5
HRes	640	Sharp	+2
HStart	42	AGC	Off
VLines	525	Direct	Off
VStart	32	Dither	Off
VRes	476		
HSize	100		
VSize	76		

CCIR, PAL Systems

USER		USER	
HWidth	960	B-Lev:	-3
HTRC	0	W-Lev:	+4 or +5
HRes	767	Sharp	+2
HStart	75	AGC	Off
VLines	625	Direct	Off
VStart	80	Dither	Off
VRes	508		
HSize	100		
VSize	79		

8. Remove the BNC cable (that came with the printer) and reconnect the integrating cable to its appropriate place as noted on each cable on the integrating cable.
9. “Video Print” a saved image to test the printer configuration. If the printout is too light or too dark, repeat the above steps and adjust the “W-Lev” appropriately.

16.5 Appendix E Settings for Mitsubishi P90W Thermal Printer

There are three settings for the P90. Set Contrast to zero, brightness to zero and gamma to 5.

The dip switches should stay in the orientation in which they are shipped. This is so that Pin 1 is up (on) and 2 through 10 are down (off).

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